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EMERGENCY MEDICAL SERVICE PROTOCOLS

2026 REVISION V3

08/17/2026

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PROFESSIONALISM

EMS is really all about service. People like us, who have chosen to serve in EMS, have a moral and ethical obligation to behave in ways that serve the needs of others and reflect positively upon our organization, our community, and the whole discipline of EMS. This is not easy. In fact, it's a constant struggle that takes significant time, dedication, and effort. The choices we make and the manner in which we behave are a constant reflection of our professionalism. The following paragraphs were taken from Dr. Brain Bledsoe's "Paramedic Emergency Care Second Edition," and establish the foundation that all providers (volunteer or career) should strive to achieve and maintain.

"Professionalism describes the conduct or qualities that characterize a practitioner in a particular field or occupation. Health care professionals promote quality patient care and take pride in their profession. They earn the respect and confidence of team members by performing their duties to the best of their abilities and by exhibiting a high level of respect for their profession. Attaining professionalism is not easy. It requires an understanding of what distinguishes the professional from the non-professional. To develop this skill, keep the following points in mind. Professionals place the patient first; non-professionals place their egos first. Professionals practice their skills to the point of mastery, and then keep practicing them to improve and remain sharp. Non-professionals do not believe their skills will fade and see no reason to constantly strive for improvement. Professionals understand the importance of response times; non-professionals get to an accident when it's convenient. Professionals take refresher courses seriously, because they know they have forgotten a lot and because they are eager for new information. Non-professionals believe they don't need training sessions and dislike being required to attend them. Professionals critically review their performance, always seeking a way to improve. Non-professionals look to protect themselves, to hide inadequacies, and to place blame on others. Professionals check out their equipment prior to the emergency response. Non-professionals hope that everything will work, supplies will be in place, batteries will be charged, and oxygen levels will be adequate. Maintaining professionalism requires effort. But, the result of that effort - the admiration and respect of one's peers - is the highest compliment a person can receive."

The NHTSA's National Standard Curriculum lists the following as **professional behaviors**:

- Integrity
- Empathy
- Self-motivation
- Appearance and personal hygiene
- Self-confidence
- Communications
- Time management
- Teamwork and diplomacy
- Respect
- Patient advocacy
- Careful delivery of services.

Being a professional has nothing to do with pay or rank or the level of certification that you hold. It is the goal that every member of our practice, from EMT-Basic to the Medical Director, to constantly strives for and remain a comprehensive, clinically sophisticated, and compassionate EMS System.

PROTOCOL AUTHORIZATION

The Gladstone Fire Department has developed the following set of medical protocols. These protocols have been reviewed and approved for use by Dr. Jared White, who serves as Medical Director.

Medical Directors Statement of Approval and Compliance

It is to our knowledge, all Gladstone Fire Department personnel meet or exceed the requirements as prescribed by the Missouri Bureau of EMS, and are licensed in good standing as required through policy, protocol, continuing education and skills competency evaluation.

I have reviewed and approve the medical protocols for the Gladstone Fire Department.

Approval Date: 08/17/2026

Revision Date: 08/17/2026

On file

Dr. Jared White, Medical Director

On file

Mike Desautels, Fire Chief

SYSTEMS DESIGN

The City of Gladstone shall provide the Gladstone Fire Department an Advanced Life Support (ALS) ambulance service. This service shall be ordered on calls for service within the City of Gladstone, as well requests from outside agencies for mutual aid. This service shall provide for a timely response, quality patient care and patient transports to an appropriate medical facility. In the event the fire department is unable to respond, a mutual aid provider shall be requested.

Non-emergency response shall include but not necessarily limited to the following:

1. Scheduled direct admissions to a hospital.
2. Hospital to home transfers, or nursing home to extended care facility transfers.
3. Home to hospital for medical tests.

Emergency responses shall include but not limited to the following:

1. Any call involving trauma resulting from an accident, fall, sports injury, burn, or etc., incident that results in injury requiring immediate medical attention.
2. Any call involving an illness resulting from diabetes, heart condition, strokes, seizures, or etc., any serious illness requiring immediate medical attention.

MEDICAL CONTROL FORMAT

All patient treatment and transport shall follow guidelines as listed in current protocol. [Medical Control](#) shall be contacted when indicated in the individual protocol, or when any problem or question exists. This is to ensure quality patient care on-scene, enroute to and upon arrival at the destination hospital.

When transporting a patient to a hospital other than the medical control hospital and orders are to be requested, the “other” receiving hospital should be contacted directly.

The following information should be transmitted to the medical control hospital in all emergency situations:

1. Paramedic’s name and unit/service.
2. Expected arrival time.
3. Patient’s chief complaint and brief history.
4. Patient’s condition (including GCS).
5. Patient’s vital signs.
6. Findings of exam and treatment.
7. Request for additional orders or questions.

MEDICAL CONTROL

North Kansas City Hospital (NKCH) shall serve as the primary medical control hospital for the Gladstone Fire Department. All references within medical protocols unless otherwise stated, shall be directed toward North Kansas City Hospital.

Liberty Hospital (LH) shall serve as a secondary medical control hospital for the City of Gladstone, emergency medical service. If unable to contact North Kansas City Hospital, requests for orders shall be obtained from Liberty Hospital.

If the patient requests transport to an alternate hospital, medical control at the receiving hospital should be contacted for orders. If unable to obtain orders from the receiving facility, contact North Kansas City Hospital medical control for orders.

For Pediatric patients (under the age of 18), Children's Mercy Hospital (CMH) shall serve as the primary medical control hospital.

DRUG ACCOUNTABILITY

The purpose of this protocol is to provide guidelines for the security, inventory and record keeping procedures for all scheduled narcotic medications in accordance with Fire / EMS Division policy F458 - Schedule Drug.

1. The following are considered scheduled narcotic drugs requiring strict accountability:
 - A. Morphine Sulfate
 - B. Versed - 2mg vials
 - C. Versed - 5mg vials - Nasal
 - D. Fentanyl - 100 mcg vials
 - E. Ketamine - 500mg vials
2. The drugs listed above shall be kept in the following manner:
 - A. Limited quantities shall be maintained in a sealed/tagged tamper proof container. This container shall be stored inside the Narcotic Safe located within the patient compartment of the ambulance.
 - B. The following shall be considered **minimum** quantities to be carried in the ambulance at all times. Supply shall be restocked as soon as possible.
 1. 4 Morphine Sulfate
 2. 4 Versed 2mg
 3. 4 Versed 5mg - Nasal
 4. 4 Fentanyl - 100mcg
 5. 2 Ketamine vials - 500mg
3. Daily inventory and drug accountability shall be the responsibility of the off-going and on-coming paramedic. Inventory and accountability shall be completed in the following manner:
 - A. At the beginning of each shift the off going and the on coming paramedics shall conduct a joint inventory of the scheduled drugs on the ambulance.
 - B. As the inventory is completed, each paramedic shall sign the logbook. This log shall contain the following:
 1. Date and time
 2. Quantity of drug
 3. Lock identification #
 4. Paramedics' signatures and employee #
 - C. The inventory procedure shall be completed at any time there is a change in the on duty paramedic regardless of the time of day.

- D. On the 1st day of every month in the presence of an authorized officer the on duty paramedic shall open the sealed/tagged narcotic box and perform a physical inventory; also inspecting for signs of tampering and expired narcotics (see below). The unit container shall be restocked / resealed and documented in the logbook accordingly.

Broken Seal Procedure:

- A. If a seal is found to be broken, the paramedic is to immediately notify a supervisor.
- B. The supervisor and paramedic will together conduct an audit of the narcotics box and verify the contents.
- C. The supervisor will make an entry in the narcotic log to document the broken seal; both the supervisor and medic will sign the log.
- D. The supervisor will draft a memo to the Fire Chief going through the Chain of Command for his/her shift.(CC Narcotics Officer) explaining the incident.
- E. At the discretion of the Fire Chief, a sample of a narcotic(s) may be sent to the Missouri State Highway Patrol lab for content verification.

Expired Medication Procedure

- A. In the event expired narcotics are found during off hours, they shall be taped and marked as “expired do not use” placed back in the tagged narcotic container until they can be placed into the Supply Safe expired inventory.
 - B. In the presence of an authorized officer the expired narcotics will be removed and deducted from the ambulance inventory and noted as placed into the expired inventory.
 - C. The ambulance inventory shall then be restocked, re-tagged and logged, a piece of tape will be placed on the outside of the container with the name of the medications and expiration dates clearly listed.
 - D. Expired narcotics will then be placed into the expired container located in the Supply Safe and logged in the “Expired Narcotics Log,” this is located in the back of the Supply Safe log. The expired container shall be re-tagged, logged and re-secured.
4. Drug administration records shall be maintained on all scheduled drugs. These records shall be completed at the time of care and shall indicate the following:
- A. Date and time of administration.
 - B. Name of patient.
 - C. Physician giving order for administration if applicable.
 - D. Quantity administered.
 - E. Paramedic’s signature.
 - F. All unused portions shall be discarded at the hospital prior to returning to service. The amount discarded and the signature of the witness must be on the usage form. It is **preferred HOSPITAL STAFF WITNESS DISPOSAL**, If not possible have a supervisor witness.

HELICOPTER UTILIZATION

1. Indications

- A. Need for optimum speed of response and transport. (Keep in mind helicopter response and loading time).
- B. Impaired ground accessibility due to location, traffic or condition of roads.
- C. Shortage of ground transport units and there is a need for rapid transit.
- D. Need to achieve distribution of patients among area hospitals. (Mass casualty Incident).

2. Procedures

- A. It is the responsibility of the attending Paramedic and the Incident Commander to assess the situation and determine the need for air ambulance transport.
- B. If an air ambulance is needed it is the responsibility of the Incident Commander to contact dispatch and request it.
- C. It shall be the responsibility of the Incident Commander or designee along with law enforcement to establish a suitable landing site. This site should be:
 - 1. 100 feet square.
 - 2. Free of overhead obstructions.
 - 3. Accessible to aircraft for patient transfer.
 - 4. A fire apparatus with foam application capabilities should be available.

PATIENT DESTINATION CRITERIA

1. General

- A. For all life-threatening emergencies transport the patient to the nearest appropriate facility.
- B. **When the nearest appropriate facility conflicts with the choice of the patient or a person acting in the behalf of the patient, contact Medical Control.**
- C. For all non-life-threatening emergencies, transport may be made to the following facilities: North Kansas City Hospital, St. Luke's Northland, Liberty Hospital, Truman/Truman West, Veterans Medical Center- Kansas City, St. Luke's-Plaza, K.U. Medical Center, and Research.

2. Trauma Criteria

- A. Any trauma patient meeting the following criteria should be considered for transport to a Level I/II trauma center (NKCH, Liberty, KUMC or Truman)(Children's Mercy Hospital for patients under the age of 18) even if not the nearest facility, a high index of suspicion should be maintained regardless of time lapsed from point of injury to EMS patient contact:

Trauma Activation

- Trauma code
- GCS < 8 with MOI or other signs of shock.
- Confirmed & sustained systolic BP <90 & age specific for pediatrics.
- Respiratory compromise/obstruction and/or intubation Resp rate >29 or <10.
- GSW to head, neck, or torso.
- BP <90, Absent peripheral pulses.
- Fall with head injury of patient taking anticoagulants.

Trauma Alert

- Confirmed active hemorrhage
- GCS <14
- Extremity trauma with loss of distal pulse or sensation.
- MVC with:
 - Major auto deformity >1 foot
 - Unrestrained rollover
 - Death in same passenger compartment
 - Extrication >20 minutes
 - Ejection from vehicle
- MCC or ATV crash >20 mph or separation of rider from vehicle.
- Auto vs. pedestrian/bicycle
- Penetrating injury (i.e. stab wound) to head, neck, or torso.
- Major burns >20% BSA or involving inhalation injury.
- Falls > 3 feet.
- Pelvic fractures
- Open fractures
- Pregnant trauma patients >20 weeks gestation

PATIENT DESTINATION CRITERIA CONT:

- B. If NKCH, Liberty, and Truman are all on diversion, consider KU or St. Luke's on the Plaza for patients meeting criteria.
- C. Trauma codes should be taken to the nearest Level I/II trauma center whether the trauma center is on diversion status or not.
- D. Patients who meet the trauma routing criteria and are less than 18 years of age shall be transported to a pediatric trauma center (Children's Mercy Hospital).

HOSPITAL TCD (Time Critical Diagnosis) DESIGNATION

LEVEL 1 TRAUMA CENTERS:

*KU Medical Center (PRIMARY Burn Center and Hyperbaric Chamber)

*St. Luke's - Plaza

*Research Medical Center (Secondary Burn Center)

*Truman Medical Center

LEVEL 2 TRAUMA CENTERS:

*North Kansas City Hospital (Primary)

*Liberty Hospital (Secondary)

ACUTE STEMI

*North Kansas City Hospital (Primary)

*St. Luke's - Plaza (Secondary)

*KU Medical Center (Secondary)

*Liberty Hospital (

*St. Luke's Northland

ACUTE CVA

*St. Luke's - Plaza (Primary)

*KU Medical Center (Secondary)

*Research Medical Center (Secondary)

**St. Lukes is the definitive stroke treatment center, however if a pt refuses to go there then they may be taken to a stroke center of their choice. If they wish to go to NKCH educate them that there may be the possibility they could be transferred out of if they have certain problems. **

Note: Patients meeting routing criteria SHALL NOT be taken to undesignated hospitals w/o medical control (physician) approval and/or contact with the receiving physician.

Note: Patient preference and continuity of care should always be considered, for example the patient has received all cardiac care at KU Medical Center, it would be beneficial for the patient to be transported to KU even though it is not the closest facility. When in doubt contact medical control.

BIOHAZARD EXPOSURE PROTECTION

See Department SOG F-311

PRE-HOSPITAL DEATH DETERMINATION

Patients who have obviously expired will not be resuscitated or transported by Gladstone EMS personnel if any of the following obvious signs of death are present:

Trauma Related

- A. If there is any doubt the arrest state is traumatic or any question concerning any sign of life at the scene, then resuscitation should be initiated and Medical Control contacted.
- B. A patient with obvious traumatic injuries not consistent with human life, for example decapitation, incineration, transection of thorax (hemicorpectomy), massive trauma with visible organ damage; who is asystolic (confirmed in three leads) or has no audible heart sounds and who has not been observed at the scene to possess any sign of life (verbal or motor response, respiratory effort, or palpable pulse) may be pronounced DOA.
- C. If unable to begin CPR in a reasonable amount of time due to patient accessibility or location, contact Medical Control and notify them of patient's condition and circumstances that prohibit appropriate resuscitation.
- D. Documentation of any of the above findings shall occur in the ambulance run report.

Medical Related

- A. A patient in cardiopulmonary arrest due to medical causes may be pronounced DOA if they are asystolic (confirmed in three leads) or have no audible heart sounds and also have one of the following signs:
 - 1. Rigor mortis
 - 2. Dependent lividity
 - 3. Decomposition
- B. Rigor mortis and asystole are not reliable signs of death in the hypothermic. If hypothermia is a consideration, resuscitation should be initiated.
- C. Generalized tissue decomposition is not compatible with life; the patient may be pronounced DOA without cardiac monitor application.
- D. **If rigor mortis, lividity, or tissue decomposition is present, medical control contact is not necessary.**
- E. Any of the above findings shall be documented in the ambulance run report.

For all emergency scenes where patient needs exceed available EMS resources, initial assessment and treatment shall be in accordance with an approved triage methodology.

REPORT WRITING

Documentation is one of the most important tasks that we perform as pre-hospital providers. While there are many reasons to document, the main purpose of documentation is to provide an accurate, comprehensive, and permanent record of each patient's condition and the treatment rendered. Always remember if it's not written down it didn't happen....

Any pertinent notes and information should be left with the receiving hospital. This would include any cardiac strips in cases of a STEMI. Signatures from the receiving hospital should be captured electronically on currently RMS System. The formal EMS report shall be completed prior to the end of shift.

Listed below is the minimum criterion required for Gladstone EMS reports.

- A. How the call was received from dispatch, pre-arrival information, units responding location and mode of response.
- B. Nature of the call on arrival.
- C. How the patient was found, presenting signs, appearance and pertinent scene factors.
- D. The patient's chief complaint.
- E. HPI & PPMH
- F. Two sets of vital signs. This includes: blood pressure, pulse rate, respiratory rate, and SpO2 and EtCO2 readings if applicable.
- G. IV attempts, location, patency and who initiates.
- H. Intubation attempts, ET tube location measured from the teeth in cm, and confirmation methods.
- I. Medication administration, name, dose, route and results.
- J. Procedures performed, by whom and results.
- K. The patient's overall response to treatment and transport.
- L. Orders requested/received, and given by whom.
- M. Attending Paramedic shall end the report with their full name, title and state license number (P#).

All calls shall be documented in the first person, past tense and in chronological order, from the time of dispatch through return to service.

All reports shall be written utilizing proper grammar, spelling and only Department approved abbreviations.

05/23/2025

GFD RECOGNIZED ABBREVIATIONS

AAA	Abdominal Aortic Aneurysm
AIDS	Acquired Immune Deficiency Syndrome
AMI	Acute myocardial infarction
ACLS	Advanced cardiac life support
ALS	Advanced life support
ABC	Airway, Breathing, Circulation
AC or ACF	Antecubital or Antecubital Fossa
A+Ox3	Alert and oriented to person, place and time
APGAR	Appearance, Pulse, Grimace, Activity, Respiration
ASA	Aspirin
ASAP	As soon as possible
A-FIB	Atrial fibrillation
BVM	Bag valve mask
BLS	Basic life support
Bicarb	Sodium Bicarbonate
BP	Blood pressure
BM	Bowel movement
CA	Cancer
CPR	Cardiopulmonary resuscitation
CATH	Catheter/Catheterization
cm	Centimeter
CNS	Central nervous system
CVA	Cerebral vascular accident
CSF	Cerebrospinal fluid
C-Collar	Cervical collar
C-Spine	Cervical spine
CP	Chest pain
COPD	Chronic obstructive pulmonary disease
C/O	Complains of
CHF	Congestive heart failure
CPAP	Continuous positive airway pressure
CABG	Coronary artery bypass graft
CAD	Coronary artery disease
DOA	Dead on arrival
DVT	Deep vein thrombosis
DTs	Delirium tremors
DM	Diabetes mellitus
DC	Discontinue
mg/dL	Milligrams per Deciliter
DO	Doctor of Osteopathy
DNR	Do not resuscitate
DX	Diagnosis
Gtts	Drops
EKG/ECG	Electrocardiogram
EMS	Emergency medical services
EMT	Emergency medical technician
EMT-P	Emergency medical technician paramedic
ER	Emergency room
ET	Endotracheal
EPI	Epinephrine

7/25/2023

=	Equal
EOA	Esophageal obturator airway
ETA	Estimated time of arrival
EtCO ₂	End tidal CO ₂
ETOH	Ethyl alcohol
EJ	External jugular
FD	Fire department
FX	Fracture
GI	Gastrointestinal
Ga	Gauge
GU	Genitourinary
G#P#	Gravida # Para #
GYN	Gynecologist
HX	History
HPI	History of present illness
HIV	Human immunodeficiency virus
HTN	Hypertension
ID	Identification
ICU	Intensive care unit
ICP	Intracranial pressure
IM	Intramuscular
IN	Intranasal
IO	Intraosseous
IV	Intravenous
IVP	IV Push
JVD	Jugular vein distention
KED	Kendrick extrication device
kg	Kilogram
LR	Lactated ringers
LMP	Last menstrual period
<	Less than
LOC	Level of consciousness-loss of consciousness
LPM	Liters per minute
L-Spine	Lumbar spine
MD	Medical doctor
MEDS	Medications
mcg	Micrograms
meq or mEq	milliequivalent
MAST	Military anti-shock trousers
mg	Milligram
MI	Myocardial infarction
ML/ml	Milliliter
mm	Millimeter
>	More than
AM	Morning
MVC	Motor vehicle collision
NC	Nasal cannula
NG	Nasogastric
N&V	Nausea and vomiting
NTG	Nitroglycerin
NKA or NKDA	No known allergies, No Known Drug Allergies

Ø	None
NRB	Non-rebreather mask
NS	Normal saline
NSR	Normal sinus rhythm
OB	Obstetrics
PO	Orally by mouth
oz	Ounce
OD	Overdose
O2	Oxygen
SpO2	Oxygen Saturation Percentage
PMH	Past medical history
PT	Patient
PID	Pelvic inflammatory disease
PCN	Penicillin
PASG	Pneumatic anti-shock garment
PD	Police department
+	Positive
KCL	Potassium chloride
Lbs	Pound
Lpm	Liters per minute
PAC	Premature atrial contraction
PJC	Premature junctional contraction
PVC	Premature ventricular contraction
RX	Prescription
1°	Primary, first
PERL	Pupils equal, reactive to light
PO	Per Oral
ROM	Range of Motion
RN	Registered nurse
RESP	Respiration, respiratory
R/O	Rule out
RSR	Regular Sinus Rhythm
2°	Second, secondary
SOA	Shortness of air or breath
SOP	Standard operating procedure
SQ	Subcutaneous
SL	Sublingual
SIDS	Sudden infant death syndrome
SIVP	Slow IV Push
SVT	Supra ventricular tachycardia
3°	Tertiary, third
X	Times
TKO	To keep open
TIA	Transient ischemic attack
TX	Treatment
TB	Tuberculosis
URI	Upper respiratory infection
UTI	Urinary tract infection
VD	Venereal disease
V-FIB	Ventricular fibrillation
V-TACH	Ventricular tachycardia
WT	Weight
Y/O	Years old
D5W	5% dextrose in water

PRE-HOSPITAL TERMINATION OF RESUSCITATION

Numerous studies have proven that a certain population of cardiac arrest patients will not survive hospital discharge if they fail pre-hospital therapy. Patients that meet the criteria below shall be eligible for termination of resuscitative efforts in the field and do not require transport to an Emergency Department. If there is any doubt, continue resuscitative efforts and transport.

Inclusion Criteria

- A. Primary cardiac arrest.
- B. Age >21 years of age.
- C. Either asystole or PEA as presenting rhythm.
- D. Asystole must be the terminal rhythm.
- E. ETT intubation.
- F. IV or IO access.
- G. > 20 minutes of appropriate therapy.
- H. Capnography reading less than 10 mm/Hg

Exclusion Criteria

- A. Non-primary cardiac arrest (cause of arrest possibly attributed to other pathology). Including but not limited to the following:
 - 1. Hypothermia
 - 2. Drug overdose (possible acute overdose).
 - 3. Primary respiratory arrest (i.e. status asthmaticus).
 - 4. Carbon monoxide poisoning
 - 5. Traumatic arrest.
 - 6. Drowning
- B. Ventricular fibrillation or ventricular tachycardia at any time during resuscitation.
- C. Any return of spontaneous circulation during resuscitation.
- D. Any neurologic activity.
- E. Family desires transport (either voiced or perceived).
- F. Other, such as unstable scene or disagreement among family members.
- G. EMS responder witnessed cardiac arrest.

Procedure

- A. Follow appropriate cardiac arrest protocol to include:
 - 1. Successful early intubation.
 - 2. Successful early IV drug therapy.
 - 3. Minimum of 20 minutes of appropriate therapy.
- B. If the patient fits the inclusion criteria and does not fit any exclusion criteria and the above procedures have been accomplished the paramedic may initiate the termination of resuscitation.
- C. **Paramedic does not need to contact medical control if all criteria is followed.**
- D. The appropriate law enforcement agency must be notified.
- E. Rhythm strips must accompany report and report must contain the all proper documentation.

PHYSICIAN ON SCENE

If a physician is found in attendance of a patient at the scene (excluding a physicians office), or a physician comes to the scene after the paramedic has initiated treatment, the paramedic will:

- A. Advise the physician of your treatment, medical protocols, and additional directives received from Medical Control while continuing patient care.
- B. If a physician on scene desires to give medical direction regarding the patient care, the paramedic is to explain to the physician that you are functioning under strict medical control protocols. In order for the physician to assume patient care, the physician must be willing to assume full responsibility and liability for the patient. This includes accompanying the patient to the hospital. **Medical Control must be contacted immediately and made aware of the situation.** Ultimate determination of physician shall be made through Medical Control.
- C. If a physician in their office is treating a patient and orders are given, they shall be relayed to Medical Control. If a conflict in orders should arise, Medical Control shall overrule all other orders.

QUALITY IMPROVEMENT PROCEDURE

The Quality Assurance/ Quality Improvement (QA/QI) Program is a systematic evaluation to ensure excellence in delivery of patient care and operations of the Gladstone EMS system.

Quality Assurance is a formal methodology designed to assess the quality of services provided and includes formal review of care, problem identification, corrective actions to remedy any deficiencies and evaluation of actions taken.

Quality Improvement is a continuous process that identifies problems in health care delivery, examines solutions to those problems, and regularly monitors the solutions for improvement. Our program will focus on ways to improve pre-hospital care rather than administering discipline and destructive criticism.

- A. To continually provide suggestions for improvement in the quality of medical care rendered.
- B. To improve patient care by adjusting protocols, treatment decisions, and procedures as needed.
- C. To ensure proper medical documentation.

Procedure

- A. The EMS Quality Assurance Manager, Medical Director or his designee shall review the following categories of calls.
 - 1. All cardiac arrests.
 - 2. Any call marked life threatening.
 - 3. All calls involving any new medication or procedure for the first six months of use.
 - 4. Any call that varies from protocol.
 - 5. Any call that the attending paramedic or supervisor desires reviewed.
 - 6. Any STEMI activations.
 - 7. Any Stroke activations.
 - 8. Any Trauma activation.
 - 9. Any Hospital diversions.
 - 10. Academic review
- B. Patient Safety Work Product (PSWP) meeting criteria for inclusion as defined in the Patient Safety Organization (PSO) Policy shall be promptly submitted to the Patient Safety Evaluation System (PSES) for evaluation and reporting to the Missouri Center for Patient Safety (MOCPS).
- C. Upon the report return from the Medical Director, the attending Paramedic and Supervisors will review, comment (if needed) and sign the QA Worksheet.

PARAMEDIC COMPETENCIES

Competencies are completed to ensure personnel are up to date on skills, service protocols and to familiarize employees with new equipment or procedures. This is just another way Gladstone Fire / EMS sets itself apart from other agencies by ensuring employees are highly trained and up-to-date with current equipment and procedures.

Minimum Performance Requirements (MPR's) Bi-Annual Certification

Advanced Airway Management

- Intubation (Nasal and Oral)
- Invasive surgical airway
- EtCO2 monitoring
- CPAP Operation including in-line nebulizer Tx.

IV Access

IV ACCESS Proficiency - Difficult access

Must maintain a 70% success rate

Needle Decompression of the Chest

Cardiovascular Skills

- EKG/ACS injury pattern
- BLS
- ACLS MegaCode
 - AutoPulse Operation
 - Cardiac Monitor Operations

Continuous Infusion Medications

- Magnesium Sulfate
- Cardizem
- Epinephrine
- Amiodarone

Bi-Annual Written Protocol Examination

Minimum Performance Standards (MPS) include application of knowledge / skills to approved protocols. This includes trauma, medical, environmental, cardiac arrest – adult/pediatric, scene management and any procedure used within the approved EMS protocols.

ADULT/CHILD ABUSE REPORTING

Includes physical abuse, emotional abuse, sexual abuse, or neglect.

Duty to Report

Any person having reasonable cause to suspect that any child, a handicapped person, or elderly adult has been or may be abused or neglected shall make a report to the appropriate law enforcement agency and contact the following agencies:

Missouri Child Abuse Hotline – 1-800-392-3738

Missouri Adult Abuse Hotline – 1-800-392-0210

Assessment/Treatment

- A. Provide necessary medical care according to the appropriate protocol.
- B. Although a preliminary assessment of abuse or neglect is made in the field, appropriate personnel make the final determination upon review of the incident. Accusations of abuse or neglect should not be made in the field.

Notification

- A. In a private and confidential environment notify the physician at the receiving hospital of the facts upon which neglect or abuse is suspected.
- B. Notification of the shift supervisor of the facts of the neglect or abuse is encouraged.

Documentation

- A. The following information shall be included in the attending paramedics report:
 - 1. All facts upon which abuse or neglect is suspected.
 - 2. Identity of physician notified.
 - 3. Whether physician or hospital staff is going to make proper notification.
 - 4. If notification is made by the attending paramedic.

PARENT-CHILD CONSIDERATIONS

- A. In general, a parent should accompany most conscious children during transport unless the parent's presence interferes with the care of the child.
- B. In situations of major trauma and cardiopulmonary arrest the paramedic should exercise his or her judgment as to the status of the parent riding in the ambulance.
- C. Allow the parent to hold and touch the child whenever possible.
- D. Both the parent and child respond best to open and honest dialogue.

Do-Not-Resuscitate / Advanced Directives / Emergency Order Sets

Missouri law provides for the observance of: Advance Directives, Do-Not-Resuscitate orders and Emergency Order Sets when executed in compliance with Missouri state statute.

Advanced Directives

The Advanced Directive becomes effective only when the declarant's condition is terminal and the declarant does not have the capacity to make health-care decisions. [Sec. 459.025, RSMo](#). It is designed to allow a patient to forgo “death prolonging procedures,” which are defined as “any medical procedure or intervention which, when applied to a patient, would serve only to prolong the dying process and where, in the judgment of the attending physician pursuant to usual and customary medical standards, death will occur within a short time whether or not such procedure or intervention is utilized. Death-prolonging procedures shall not include the administration of medication or the performance of medical procedures deemed necessary to provide comfort, care or to alleviate pain nor the performance of any procedure to provide nutrition or hydration.” [Sec. 459.010\(3\), RSMo](#).

Transportable Physician Orders for Patient Preferences (TPOPP)

The Transportable Physician Orders for Patient Preferences (TPOPP) aims to improve the communication of a person's preferences regarding life-sustaining treatments and, thereby, generate high quality medical care in accordance with [Sec. 459.010\(3\), RSMo](#). In order to accomplish this aim, TPOPP was designed to:

- Document a person's treatment preferences regarding CPR and the use of mechanical intervention for breathing/ventilation along with other life-sustaining treatments such as tube feedings, etc.
- Translate those preferences into an actionable portable set of physician orders
- Communicate an individual's care preferences across health settings
- Reduce repetitive documentation while complying with state laws and the **Federal Patient Self-Determination Act**.

When properly executed in accordance with [Sec. 459.025, RSMo](#), the TPOPP shall be accepted and honored by the attending paramedic.

Do-Not-Resuscitate (DNR)

Outside the hospital “Do-Not-Resuscitate” (DNR) requests are orders provided by a patient's physician to refrain from initiating cardiopulmonary resuscitative measures in the event of a cardiac or respiratory arrest. DNR requests are compatible with maximal therapeutic care and the patient may receive vigorous support (i.e. IV's & drugs) until the point of cardiac or respiratory arrest.

Outside the hospital “Do-Not-Resuscitate” orders, are regulated by RSMo sections [190.600](#) to [190.621](#). When properly executed, the DNR shall be accepted and honored by the attending paramedic. Missouri Bureau of EMS has ruled a photocopy / fax is also acceptable. [DNR FAQ's](#)

Guidelines

Guidelines for acceptance of a DNR request include:

- A. If any doubt exists about the validity of the DNR form or the intent of the patient or the appropriate surrogate, resuscitation must be initiated and Medical Control contacted.
- B. Only the approved Bureau of EMS [DNR form \(MO580-1936\)](#) shall be used.
- C. The DNR form must be signed and dated by the patient or appropriate surrogate, a witness, and the patient's physician.
- D. The revocation provision must remain unsigned in order for this DNR form to remain valid.
- E. The DNR form must be with the patient at the time of the EMS providers' arrival.
- F. The EMS provider must see the form.
- G. No cardiac monitoring is necessary. No Medical Control contact is necessary.
- H. If this form is presented after care has begun, all resuscitative efforts shall cease and Medical Control shall be notified.

Documentation

The EMS report shall include the existence of a (DNR, DPA, TPOPP or other forms of Advanced Directives). A copy shall be attached to the EMS report. The original form shall remain at the facility or home. If the patient is transported, the form(s) will accompany the patient.

Emergency Order Sets

Pediatric Emergency Form

Children's Mercy Hospital (CMH) has developed a form for special medical needs pediatric patients who require very specific emergency medical care.

Upon arrival of EMS providers, the parent / caregiver shall provide a "[Pediatric Emergency Form](#)" to EMS personnel, which contains detailed emergency instructions, medical history and hospital emergency contact information. **The Pediatric Emergency Form shall supercede existing protocol and shall be followed by the attending paramedic.**

If questions arise regarding the specific orders listed on the form, Children's Mercy Hospital can be contacted at the phone numbers listed on the Pediatric Emergency Form. When contacted, the attending paramedic shall provide the hospital with the Medical Record (MR) number listed on the top right corner of the form. Children's Mercy Hospital personnel will utilize the MR number to confirm the specific orders.

In addition, CMH has advised parents / caregivers to have a "File of Life" information card located on the refrigerator in the patients' residence.

Documentation

The EMS report shall include the existence of a "Pediatric Emergency Form." A copy shall be attached to the EMS report. The original form(s) shall be transported with the patient and given to CMH staff.

PUBLIC SERVICE CALLS - DOCUMENTATION

Purpose

To provide for adequate and uniform documentation of calls for public service or assistance. These include but are not limited to:

- a. Lifting assistance.
- b. Assistance to or from a vehicle.
- c. Assistance with medical equipment.

Procedure for documentation

- a. All public service calls shall be documented.
- b. These reports shall be completed using the NIFRS section of ESO.
- c. All reports shall include the patient's name and contact information
- d. The narrative shall be completed in accordance with EMS report standards **(auto-generated reports are NOT acceptable forms of documentation)**.
- e. Any ambulance request that is dispatched with a nature other than Public Service requires the Gladstone Ambulance EMS Report be filled out along with the appropriate accompanying documents.

ALS ENGINE REFUSALS

To allow Paramedics responding on ALS equipped fire apparatus to cancel an ambulance response and obtain a patient refusal.

Procedure

Gladstone ALS apparatus staffed by paramedic personnel may assess patient condition and need for ambulance transport. If it is determined, based on the assessment of the patient, a transport ambulance is not needed and/or the patient does not desire ambulance transport, the responding ambulance may be cancelled and a patient refusal accepted. It is the responsibility of the attending paramedic to explain all options for transport if medical attention is deemed necessary.

Documentation

Listed below is the minimum criterion required for Gladstone Fire Department reports.

How the call was received from dispatch, pre-arrival information, units responding location and mode of response:

1. Nature of the call on arrival.
2. How the patient was found, presenting signs, appearance and pertinent scene factors.
3. The patient's chief complaint.
4. HPI & PPMH
5. Vital signs. This includes: blood pressure, pulse rate, respiratory rate, and SpO2 and EtCO2 readings if applicable.
6. IV attempts, location, patency and who initiates.
7. Intubation attempts, ET tube location measured from the teeth in cm, and confirmation methods.
8. Medication administration, name, dose, route and results.
9. Procedures performed, by whom and results.
10. The patient's overall response to treatment.
11. Attending Paramedic shall end the report with their full name, title and state license number (P#).

All calls shall be documented in the first person, past tense and in chronological order, from the time of dispatch through return to service.

All reports shall be written utilizing proper grammar, spelling and only Department approved abbreviations.

PATIENT REFUSALS

Purpose

To establish guidelines and authorization for allowing a patient to refuse procedures, treatment and or transport.

Procedure

The paramedic will be allowed to accept patient refusals provided sufficient information about the patient's condition have been given, allowing the patient or responsible person to make an informed decision, and:

1. Patient is a mentally competent adult, 18 years of age or older.
2. If the patient is not mentally competent, an authorized caregiver with power of attorney or guardianship may refuse in lieu of.
3. If the patient is under the age of 18, a parent or guardian must sign the refusal unless:
 - a. The minor is emancipated
 - b. Married
 - c. Pregnant
 - d. A Parent
 - e. A member of the armed forces
 - f. Financially independent, living away from home
4. In those incidents where the minor is 16 to 17 years of age and a parent or guardian is not available to sign a refusal, verbal consent by phone is acceptable provided sufficient information about the condition of the patient has been given, allowing the parent or guardian to make an informed decision. The name of the individual giving authorization for the refusal must be included in the narrative of the EMS report.
5. If the patient refuses care and/or transport and will not sign the refusal of care form, mark the form refused to sign, and document the refusal to sign in the narrative of the EMS report.
6. **If there are any questions or problems regarding the appropriateness of a refusal, Medical Control shall be contacted.**
7. All patient refusal reports shall be reviewed and initialed by the shift Captain and Battalion Chief prior to sending for final disposition.

ADVANCE BENEFICIARY NOTICE

Purpose

To establish a guideline for the use of the Advanced Beneficiary Notice (ABN).

Procedure

1. Medicare requires the Advanced Beneficiary Notice (ABN) when it is likely that costs for treatment and transport will not be covered.
2. The form is designed to help patients make an informed choice about whether or not to receive services, knowing that services may not be covered.
2. The form is to be filled out for all **Medicare and Blue Cross** insured patients.
3. The form is not intended to discourage treatment and/or transport; it is to help patients make an informed decision.
4. The attending paramedic shall inform Medicare and Blue Cross patients when it is possible that services will not be covered. The patient shall be given the form to read, check yes, services are wanted, or no, services are not desired. The form is to be signed and dated by the patient or a person acting on behalf of the patient.
5. Should the patient refuse to sign, date it and mark the form refused to sign.
6. The completed form shall be attached to the hospital face sheet and forwarded to the City billing clerk.

CHEST PAIN/PRESSURE

Chest pain/pressure can be caused by a variety of different problems including: MI, angina pectoris, pulmonary embolism, pneumonia and musculoskeletal chest wall pain.

Assessment

1. Obtain brief history characterizing pain as to the nature, severity, onset, duration, location, and radiation.
2. **INITIAL 12/15 LEAD EKG SHOULD BE COMPLETED WITHIN 5 min OF PATIENT CONTACT**
3. Obtain vitals, **A BP SHOULD BE TAKEN ON BOTH ARMS, a difference of 15mmhg, systolic or diastolic and a history of hypertension could indicate an Aortic DISSECTION AND SHOULD BE REPORTED to receiving staff.**
4. Perform a brief cardiopulmonary assessment to include inspection, palpation, and auscultation of the chest. Obtain current medications, past medical history and allergies.

Obtain a 12/15 lead EKG on chest pain of any kind, regardless if you think it's cardiac or not.

STEMI SCENE TIMES SHOULD BE MINIMIZED

Secure Airway

Administer appropriate O2 concentration

Obtain IV access

If a STEMI is confirmed, refer to Code STEMI Activation

Consider, [ASPIRIN](#) 324 mg to be chewed or swallowed if no history of asthma or sensitivity to aspirin.

Consider [Nitroglycerin](#), as long as systolic BP remains >100 and pulse rate >50 and <140.

0.4 mg SL tablet every 5 minutes until chest pain is relieved to a total of 3 doses or hypotension develops.

If there is no relief of chest pain after 1st dose of NTG, you may administer Morphine.

WARNING

NITRATE ADMINISTRATION WITH SIGNS AND SYMPTOMS OF INFERIOR MI / RVI MAY CAUSE HEMODYNAMIC COLLAPSE.

V4R lead is the best lead for determining RVI, with approximately 90 percent specificity.

Consider a fluid bolus and or use of vasopressors.

-Consider [Morphine](#) if chest pain is unresolved with NTG, and systolic BP > 100.

-Give 4 mg initially, slow IV push, followed by 2 mg increments every 5 minutes until the pain is resolved or a total of 10 mg is given.

*Consider [Zofran](#) 4mg SIVP or PO for nausea secondary Morphine administration.

***Monitor BP, SpO₂ and EtCO₂ if indicated.**

Treat arrhythmias per protocol or physician's orders.

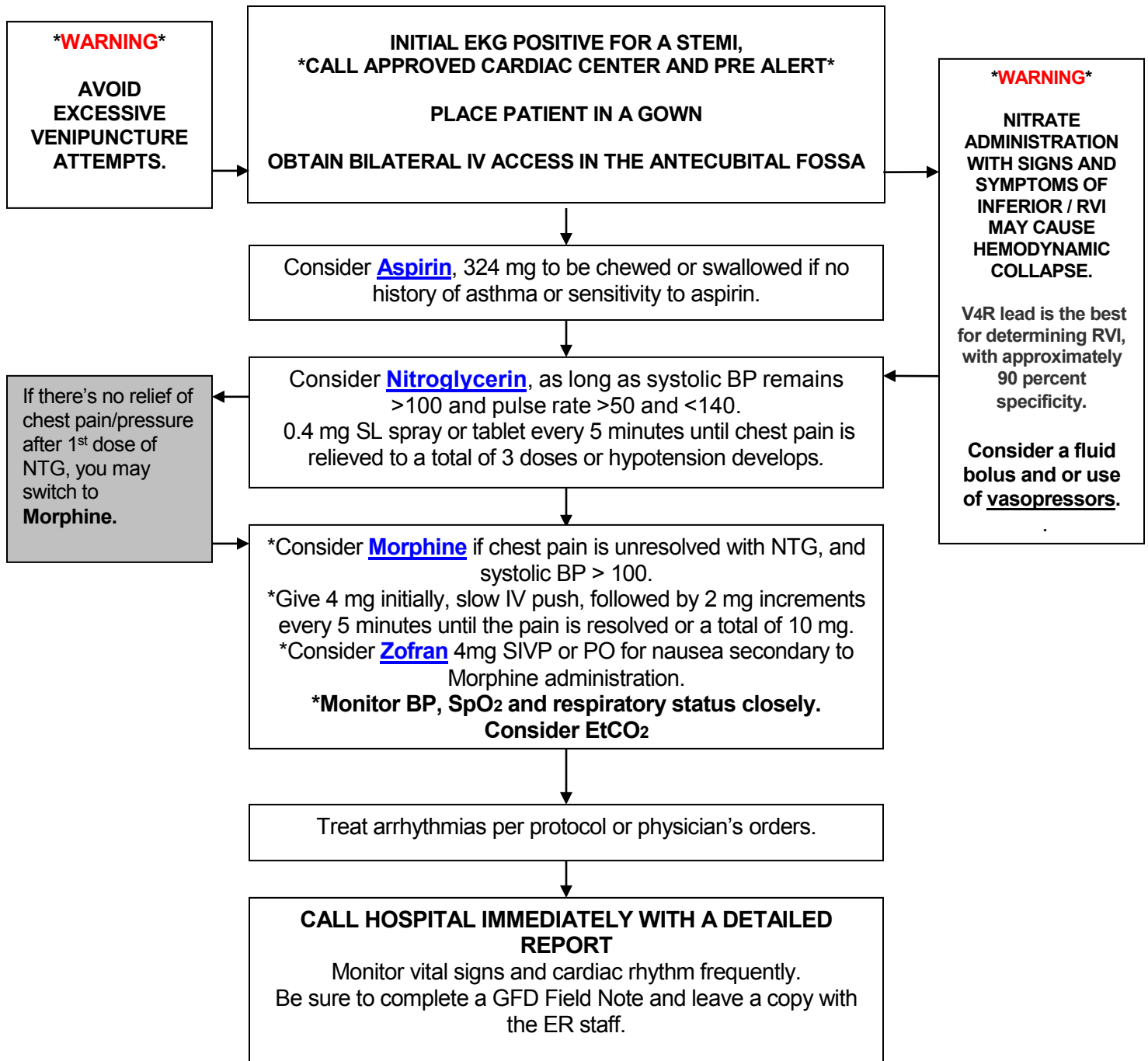
Transport monitoring vital signs and cardiac rhythm frequently.

CODE STEMI

MUST HAVES FOR ED ACTIVATION OF CODE STEMI:

- *Ongoing cardiac symptoms.
- *12/15 lead EKG show ST-elevation greater than or equal to 1 mm, in two or more Contiguous leads.
- *12/15 lead EKG **without** evidence of **LBBB** or **Paced** rhythm.
- *No obvious evidence of acute GI bleed or CVA.

STEMI SCENE TIMES SHOULD BE MINIMIZED, CONSIDER CODE 1 TRANSPORT WITH A SECOND PARAMEDIC.



CARDIAC DYSRHYTHMIAS

Assessment

1. Obtain brief history of current symptoms, especially any chest pain or shortness of breath, "tearing" back pain and/or past medical history.
2. **INITIAL 12/15 LEAD EKG SHOULD BE COMPLETED WITHIN 5 min OF PATIENT CONTACT**
3. Obtain vitals, A BP SHOULD BE TAKEN ON BOTH ARMS, a difference of 15 mmHg, systolic or diastolic and a history of hypertension could indicate an Aortic DISSECTION AND SHOULD BE REPORTED to receiving staff.
4. Briefly assess the cardiac and respiratory systems, and level of consciousness. Note any signs of poor perfusion such as altered level of consciousness, clammy skin, or hypotension.

Unstable

- A. Presence of a dysrhythmia with any of the following:
 1. Altered mental status
 2. Clinical signs of shock
 3. Shortness of breath/pulmonary edema
 4. Severe Chest pain/Pressure
 5. Cyanosis

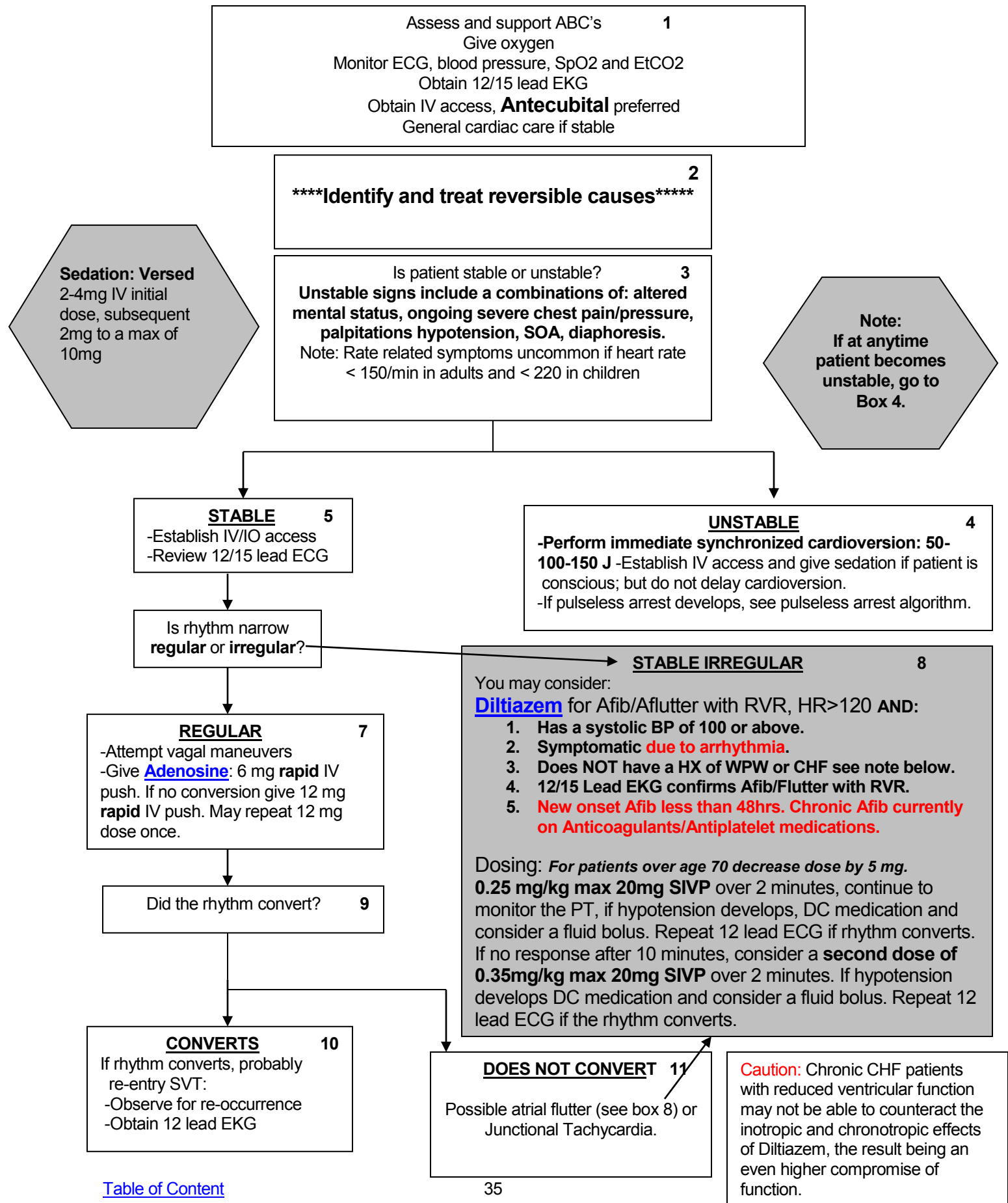
Stable

- A. Presence of a dysrhythmia and *absence of above signs and symptoms.*
- B. Transport with supportive care only.

General Treatment

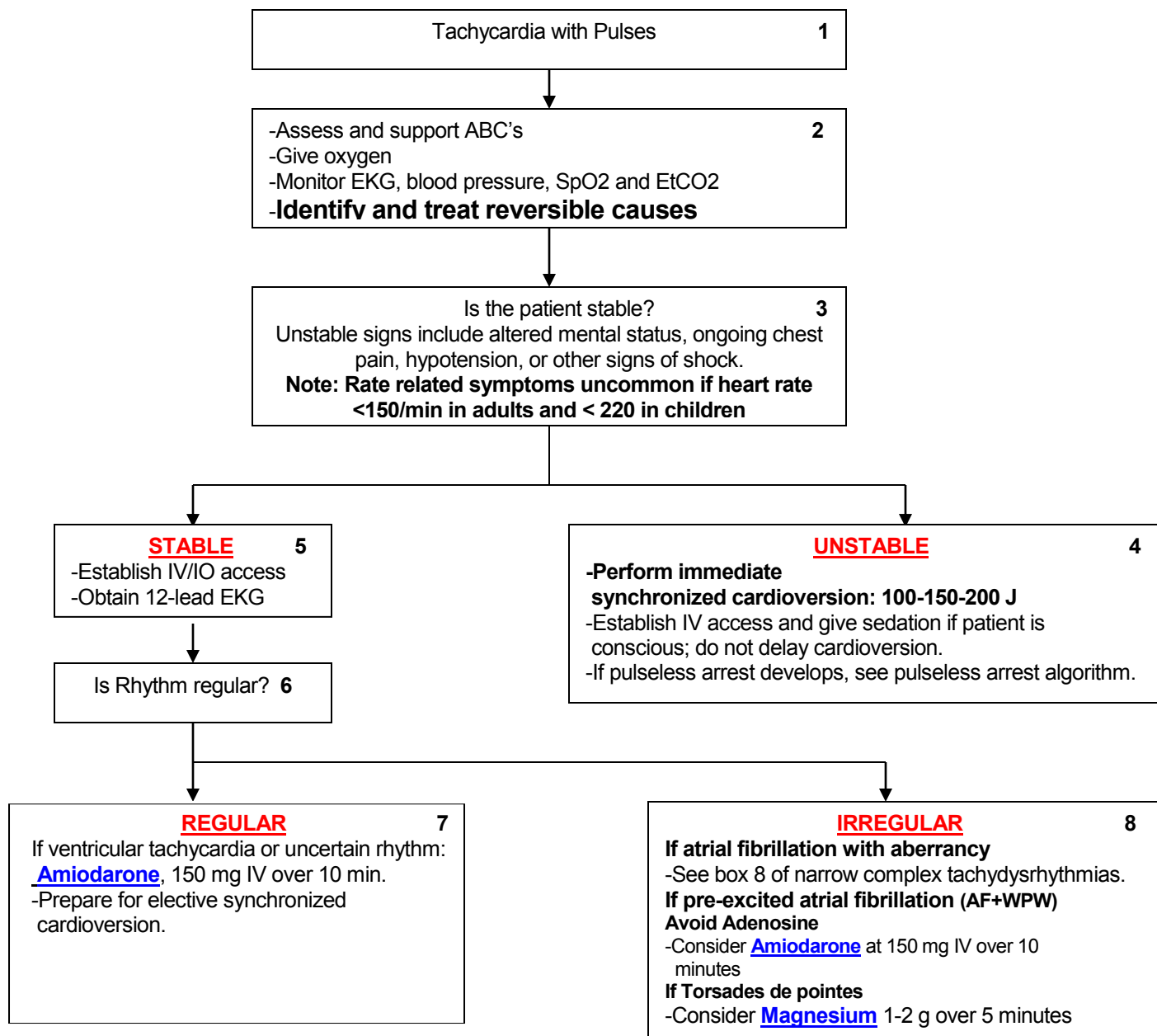
- A. Secure airway and assure ventilations.
- B. Administer O2.
- C. Start an IV. Consider an IO in severely obtunded children.
- D. Monitor cardiac rhythm and vitals en route to hospital.
- E. Proceed according to specific dysrhythmia protocol.

NARROW COMPLEX TACHYDYSRHYTHMIAS

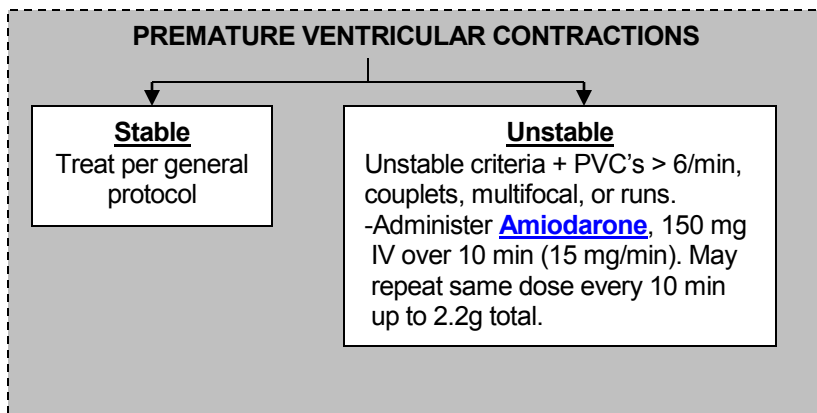


WIDE COMPLEX TACHYDYSRHYTHMIAS

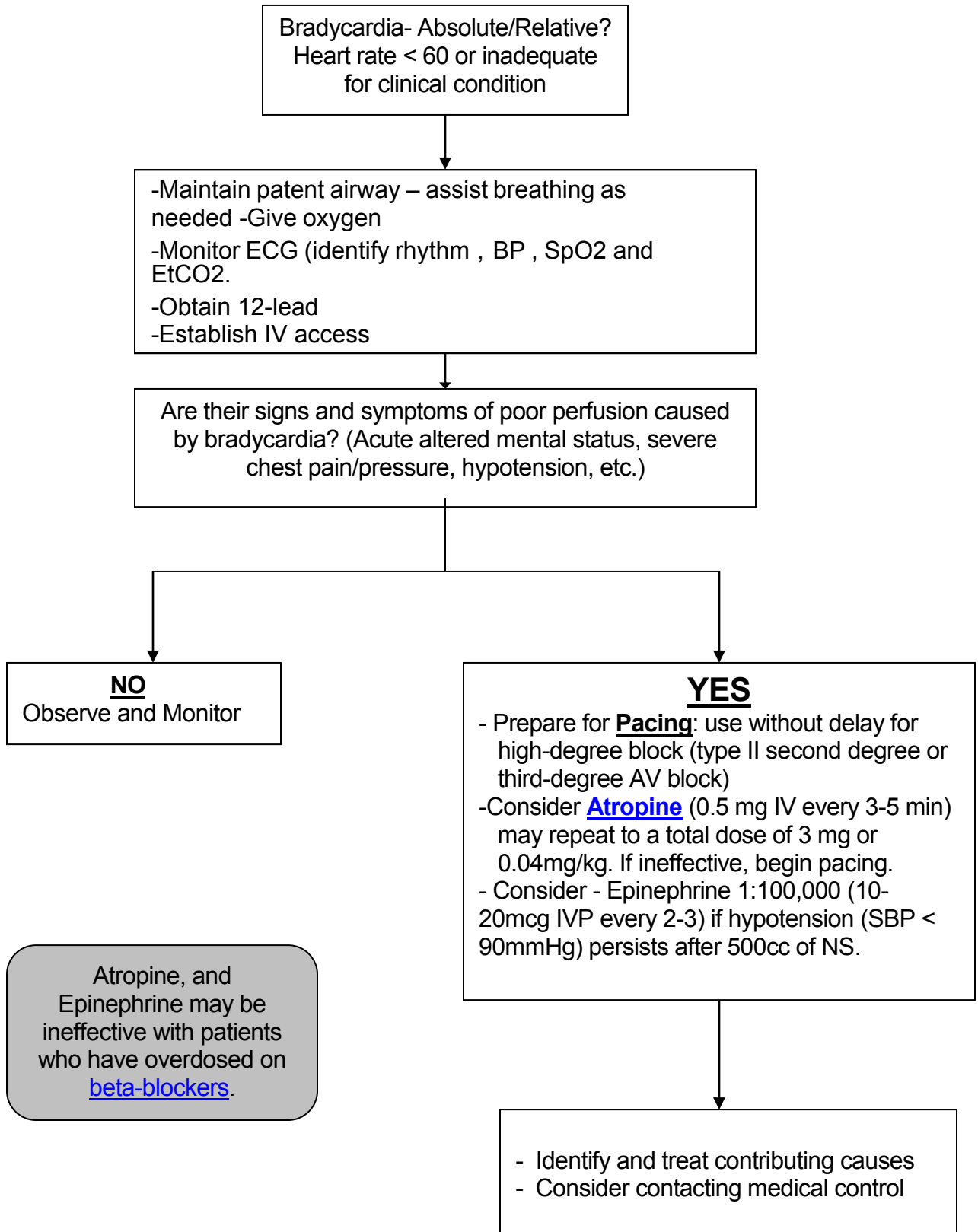
QRS > .12 seconds



Amiodarone drip mixed by placing 150 mg in a 100 ml bag of NS and given over 10 minutes (15 mg/min).



BRADYCARDIA ALGORITHM



ASYSTOLE/PULSELESS ELECTRICAL ACTIVITY

1
Begin CPR/Autopulse
-Attach monitor and confirm Asystole or PEA
Monitor EtCO₂

2
-Obtain IV or IO access
-Secure airway

3
-Administer **Epinephrine** 1 mg IV every 3-5 minutes.
Epinephrine 1:10,000

Epinephrine may be given via ET tube at twice the dose

4
-Give 5 cycles of CPR

Check rhythm.
Is it shockable?

YES
Go to VF/VT algorithm

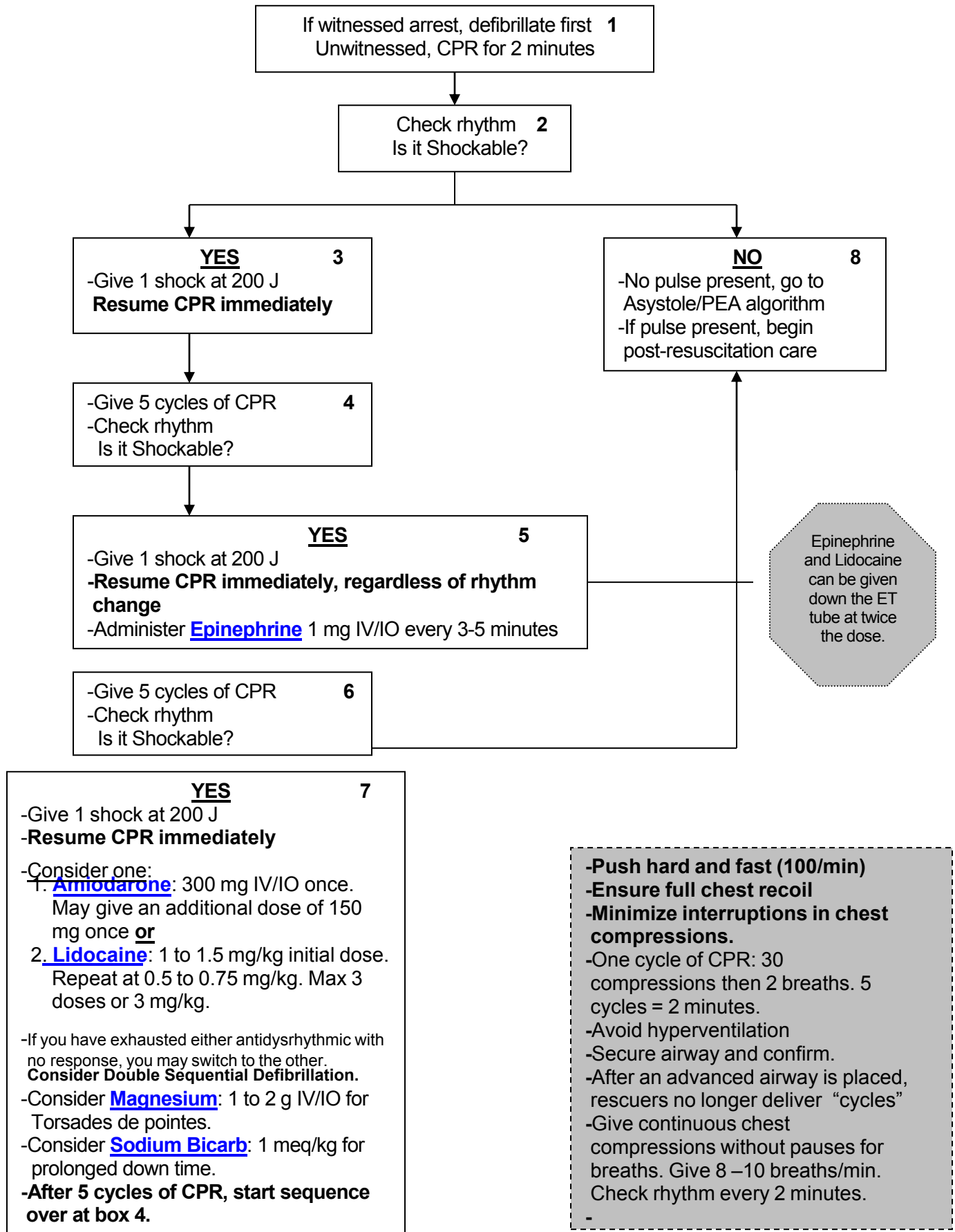
NO
-If no pulse go to box 3
-If pulse present, begin post-resuscitation care

-Push hard and fast (100/min)
-Ensure full chest recoil
-Minimize interruptions in chest compressions.
-One cycle of CPR: 30 compressions then 2 breaths. 5 cycles = 2 minutes.
-Avoid hyperventilation.
-After a secure airway is placed and confirmed, rescuers no longer deliver "cycles."
-Give continuous chest compressions without pauses for breaths. Give 8-10 breaths/min. Check rhythm every 2 minutes.

-Search for and treat possible factors:

Hypovolemia
Hypoxia
Hypoglycemia
Hypo/Hyperkalemia
Acidosis
Toxins
Tamponade
Thrombosis
Trauma

VENTRICULAR FIBRILLATION/PULSELESS VENTRICULAR TACHYCARDIA



POST RESUSCITATION (ROSC) PROTOCOL ADULT OR PEDIATRIC CARDIAC ARREST

ASSESSMENT:

- *ROSC=pulse and measurable BP with increasing ETCO₂
- * Continue to address specific differentials associated with original dysrhythmia
- *Focus on prevention of hypoxia, hypo/hypercapnia, hypotension(Survival and neurologic outcome worsen in their presence)

↓
Optimize oxygenation/ventilation
ETCO₂ 35-45mmHg
DO NOT HYPERVENTILATE

↓
Monitor vital signs and EKG rhythm

↓
Obtain 12 lead ECG

↓
SBP < 90 mmHg →
NS BOLUS
ADULTS: 500mL IV/IO MAX 1L
PEDS: 20mL/kg REPEAT X 1

↓
ADULTS: CONSIDER EPI 1:100,000 (PUSH DOSE)
10-20mcg EVERY 2-3 MINUTES IF SBP < 90mmHg

↓
IF SBP > 90mmHg AND ETT/King are In PLACE:
CONSIDER MIDAZOLAM FOR SEDATION: ADULTS: 1-2mg IV/
IO REPEAT AS NEEDED. MAX GIVEN 5mg PEDS: 0.1mg/kg UP
TO 2mg

RESPIRATORY DISTRESS

Respiratory distress can be divided into processes involving the upper or lower airway. Examples of upper airway obstruction include: foreign body, epiglottitis, croup, and trauma. Diseases that cause lower airway obstruction include: asthma, COPD, pneumonia, CHF, and pneumothorax. If unable to differentiate the cause of respiratory distress, open and maintain airway, administer oxygen, ensure adequate ventilation, and transport rapidly.

Assessment:

1. Assess adequacy of airway and ventilation to include: vocalization, rate and depth, breathe sounds, cyanosis, and signs of respiratory distress.
2. Perform brief cardiovascular assessment.
3. Check vital signs to include **SpO2** and **EtCO2**.
4. Obtain brief history to include: onset, presence of chest pain, potential aspiration, fever, and past medical history.

Upper Airway Obstruction

1. Open airway and assess for breathing.
2. If no breathing present follow AHA guidelines for obstructed airway. Attempt ventilation.
3. If unsuccessful, use direct laryngoscopy to check for foreign bodies. If present, remove with magill forceps. Assist ventilations as needed.
4. **If still unsuccessful, consider surgical cricothyrotomy airway or commercial cricothyrotomy device. All other airway techniques must be attempted prior to surgical airway.**
5. If resolved or initially only partially obstructed, do the following:
 - a. Transport in upright position.
 - b. Administer O2.
 - c. Establish IV access.
 - d. Monitor respiratory status.

Note: Surgical airway may be necessary in severe maxo-facial trauma as a first line intervention

Asthma

1. Ensure airway.
2. Establish IV access.
3. Administer O2 using appropriate administration set, for **Status Asthmaticus** consider early CPAP with inline breathing treatment. Intubation should be considered in patients exhibiting signs of physical exhaustion/respiratory failure. If the patient is conscious and needs to be intubated, consider **Versed** 2-4 mg SIVP up to a total dose of 10 mg. If unable to obtain IV access, consider IN **Versed** 2 – 4 mg, max 10 mg. See [intranasal procedure](#).
4. **Duoneb** 3 ml nebulized. Set O2 at 6-8 lpm. Consider if needed **Albuterol** 2.5 mg x 2, heart rate should not exceeds 150 bpm. **Xoponex** may also be considered if other medications are ineffective or contra-indicated (**Pediatric dosing of Xoponex, dilute with 3ml of NS**)
5. Consider **Solu-Medrol** 62.5 mg IV. Pediatric 2mg/kg **USE WITH EXTREME CAUTION IN DIABETIC PATIENTS AS MAY INCREASE INSULIN METABOLIC REQUIREMENTS**
6. For severe cases consider **Magnesium Sulfate** 1g drip over 10 min, re evaluate recheck vitals and then give the second 1g dose over 10 min. Consider **Epinephrine** 1:1,000 concentration, 0.3 mg IM. Contact medical control for orders if age is over 40 **AND** pulse rate > 120, or HX of CAD. Consider **Terbutaline** 0.25mg SQ.
7. Consider BiPap.

RESPIRATORY DISTRESS (CONT)

COPD

1. Ensure airway.
2. Administer O₂. Consider [CPAP](#) for severe respiratory distress. Intubation should be considered in patients exhibiting signs of physical exhaustion/respiratory failure. If the patient is conscious and needs to be intubated, consider [Versed](#) 2-4 mg SIVP up to a total dose. If unable to obtain IV access, consider IN [Versed](#) 2-4 mg, max 10 mg. See [intranasal procedure](#)
3. Start an IV at a TKO rate.
4. Apply cardiac monitor and obtain 12/15 lead ECG.
5. [Duoneb](#) 3 ml nebulized. Set O₂ at 6-8 Lpm.
6. Consider if needed [Albuterol](#) 2.5 mg x 2, heart rate should not exceeds 150 bpm. [Xoponex](#) may also be considered if other medications are ineffective or contra-indicated.
7. For severe cases consider [Magnesium Sulfate](#) 1g drip over 10min, re-evaluate recheck vitals and then give the second 1g dose over 4-5 minutes.
8. Consider [Solu-Medrol](#) 62.5 mg IV, Pediatric 2mg/kg. **USE WITH EXTREME CAUTION IN DIABETIC PATIENTS AS MAY INCREASE INSULIN METOBOLIC REQUIREMENTS**
9. Consider BiPap.

Congestive Heart Failure- (see CHF Protocol)

1. Ensure airway.
2. [CPAP](#) for severe respiratory distress; Administer O₂ and assist ventilations as needed. Intubation should be considered in patients exhibiting signs of physical exhaustion/respiratory failure. If the patient is conscious and needs to be intubated, consider [Versed](#) 2-4 mg SIVP up to a total dose of 10 mg. If unable to obtain IV access, consider IN [Versed](#) 2-4 mg, max 10 mg. [See intranasal procedure.](#)
3. Monitor cardiac rhythm and obtain 12/15 lead ECG.
4. Start an IV at a TKO rate. Carefully monitor fluid.
5. [Nitroglycerin](#) 0.4mg SL if the patient maintains a systolic blood pressure > 100, and a pulse rate >50 and <140. May be repeated every 5 minutes until distress is relieved, or systolic blood pressure < 100, or 3 doses.
6. **If EtCO₂ indicates constrictive pattern you may consider** [Duoneb](#) 3 ml nebulized. Set O₂ at 6-8 Lpm. Consider if needed [Albuterol](#) 2.5mg x 2, heart rate should not exceeds 150 bpm. [Xoponex](#) may also be considered if other medications are ineffective or contra-indicated.
7. Transport monitoring vital signs.

Tension Pneumothorax

1. Secure airway.
2. Administer O₂.
3. Monitor cardiac rhythm.
4. [Needle decompression](#) as per decompression procedure.
5. Transport monitoring vital signs and respiratory status.

RESPIRATORY DISTRESS (CONT)

Pediatric Respiratory Distress with Inspiratory Stridor/Croup

Treatment:

1. Humidified O₂, as appropriate, with upright posture



2. If needed, assist ventilations with PPV using 100% O₂



3. For signs of severe croup, administer inhalation of nebulized solution of Epinephrine 1:1,000 (1mg/mL) at the following dose:

5-10 kg: 2.5mL

Greater than 10kg: 5mL

Signs of severe croup include stridor at rest AND one of the following: tachypnea, moderate inter costal retractions (including suprasternal retractions), agitation/restlessness/tired appearing, difficulty talking or feeding.



4. Contact Medical Control for the second dose of nebulized Epinephrine if symptoms do not improve consider alternate diagnosis such as aspiration of foreign body, bacterial tracheitis, epiglottitis.

*** Nebulized Epinephrine may be contraindicated in children with a history of congenital heart disease.**

***Albuterol is not an appropriate treatment for croup**

5-10 kg: Nebulized Epinephrine Mix 0.5 mg (0.5 ml) Epi 1:1000 with 2ml of normal saline

>10 kg: Mix 1mg(1ml) Epi 1:1000 with 4ml of normal saline

ALLERGIC REACTION/ANAPHYLAXIS

Allergic reactions can be mild to life threatening. Potential routes for exposure include inhalation, ingestion, injection, or direct contact. Treatment addresses mild anaphylaxis, anaphylaxis with respiratory distress, and anaphylactic shock. If cardiopulmonary arrest should occur, proceed to the cardiac arrest protocol.

Assessment

1. Check vital signs.
2. Obtain brief history with attention paid to known allergies or any recent exposures.
3. Assess cardiac and pulmonary systems.

-Secure Airway.
-Administer O2.
-Place on cardiac monitor, SpO2 and EtCO2.
-Establish IV access.
-Transport immediately

Localized Allergic Reaction

(normal BP & pulse with no stridor or respiratory distress)

Supportive care, watch for signs of decompensation

Consider **Benadryl**:
25-50 mg IV, Pediatric
1 mg/kg IV, up to a total
single dose of 50 mg.

Transport

Contact **medical control**
for orders for Epinephrine
in any of the columns, for
patients over age 40 with
a HX of CAD.

For a pulse rate > 120.

You may consider

Terbutaline 0.25mg SQ

Mild Anaphylaxis with Respiratory Distress

Duoneb 3 ml nebulized. Set O2 at
6-8 Lpm. Consider if needed
Albuterol 2.5mg x 2, heart rate
should not exceeds 150 bpm.

Xoponex may also be considered
if other medications are contra-
indicated or ineffective. Pediatric
dosing, dilute in 3ml of NS.

Consider **Benadryl**:
25-50 mg IV, Pediatric
1 mg/kg IV, up to a total
single dose of 50 mg.

Consider **SoluMedrol** 62.5 mg
IVP. Pediatric- 2mg/kg

If no improvement proceed to
Anaphylactic Shock:

-Consider **Epinephrine 1:1,000**
0.3 to 0.5mg, Sub Lingual injection or IM

-Peds 1:10,000 Epinephrine 0.01 mg/kg
IM

Anaphylactic Shock

Give a fluid bolus. 20 ml/kg up to a
maximum of 500 ml. Consider a
repeat bolus if needed to maintain
blood pressure.
SUPPORT RESPIRATORY,
CONSIDER BRONCHODILATORS

-Administration of **Epinephrine 1:1,000**
0.3 to 0.5mg, Sub Lingual injection or IM

-**Preferred** - venous plexus at the base
of the tongue.

-Peds 1:10,000 Epinephrine 0.01 mg/kg
IM

Consider **Benadryl**:
25-50 mg IV, Pediatric
1 mg/kg IV, up to a total
single dose of 50 mg.

Administer **SoluMedrol** 62.5 mg
IVP. Pediatric 2mg/kg

******For CRITICAL Patients******

Consider **Epinephrine Drip** 2-
10mcq/min, .5mg of Epi mixed in
500cc admin thru 60gtt set, drip at
2ml/min titrate to effect.

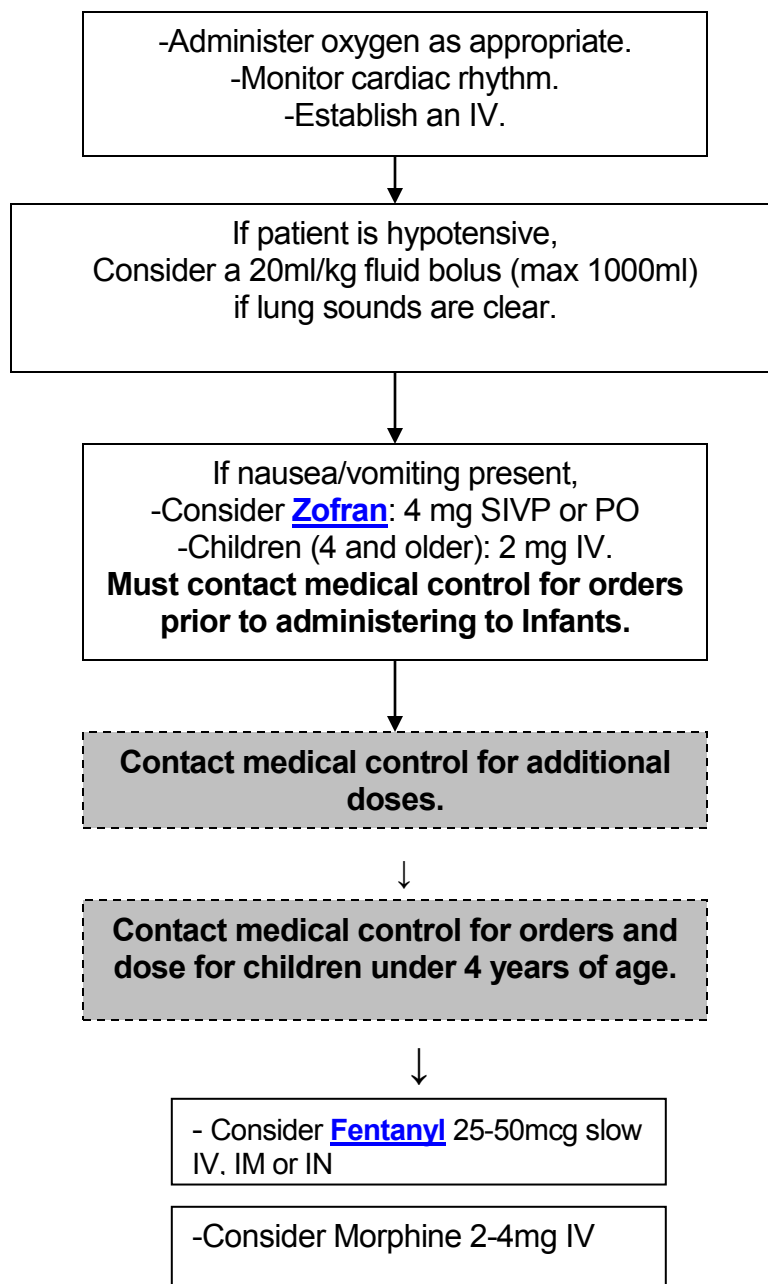
1:1,000 concentration – 0.3 to 0.5 ML = .3 - .5 mg
1:10,000 concentration – 3 to 5 ML = .3 – .5 mg

05/23/2025

ABDOMINAL PAIN

Assessment

1. Take vital signs.
2. Obtain brief history including onset, duration, and past medical history.
3. Briefly assess cardiac, pulmonary, and neurological systems.
4. Identify possible causes.
5. Consider gynecological emergencies in females. Refer to gynecological emergencies protocol.



BEHAVIORAL/PSYCHIATRIC DISORDERS

Assessment

1. Check vital signs.
2. Obtain brief history noting any bizarre or abrupt changes in behavior, suicidal intentions, alcohol or drug ingestion, and any significant past medical history.
3. Perform a physical exam if possible.
4. Perform a brief neurological exam including mental status.
5. If restraints are used a minimum of two people will accompany patient until care has been transferred.

Probable Psychiatric Problem Only

-Attempt to establish rapport with patient.
-If patient is dangerous to self or others, consider law enforcement assistance.

Restrain in lateral recumbent, supine, or prone position only if absolutely necessary.

If suicidal, do not leave patient alone and if possible remove any dangerous objects.

If emergency transport is unnecessary, do as little as possible except reassure the patient during transport.

Lethargic, Unstable Vital Signs, or Hypoglycemia Suspected

-Administer O2.
-Apply EKG
-Start an IV, TKO.
-Do a glucose check.

Consider administration of Dextrose.

Adults: D10 250mL IV
Ped: >8 yrs - D10 2mL/kg IV
Ped: <8 yrs - D10 1mL/kg IV

Consider Narcan: 0.4-2 mg SIVP, titrate to respiratory effort, monitor EtCO2, IM, ET, IO, or IN.

Transport monitoring vital signs.

If trauma or dysrhythmia is noted proceed to appropriate protocol.

Agitated See Excited Delirium protocol

Consider own safety and limitations.

If patient is dangerous to self or others, request law enforcement assistance.

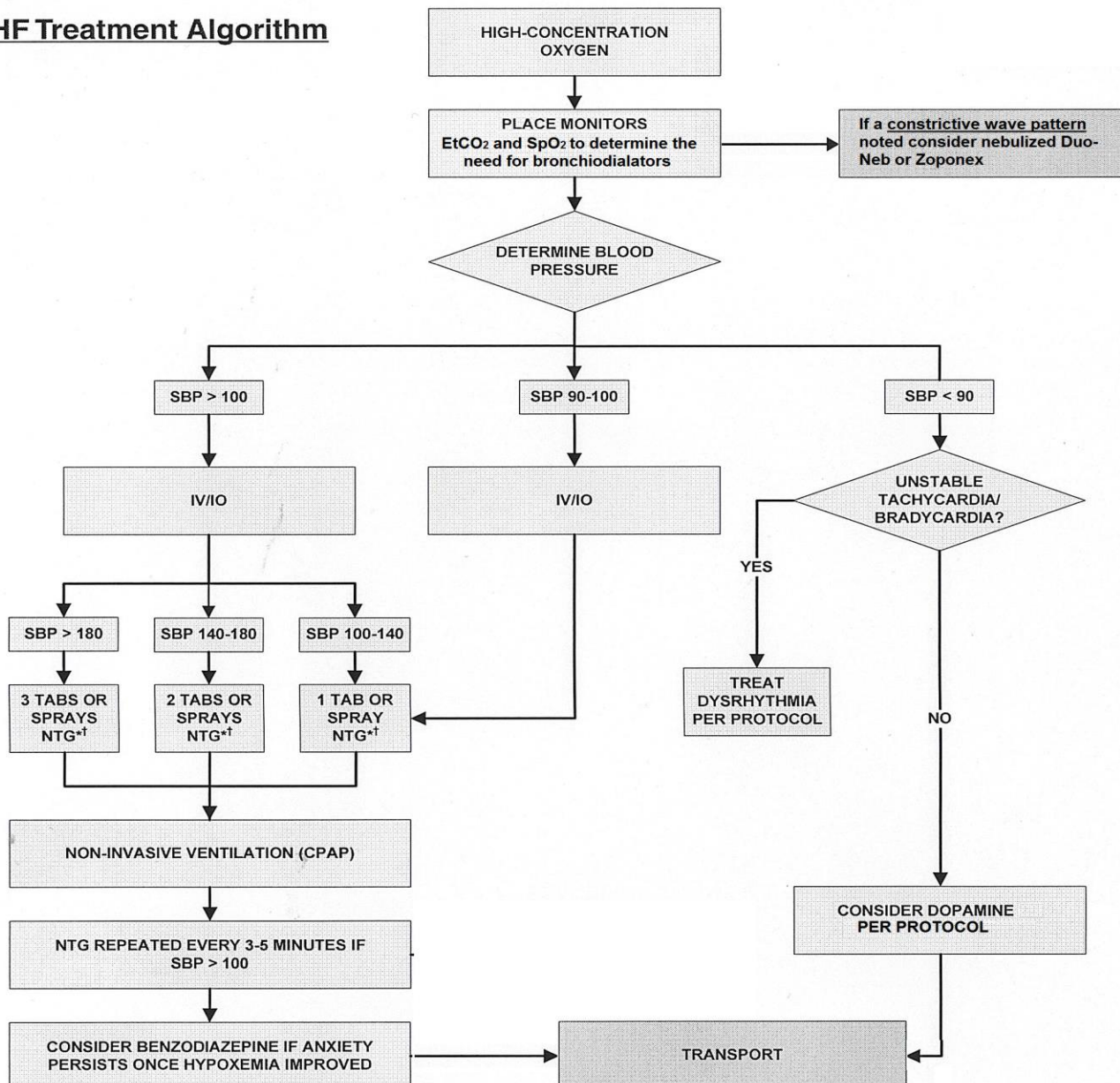
Consider Versed 5-10mg
Restrain in lateral recumbent, supine, or prone position only if absolutely necessary.

If patient is restrained, continuously monitor airway and breathing.

CONGESTIVE HEART FAILURE (CHF)

1. Ensure airway.
2. **Rapid CPAP for severe respiratory distress**; Administer O2 and assist ventilations as needed. Intubation should be considered in patients exhibiting signs of physical exhaustion/respiratory failure. If the patient is conscious and needs to be intubated, consider [Versed](#) 2-4 mg SIVP up to a total dose of 10mg. If unable to obtain IV access, consider IN [Versed](#) 2-4 mg, max 10 mg. [See intranasal procedure.](#)
3. Monitor cardiac rhythm and obtain 12/15 lead ECG.
4. Start an IV at a TKO rate. Carefully monitor fluid.
5. [Nitroglycerin](#) 0.4mg SL if the patient maintains a systolic blood pressure > 100, and a pulse rate >50 and <140. May be repeated every 5 minutes until distress is relieved, or systolic blood pressure < 100, or 3 doses.
6. **If EtCO2 indicates constrictive pattern you may consider [Duoneb](#) 3 ml nebulized.** Set O2 at 6-8 Lpm. Consider if needed [Albuterol](#) 2.5mg x 2, heart rate should not exceeds 150 bpm. [Xoponex](#) may also be considered if other medications are ineffective or contra-indicated.

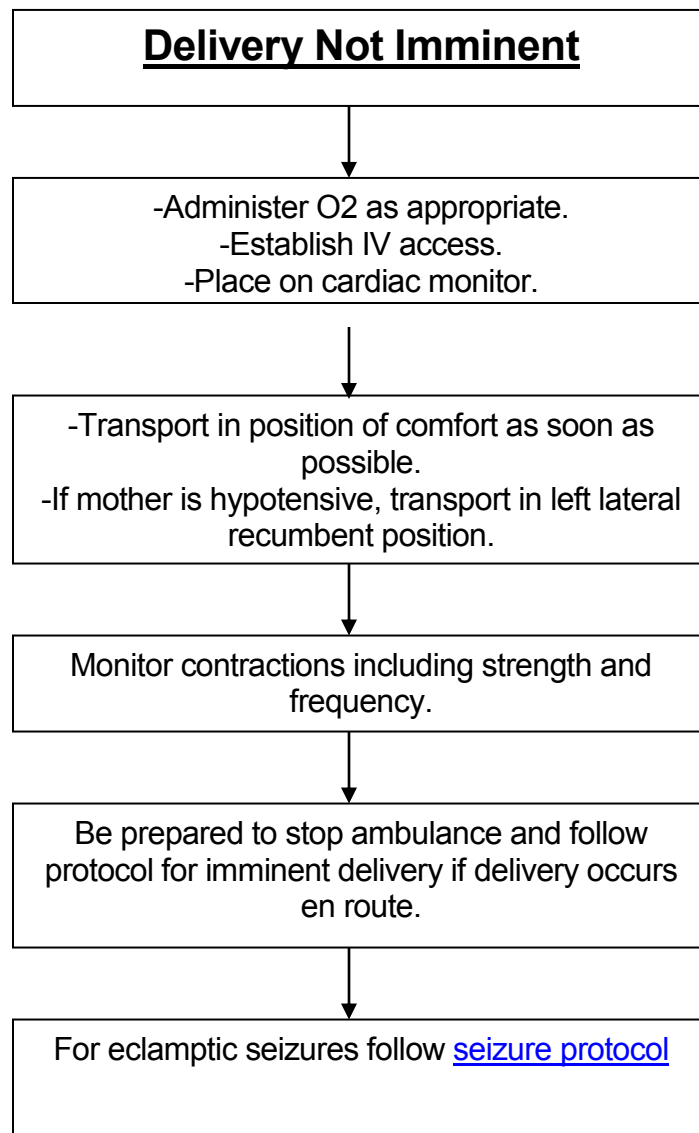
CHF Treatment Algorithm



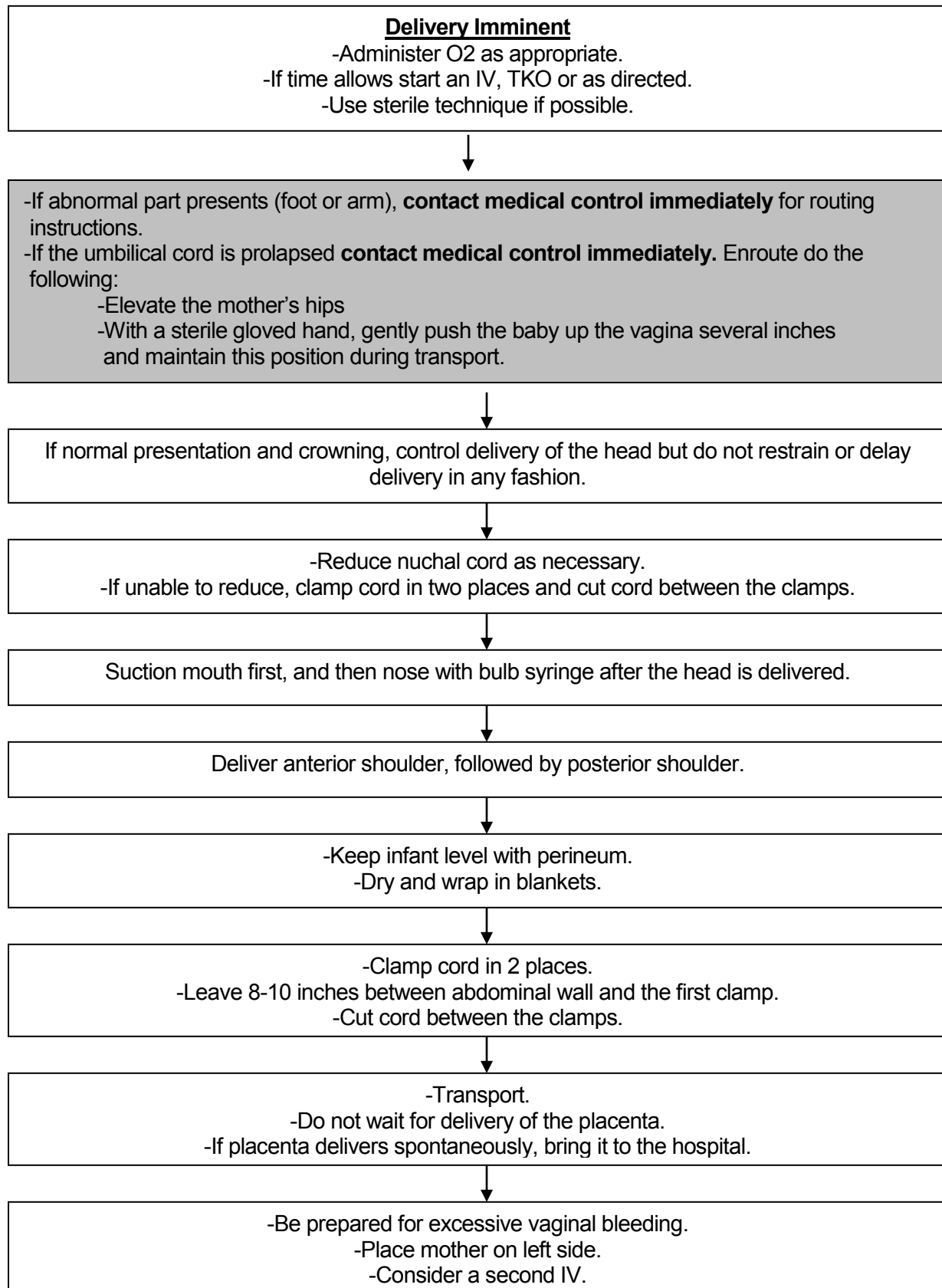
CHILDBIRTH

Assessment

1. Check vital signs.
2. Obtain brief history noting due date, multiple births, onset of labor, timing of contractions, and history of ruptured membranes including color of fluid (meconium).
3. As appropriate, assess for crowning, abnormal presentations, or significant vaginal bleeding.



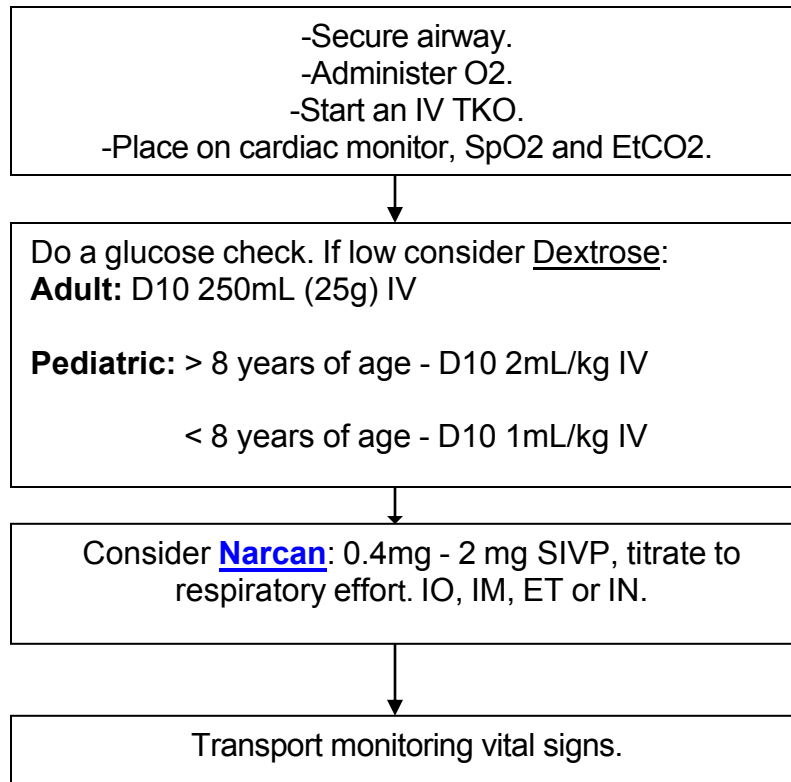
CHILDBIRTH (CONT)



COMA OR CONFUSION OF UNKNOWN ORIGIN

Assessment

1. Take vital signs.
2. Obtain brief history at scene with attention to onset and progression of present state. If this information is unobtainable, try to establish when the patient was last seen alert and/or appropriate.
3. Briefly assess cardiac, pulmonary, and neurological systems.
4. Look for signs of trauma, empty pill bottles, alcohol, etc.



If IV access is unavailable or patient's condition is determined to be life threatening, consider immediate IN administration of Narcan. See [intranasal](#) procedure

CVA/STROKES

Consider stroke in any patient presenting with the following:

- Altered mental status
- Dysarthria (slurred speech)
- Aphasia (difficulty understanding or expressing speech)
- Weakness or Numbness in the face, arm, or leg
- Vision (blurry vision, double vision, Blindness)
- Dizziness, loss of balance, loss of coordination
- Headache (especially if sudden or severe)

Assessment

1. Check vital signs
2. Briefly assess pulmonary, cardiac, and neurological systems.
3. Determine time patient was last known normal
4. Administer Cincinnati Stroke Scale
5. Administer VAN Score
6. Determine Blood Glucose Level
7. Other specific history of importance includes
 - a. medications (especially anticoagulants),
 - b. recent surgeries (3 month brain or spine, 21 days major surgeries)
 - c. history of brain tumor or aneurysm, history of bleeding disorders.

Treatment

1. Administer O2 as appropriate to maintain O2 saturation >94%
2. Place on Cardiac Monitor and pulse oximeter
3. Start an IV TKO
4. Treat blood glucose per protocol

Cincinnati Prehospital Stroke Scale (CPSS)			
Sign/Symptom	How tested	Normal	Abnormal
Facial Droop	Have the patient show their teeth or smile	Both sides of the face move equally	One side of the face does not move as well as the other
Arm Drift	The patient closes their eyes and extends both arms straight out for 10 seconds	Both arms move the same, or both do not move at all.	One arm either does not move, or one arm drifts downward compared to the other.
Speech	The patient repeats "The sky is blue in Cincinnati".	The patient says correct words with no slurring of words.	The patient slurs words, says the wrong words, or is unable to speak

Van Assessment Tool for Large Vessel Occlusion:

VAN

1 HOW WEAK IS PATIENT ON ONE SIDE OF THE BODY?

	Mild (minor drift)
	Moderate (severe drift – touches or nearly touches ground)
	Severe
	No Weakness

If patient has **any weakness PLUS any one of the below:**

Visual Disturbance (field cut, double, or blind vision)

Aphasia (inability to speak or understand)

Neglect (gaze to one side or ignoring one side)

This is likely a large artery clot (cortical symptoms) = VAN Positive

2 VISUAL DISTURBANCE?

	Field cut
	Double vision
	Blind new onset
	NONE

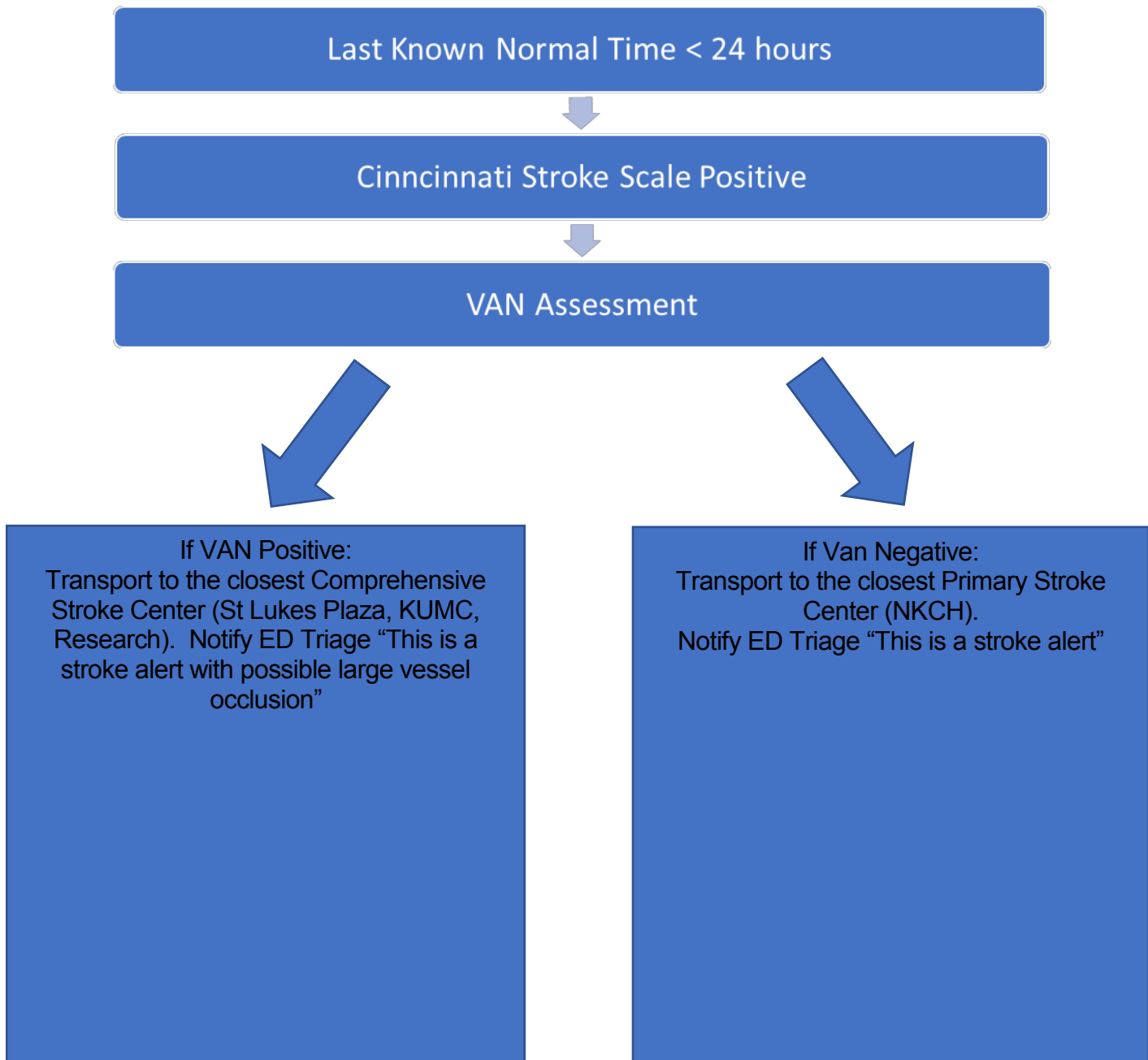
3 APHASIA?

	Expressive
	Receptive
	Mixed
	NONE

4 NEGLECT?

	Forced gazed or inability to track one side
	Unable to feel both sides at same time or unable to identify own arm
	Ignoring one side
	NONE

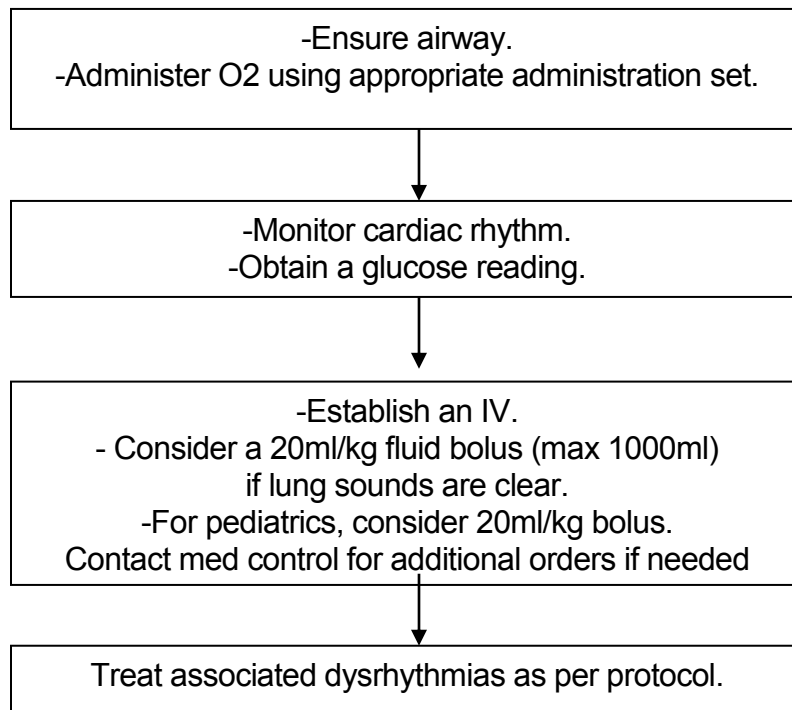
Stroke Routing Protocol:



DEHYDRATION

Assessment

1. Check vital signs; should include [Orthostatic BP](#).
2. Obtain brief history noting onset, any complaints of pain, past medical history, and medications.



Reassess
lung sounds
frequently

DIABETIC EMERGENCIES

Assessment

1. Check vital signs.
2. Obtain brief history noting onset and progression of present state, past medical history, and medications.
3. Consider other etiologies (drug use, alcohol).

-Administer O2 as appropriate.
-Establish an IV.
-**Consider 12/15 Lead EKG/**
monitor to r/o AMI with a positive
cardiac history.

<http://circ.ahajournals.org/content/97/1/12>
Diabetes/Cardiac relation

Hypoglycemia (< 60 mg/dL)

Adults: Give Dextrose - **D10 250ml (25g) IV**
Greater than 8 years of age: **2 mL/kg D10 IV**
Less than 8 years of age: **1 mL/kg D10 IV**

Refer to Broselow tape for weight estimation
as needed.

15g oral glucose may be considered for awake,
cooperative patients with a patent airway and
the ability to swallow.

Consider [Glucagon](#) if unable to obtain IV access.

Adult: **1 mg IM**

Pediatric: **Greater than 25kg - Give 1mg IM**
Less than 25kg - Give 0.5mg IM

MUST TRANSPORT AFTER
ADMINISTERING.

Glucose checks should be done prior to and
after administration of Dextrose.

Consider additional Dextrose administration if
patient remains unconscious or disoriented.

Transport monitoring vital signs.

Hyperglycemia

Consider a 20 ml/kg NS fluid bolus (max
1000ml) if blood glucose reading is **over**
400 mg/dL.

-**Use caution in CHF patients.**
-Patients must maintain clear lung sounds.

Transport monitoring vital signs.

HYPER-AGITATED DELIRIUM SYNDROME

The purpose of this protocol is to identify and treat those patients experiencing Hyper-Agitated Delirium by mitigating the effects and preventing sudden death.

TREATMENT:

- Request police assistance immediately.
- Initiate General Patient Care.
- Presentation:
Identify possible Hyper-Agitated Delirium using acronym **PRIORITY**. Presence of some or all of these symptoms is indicative of potential syndrome.
 - a. **P**: Psychological issues
 - b. **R**: Recent drug/alcohol use
 - c. **I**: Incoherent thought processes
 - d. **O**: Off (taking clothes off) and sweating, **HYPERTHERMIA**
 - e. **R**: Resistant to presence or dialogue
 - f. **I**: Inanimate objects: violent toward shiny or glass objects
 - g. **T**: Tough, unstoppable, superhuman strength
 - h. **Y**: Yelling
- **Providers should ensure crew safety at all times. (See Restraint Protocol)**
- Consider [Ketamine](#) 3-5 mg/kg IM. Maximum 500 mg. Contact Medical Control for further doses.
- Consider [Midazolam](#) "Versed"
 - a. 5-10 mg **IM/IN**
 - b. 5-10 mg **IV, 2 mg increments**
 - c. 0.1 mg/kg **IV** (Pediatric dose)
 - d. **Maximum total dosing is 10 mg, call med control for additional**
- Establish IV access **after sedation, if not already established**.
 - a. Infuse 1L of cool (room temp, not from a warmer) **Normal Saline**
 - b. Apply capnography to monitor patient respiratory status.
- Prepare to aggressively manage the airway if needed.
- Rapid Transport to the closest appropriate hospital with Medic and rider to assist.

Sudden Death Factors

- Metabolic Acidosis
 - Potentially life threatening abnormality of the blood chemistry
 - Aggravated by:
 - Increased activity, violent confrontation, dehydration
 - Blood acidity affects vital functions
 - Elevated blood potassium level
 - Survivors
 - Kidney damage due to muscle breakdown
 - May require dialysis

GYNECOLOGICAL EMERGENCIES

Assessment

1. Take vital signs.
2. Obtain brief history including onset, last menstrual cycle, past medical history and amount of blood loss if any (number of pads used in what time frame).
3. Briefly assess cardiac, pulmonary, and neurological systems.

-Administer oxygen as appropriate.
-Monitor cardiac rhythm.
-Establish an IV.

Vaginal Bleeding

Consider a 20ml/kg fluid bolus (max 1000ml) if lung sounds are clear.

-Monitor lung sounds.

Emergency Childbirth, Postpartum Care, or Postpartum Hemorrhage

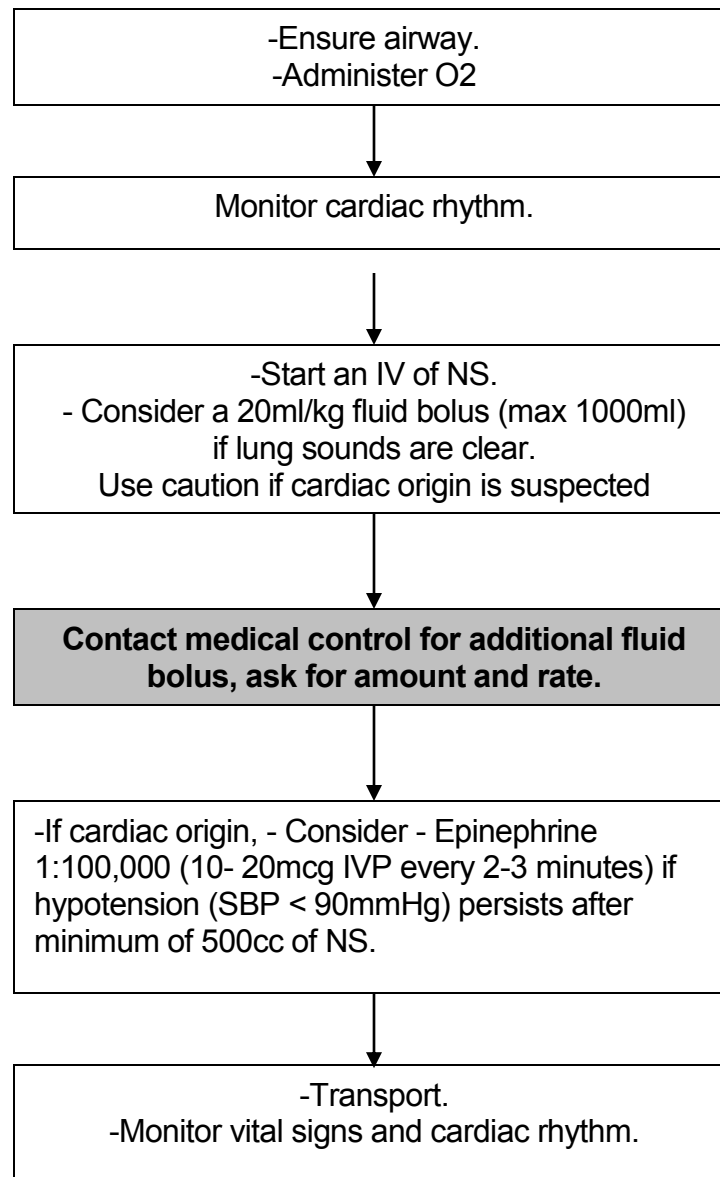
Refer to [childbirth protocol](#)

Some causes of vaginal bleeding could include: ectopic pregnancy, miscarriage, abruptio placentae, placenta previa, and pre-term labor.

HYPOTENSION

Assessment

1. Obtain vitals, A BP SHOULD BE TAKEN ON BOTH ARMS, a difference of 15mmhg, systolic or diastolic and a history of hypertension could indicate an Aortic Dissection and SHOULD BE REPORTED. Consider [Orthostatic Vitals](#). Determine if absolute or relative.
2. Obtain brief history, BP norm and noting any complaints involving the cardiovascular and respiratory systems, onset, past medical history, and medications.
3. Consider obtaining an early EKG 12/15 lead.



HYPERTHERMIA

Hyperthermia may be the result of elevations in body temperature secondary to infections, drug ingestions, chronic diseases, and environmental conditions. Elevations in body temperature may induce seizures in children. **Hyperthermia with an altered mental status is a true emergency regardless of the etiology.**

Assessment

1. Check vital signs including a temperature. Significant hyperthermia is present when core body temperature exceeds 105° F.
2. Obtain brief history noting length and type of exposure, exercise related, illness related, and environmental etiology.
3. Briefly perform cardiac, respiratory, and mental status assessments. Note the condition of the skin.
4. Remember that the elderly and pediatrics are more susceptible to heat.

Treatment without altered mental status

-Secure airway.
-Administer O2 as needed.

-Loosen clothing or remove any heavy clothing.
-Do not cover with blankets.

Consider starting an IV, TKO or as directed.

Consider cooling the patient by sponging with room temperature water. Avoid inducing shivering.

Place on cardiac monitor if the patient has an abnormal pulse rate, respiratory rate, or blood pressure.

Treatment with altered mental status

-Secure airway.
-Administer O2 as appropriate.
-Assist ventilations as needed.

-Establish an IV, TKO or as directed.
-Place on cardiac monitor.

-Remove clothing and consider cooling by sponging patient with room temperature water, and if possible fan the patient to achieve evaporation.
-Avoid inducing shivering.

*Do a glucose check. Consider administration of Dextrose if indicated:

Adults: Give Dextrose - **D10 250ml (25g) IV**
Greater than 8 years of age: **2 mL/kg D10 IV**
Younger than 8 years of age: **1 mL/kg D10 IV**

Consider administration of Narcan 0.4 -2 mg SIVP, titrate to respiratory effort, monitor EtCO2.

Transport monitoring vital signs.
Do not delay transport to perform cooling in the field.

HYPOTHERMIA

Children in general and neonates in particular are more susceptible to hypothermia than adults.

Assessment

1. Check vital signs including temperature. Severe hypothermia mimics death.
2. Obtain brief history noting length of exposure and pertinent medical history.
3. Briefly assess cardiac status, mental status, and skin for evidence of local injury such as frostbite.

-Handle the patient as gently as possible.
-Avoid any unnecessary manipulation.

Do CPR as necessary.

Secure airway with as little manipulation as possible.
Stimulation of the airway can cause ventricular fibrillation.

-Administer O2 as appropriate.
-Assist ventilations as needed.

Establish an IV, TKO or as directed.

-Place patient on cardiac monitor.
-If patient is in VF, attempt defibrillation.

Remove any wet clothing and wrap patient in blankets.

Do a glucose check.
Consider administration of **Dextrose** if indicated:

Adults: Give Dextrose - **D10 250ml (25g) IV**
Greater than 8 years of age: **2 mL/kg D10 IV**
Younger than 8 years of age: **1 mL/kg D10 IV**

Consider administration of **Narcan** 0.4-2 mg SIVP, titrate to respiratory effort, monitor EtCO2.

Transport while monitoring vital signs and cardiac rhythm.

Local (Frostbite)

Handle injured part very gently. Protect from pressure, trauma, or further environmental hazards.

Do not allow the limb to thaw if there is a chance of refreezing before evacuation and transport is complete.

-Maintain core temperature with blankets.
-Monitor vital signs.

Active rewarming should occur at the hospital, not in the field.

NAUSEA / VOMITING

For the prevention of nausea and vomiting in acute medical conditions: including headache, chest pain, abdominal pain, chemotherapy, or post-operative induced nausea and vomiting.

Assessment

1. Take vital signs.
2. Obtain brief history including: onset, duration, and past medical history.
3. Briefly assess cardiac, pulmonary, and neurological systems.

Secure airway.



-Establish an IV.
Consider a 20ml/kg fluid bolus (max 1000ml) if blood pressure is <90 mm/hg.
-Lung sounds must be clear

-Consider [Zofran](#): 4 mg SIVP or PO
-Children (4 and younger): 2 mg IV.
Must contact medical control for orders prior to administering to children.

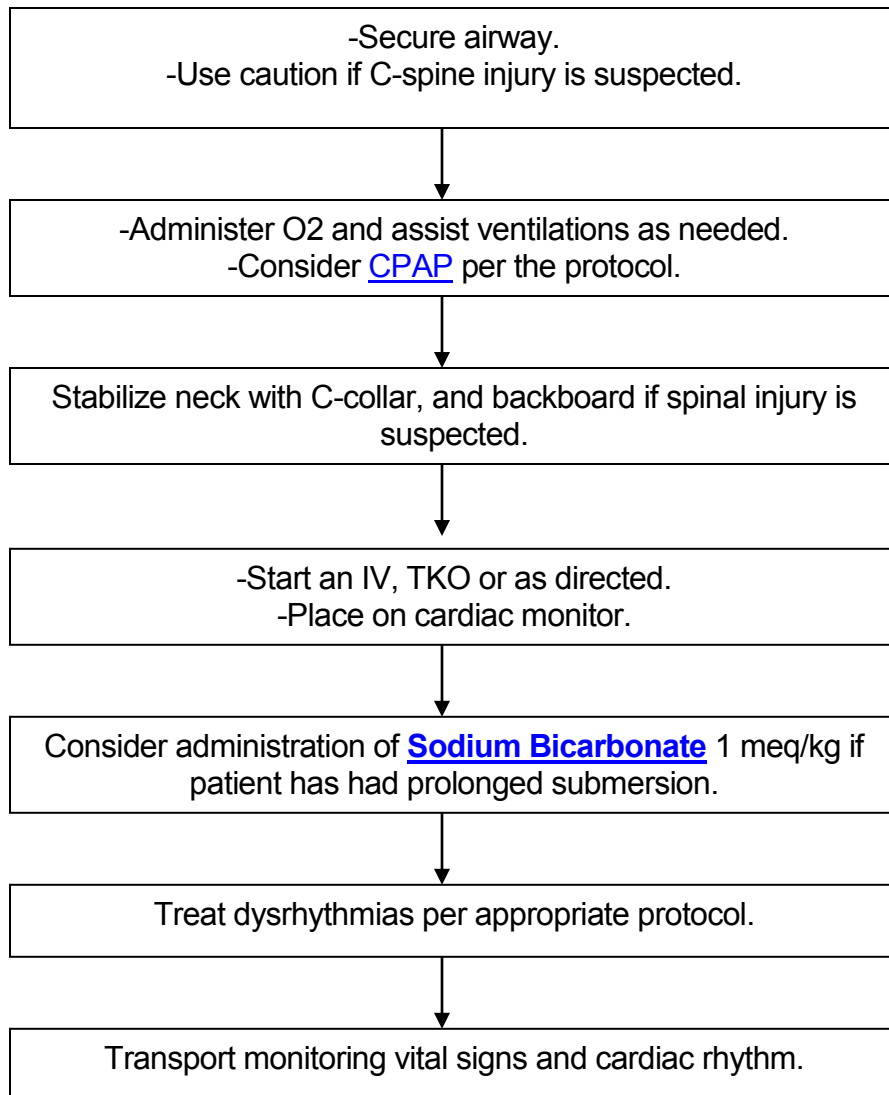
Contact medical control for additional doses.

Contact medical control for orders and dose for children under 4 years of age.

NEAR DROWNING

Assessment

1. Check vital signs. If pulseless and non-breathing, follow cardiopulmonary arrest protocol.
2. Obtain brief history noting estimated submersion time, water temperature, and mechanism of injury.
3. Assess cardiac, respiratory, and neurological status.
4. **Be alert for possible head/neck injury.**



NEONATAL RESUSCITATION

Assessment

1. Assess airway, breathing, and circulation. The need for resuscitation can be guided by heart rate, respiratory rate and effort, tone, and color.
2. Determine APGAR score.
3. Look for meconium stained fluid.
4. Resuscitation of the neonate rarely requires advanced life support. It should proceed in a stepwise fashion with continuing reassessment to avoid overly aggressive intervention.
5. **Use of Broselow™ Tape is recommended for all pediatric dosing.**

APGAR SCORING

APGAR Chart



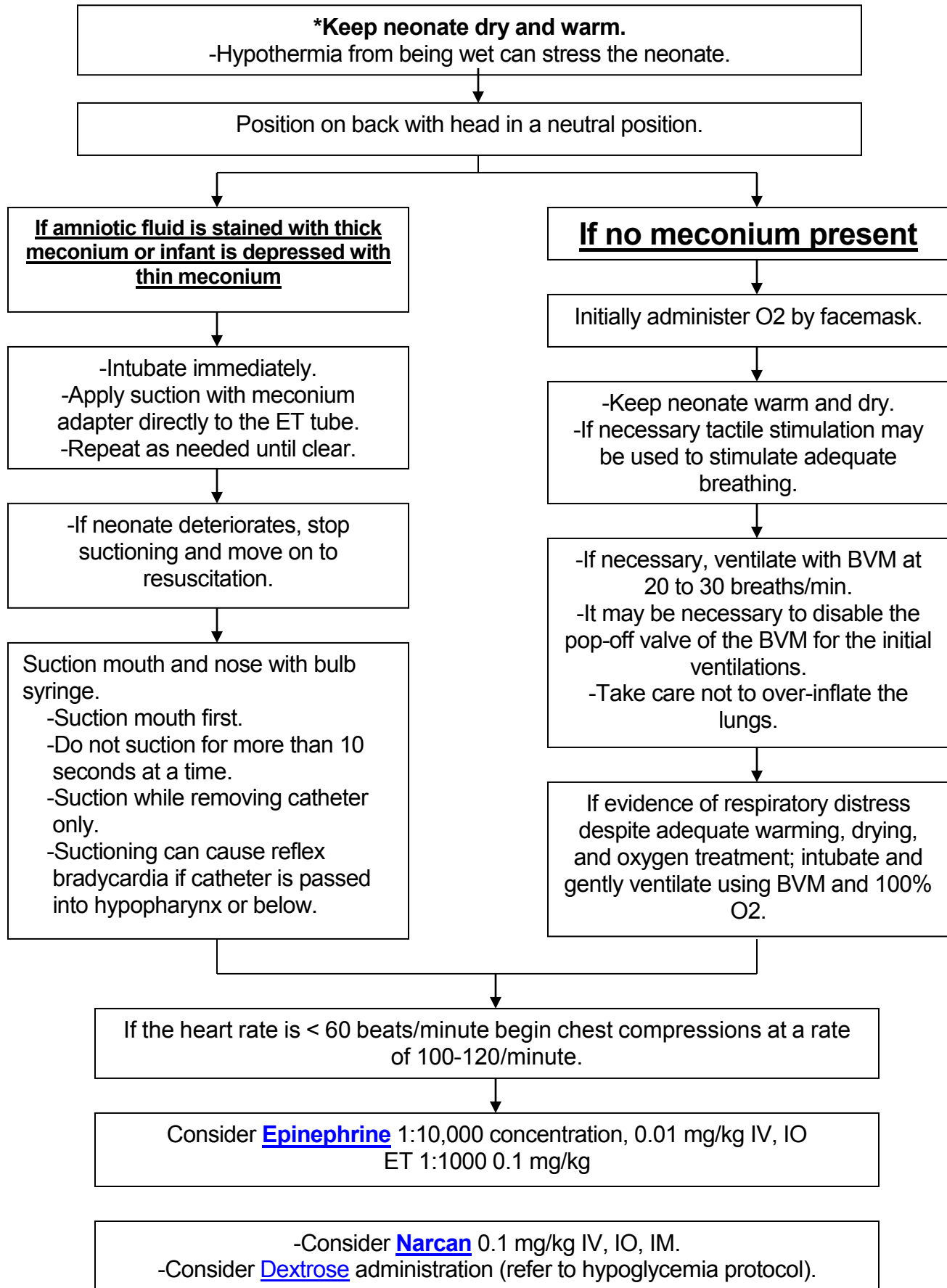
	Score 0	Score 1	Score 2
A pppearance			
P ulse	No pulse	< 100/min.	> 100/min.
G rimace			
A ctivity			
R espirations	No respirations	Weak, slow	Strong cry

Chart showing the Apgar scoring of condition of newborn soon after birth

LifeART Collection Images Copyright © 1989-2001 by Lippincott Williams & Wilkins, Baltimore, MD

<http://easycalculation.com/medical/Apgar-score.php>

NEONATAL RESUSCITATION (CONT)



POISONING

Assessment

1. Check vital signs. **Monitor SpO2 and EtCO2**
2. Obtain brief history. Bring any containers, package inserts, etc. of suspected ingestion. Small amounts of ingested alcohol may cause unconsciousness and/or hypoglycemia in children.
3. **Obtain 12 lead EKG as soon as possible.**
4. **Assess level of consciousness, cardiac, and pulmonary systems. Monitor for seizure and/or respiratory distress.**
5. Consider contacting Poison Control **1-800-222-1222** or medical control for information.

Alert patients with oral ingestion:

Transport monitoring vitals and level of consciousness.

Patients With Decreased Level of Consciousness:

- Secure airway, suction as needed.
- Administer O2 as appropriate.
- Establish an IV at a TKO rate.

- Do a glucose check.
- Consider administration of **Dextrose** per [hypoglycemia protocol](#)

- Consider administration of **Narcan** 0.4-2mg SIVP, titrate to respiratory effort, monitor EtCO2.
- If IV access is unavailable or patient's condition is determined to be life threatening, consider immediate administration of **IN Narcan**. See [intranasal procedure](#)
- Narcan** IV is preferred but also IM, IO, ET or IN.

Monitor cardiac rhythm.

Transport in lateral position.

For tricyclic antidepressant overdoses with wide complex tachycardias, consider intubation. **12 Lead EKG**, Consider administration of [Sodium Bicarbonate](#), 1 meq/kg IV (max 50 meq). **Contact Medical Control prior to administration.**

[**See commonly prescribed TCA's**](#)

For calcium channel blocker overdoses with hypotension, consider administration of [Calcium Chloride](#) 10% solution, 500 to 1000 mg or a dose of 5-10ML SIVP
Precipitates with NaHCO3
[**See list of commonly prescribed CCB's**](#)

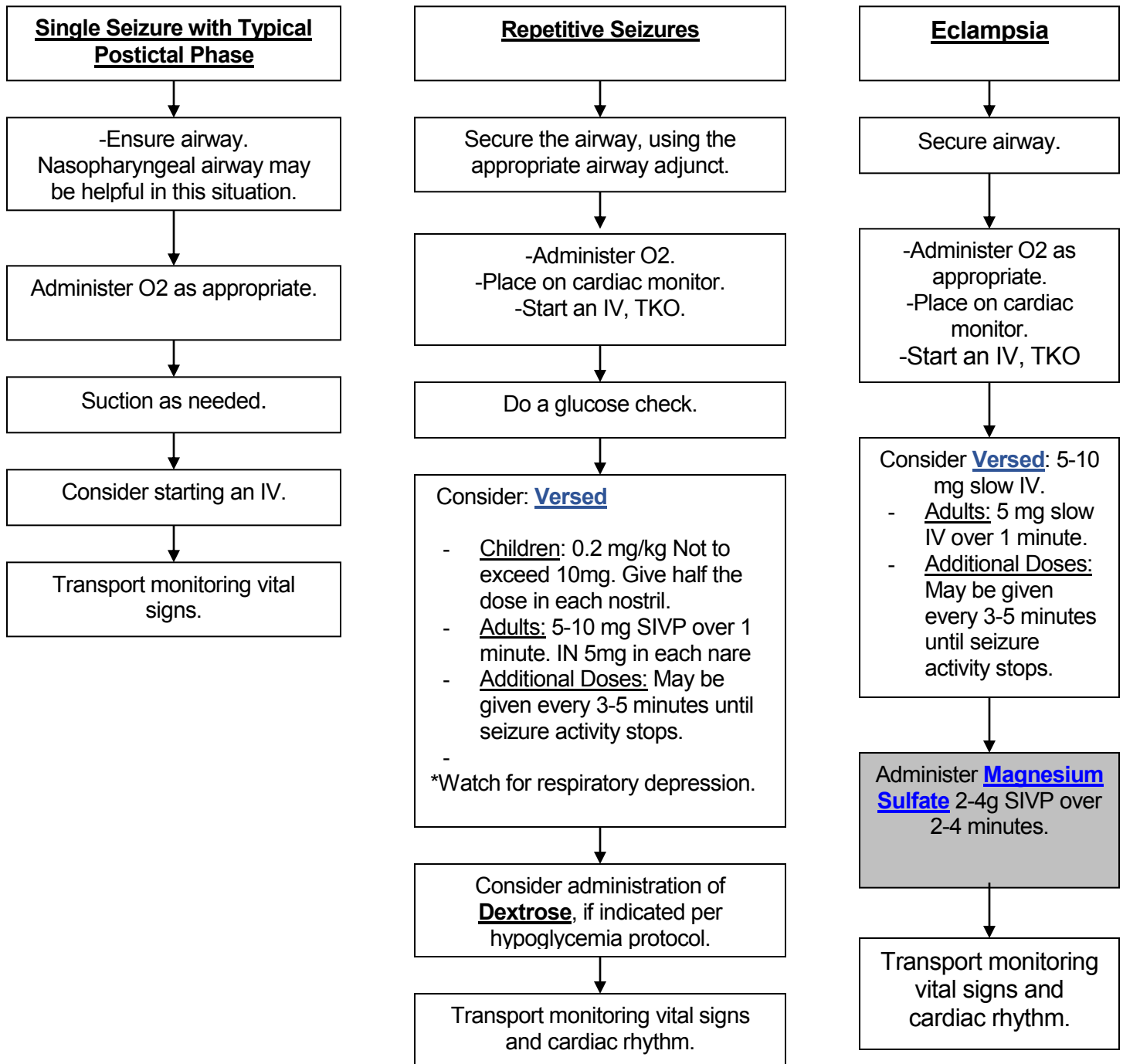
For Organophosphate or Carbamate poisoning, administer: [Atropine](#) 2 mg SIVP If no tachycardia or pupil dilation, give a second dose of 2mg SIVP.
Seizure Activity: [Versed](#) per seizure protocol

For a known benzodiazepine overdose: monitor vitals, IV, and secure airway.

SEIZURES

Assessment

1. Check vital signs.
2. Obtain a brief history including an exact description of the incident from observers. Also obtain history of medication use, presence/absence of alcohol, substance abuse, recent illness, or trauma.
3. Briefly assess cardiopulmonary and neurological system's noting any seizure activity you observe.

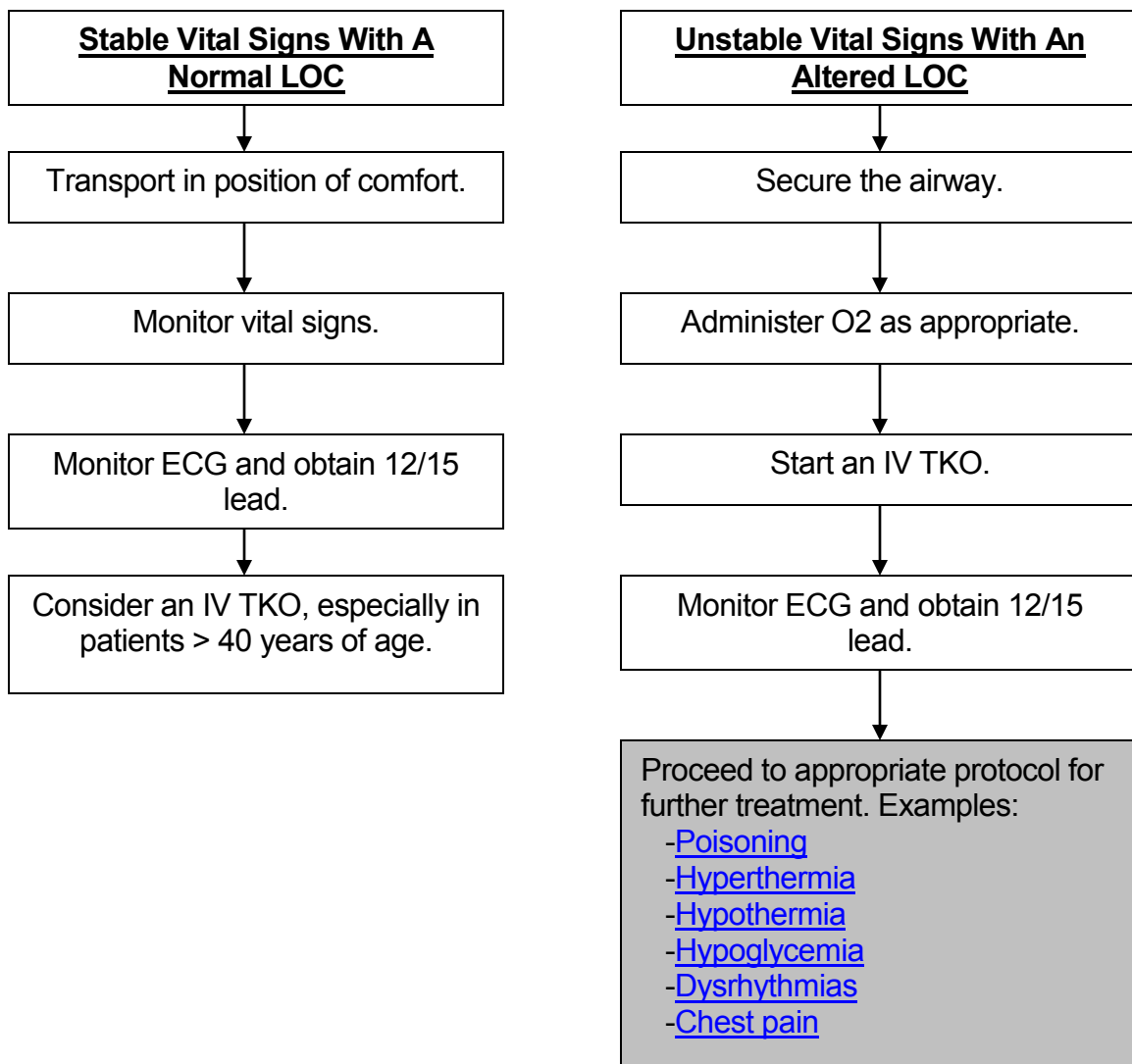


SYNCOPE

Many cases of syncope are vasovagal in nature. However other causes to be considered include dysrhythmias, poisonings, trauma, or shock of various etiologies.

Assessment

1. Check vital signs, consider [Orthostatic vitals](#)
2. Obtain brief history including onset, duration, warning symptoms (dizziness, nausea, lightheadedness), and precipitating factors (sudden change of position).
3. Perform cardiac and neurological assessments.



GENERAL TRAUMA PROTOCOL

Patients with **obvious mortal injuries considered incompatible with life.** Who are asystolic or have no audible heart sounds and who have not been observed at the scene to possess any sign of life, may be pronounced DOA per DOA protocol.

Transport of the unstable trauma patient should not be delayed by intubation attempts, unless the patient cannot be adequately ventilated with a BVM.

Assessment

1. Assess adequacy of airway and breathing with simultaneous cervical spine immobilization.
2. Assess cardiopulmonary system with attention to adequacy of perfusion (mental status, skin moisture and temperature, capillary refill, and character of pulses).
3. Use of EtCO₂ is recommended to determine and monitor compensation/perfusion.
4. Assess neurological function (LOC, pupils, motor function, sensation)
5. Obtain brief history of incident and mechanism of injury. Determine transport status.

Physiological Criteria

- Blood pressure <90 mm/hg or absence of peripheral pulses.
- Respiratory rate >29 or <10.
- GCS < 14.
-

Mechanism of Injury

- Rollover
- Occupant ejection.
- MVC with intrusion greater than 1 foot.
- Fall greater than 3 feet.
- Death of same car occupant.
- Motorcycle / ATV speed >20mph.
- Prolonged extrication time >20 minutes.
- Pedestrian struck at speed >20 mph.
- Blunt Trauma.
- Hanging.
- Unrestrained with significant MOI.
- Axial Loading / Compression injury.

Anatomic Criteria

- Pain, deformity or tenderness to neck or back.
- Neurological Deficit.
- Painfully distracting injury.
- Pain produced on unassisted rotation of the head 45 degrees in either direction.

Secure the airway and assure adequate ventilation.

- Stabilize the C-spine with manual in-line immobilization.
- A modified jaw thrust is recommended to open the airway.
- Consider the use of advanced airway adjuncts, and administer appropriate O₂ concentration.

For a severe extremity laceration or amputation with massive hemorrhage consider the early use of a tourniquet.

1. Oral/nasal tracheal intubation may be attempted if unable to adequately ventilate the patient with a BVM due to severe facial trauma, or excessive blood or secretions. Maintain in-line C-spine immobilization during intubation attempts.
2. In adults, nasotracheal intubation may be attempted if not contraindicated.
3. Cricothyrotomy may be attempted only if unable to otherwise open the airway. C-spine immobilization must be maintained during attempts.

Damage Control Resuscitation

- Control external bleeding sterile dressings and direct pressure or **tourniquet**.
- Support Airway and Breathing
- Occlusive dressing on open wounds neck to navel

Transport of the unstable patient should not be delayed to initiate IV therapy.

- Initiate 2-large bore IV's try to maintain a systolic pressure of 70-90 mmHg.

Consider 1 gram TXA via IV drip if appropriate.

Conserve body temperature!

If Cardiovascularly Unstable:

JUDICIOUSLY administer warmed IV fluids LESS IS MORE in hemorrhagic patients (THESE PATIENTS NEED BLOOD) Administer 250ml bolus reassess can administer additional 250ml.

If the patient remains in shock following the first two fluid boluses, subsequent 20 ml/kg boluses may be given with Medical Control approval.

Monitor cardiac rhythm and treat dysrhythmias per appropriate protocol.

GENERAL TRAUMA (CONT)

Spinal Immobilization

Protocol Consideration:

- Patients with penetrating injuries should **NOT** be immobilized unless a focal neurologic deficit is noted on physical examination.
- In the stable patient, spinal immobilization as indicated by mechanism and exam, using the appropriate devices and techniques.
- In the unstable patient, appropriate rapid extrication techniques should be utilized.
- USE CAUTION, look for occult fracture in >65 or <4 y/o or a HX of osteoporosis, bone or vertebral disease.
- In a pregnant patient, if the uterus is palpable above the umbilicus, place patient left lateral recumbent to prevent compression of the vena cava and maternal hypotensive syndrome.

Procedure:

1. Stabilize head and neck in neutral position unless movement causes pain, deformity, or resistance. If so, immobilize the head and neck in the position found.
2. **If the patient is ambulatory, apply cervical collar and assist / position them directly on the ambulance cot.**
3. Patients that are stable, alert and without neurological deficits may be allowed to self-extricate with assistance to the ambulance cot after placement of a cervical collar.
4. If a Long Spine Board or Scoop Stretcher is used for extrication or patient movement, **the patient should be taken off (using proper techniques) the long spine board or scoop stretcher** and placed directly on the ambulance cot.
5. Limit movement of the spine during the procedures.

Spinal immobilization for athletes and motorcyclists:

The helmet and/or shoulder pads should be removed as a unit if:

- Immobilization of the helmet does not also immobilize the head.
- The airway cannot be controlled or ventilation provided even after removal of the facemask.
- The helmet or pads prevents immobilization for transportation in an appropriate position.

Special Considerations:

- Older adults: low impact mechanisms (e.g., ground-level falls) might result in severe injury, anticoagulants/bleeding disorders.
- Patients with head injury are at high risk for rapid deterioration
- Burns:
 - Without other trauma mechanism: triage to burn facility
 - With trauma mechanism: triage to trauma center
- Pregnancy >20 weeks
- EMS provider judgment

EMERGENT Transport!!!

-Monitor vital signs and continually reassess patient status and interventions.

-Transport to the closest appropriate trauma center.

-Trauma codes should be taken to the nearest Level I or II trauma center, whether the trauma center is on diversion status or not. This applies only to trauma codes.

ABDOMINAL TRAUMA

Assessment

1. Assess per the general trauma protocol.
2. Briefly assess chest, abdomen, and genitalia.

Treat as per the general trauma protocol.

-If evisceration is present cover **exposed viscera** with a sterile saline-soaked covering and or occlusive dressing.
-Do not attempt to replace exposed viscera.

-If pelvic fracture is suspected, place patient on a long spine board, improvise padding, belt device or **PASG** for stabilization.

Transport to the appropriate trauma facility.
Conserve body temperature KEEP PATIENT WARM

BURNS

Assessment

1. Stop the burning process. Remove the patient from the source.
2. Obtain brief history noting time exposed, contact with steam, smoke or toxic gas.
3. Assess burn noting depth and extent (rule of nines). Assess cardiac and respiratory status noting any sooty sputum, singed nasal hair, or hoarseness.
4. Consider intubation or cric insertion if laryngeal swelling is present.
5. Be alert for associated injuries.

-Secure airway IMMEDIATELY.
 -Administer O2, humidified if possible.
 -Assist ventilations as needed.

-Start an IV. Consider 20 ml/kg fluid bolus initially (maximum of 1000 ml).
 -Reassess breath sounds, capillary refill, pulse rate, and blood pressure.
 *If needed, administer a second bolus of 20 ml/kg (maximum of 1000 ml) to a total bolus of 40 ml/kg (maximum of 2000 ml).

Monitor ECG, EtCO2 and CO-Oximetry

-If >20% of burns are second or third degree, cover with dry sterile burn sheet.
 -Leave blisters intact.

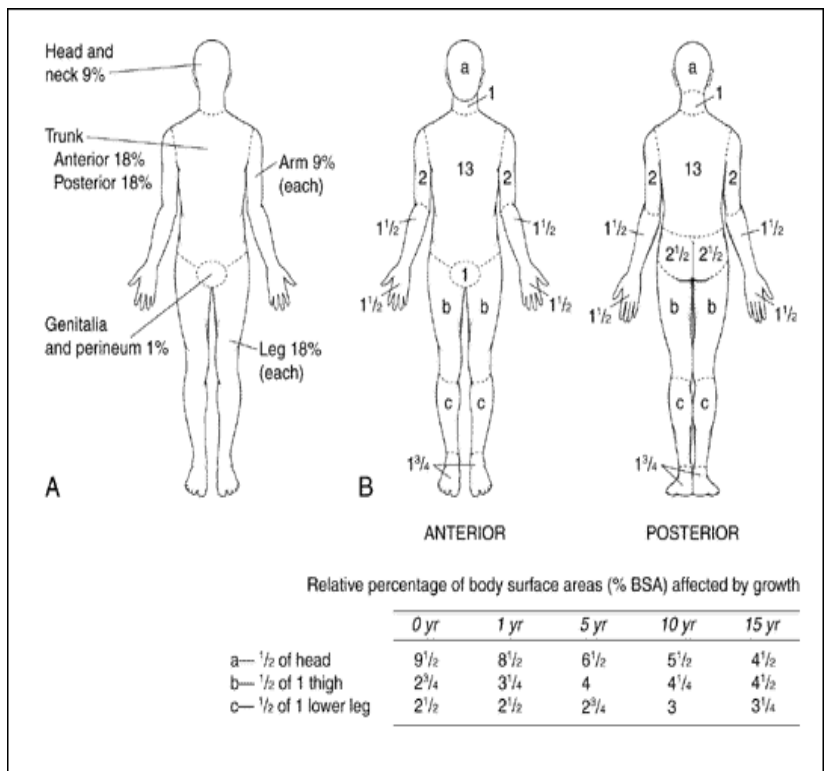
First or second degree burns with <20% involvement, cool the area with saline soaks.

Chemical burns: remove contaminated clothing. Refer to MSDS sheet, Haz Mat ERG, and medical control for further instructions.

-Consider **Morphine** 2 mg increments slow IV or IM up to a total dose of 10mg
 or
 -Consider **Fentanyl** 50-100 mcg slow IV, IM or IN.

Adult

Pediatric



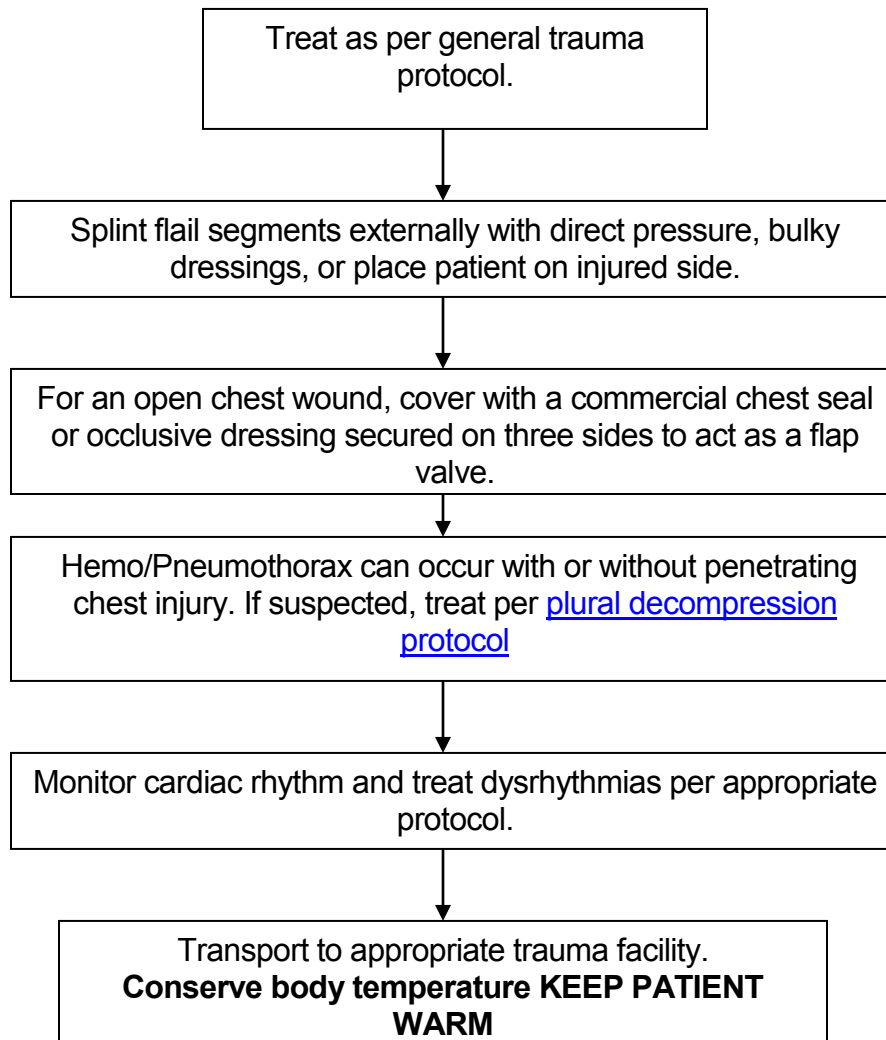
<http://easycalculation.com/medical/burn-fluids.php>

Contact medical control for transport routing to the appropriate facility. According to MARCER, EMS providers should now consider routing **Adult** burn patients directly to a burn center, regardless of the patients' age. MARCER defines a direct route patient to a burn center as one with more than 20% second degree or greater than 5% third degree burns shall be transported to KU Burn Unit. **Pediatric** burn patients shall be routed to Childrens Mercy Hospital.

CHEST TRAUMA

Assessment

1. Assess per general trauma protocol.
2. Perform brief chest exam. Specifically check for open wounds; flail segments, tracheal deviation, unequal breath sounds, subcutaneous emphysema, and adequacy of ventilation.



Use of nitrous oxide is contraindicated in chest trauma.

HEAD TRAUMA

Assessment

1. Assess per general trauma protocol.
2. Obtain brief history noting mechanism of injury and level of consciousness since the injury.
3. Check for associated injuries. Assume spinal injury in all patients with a head injury.
4. High flow oxygen is not recommended, maintain saturation levels 95% or above.

Treat as per general trauma protocol.

-Immobilize C-spine.
-Administer O2 low concentration
-Start an IV, TKO or as directed.

-Assist ventilations with bag valve mask if needed.
-If intubation is indicated: consider only oral intubation.
-You must maintain manual inline stabilization with all intubation attempts.

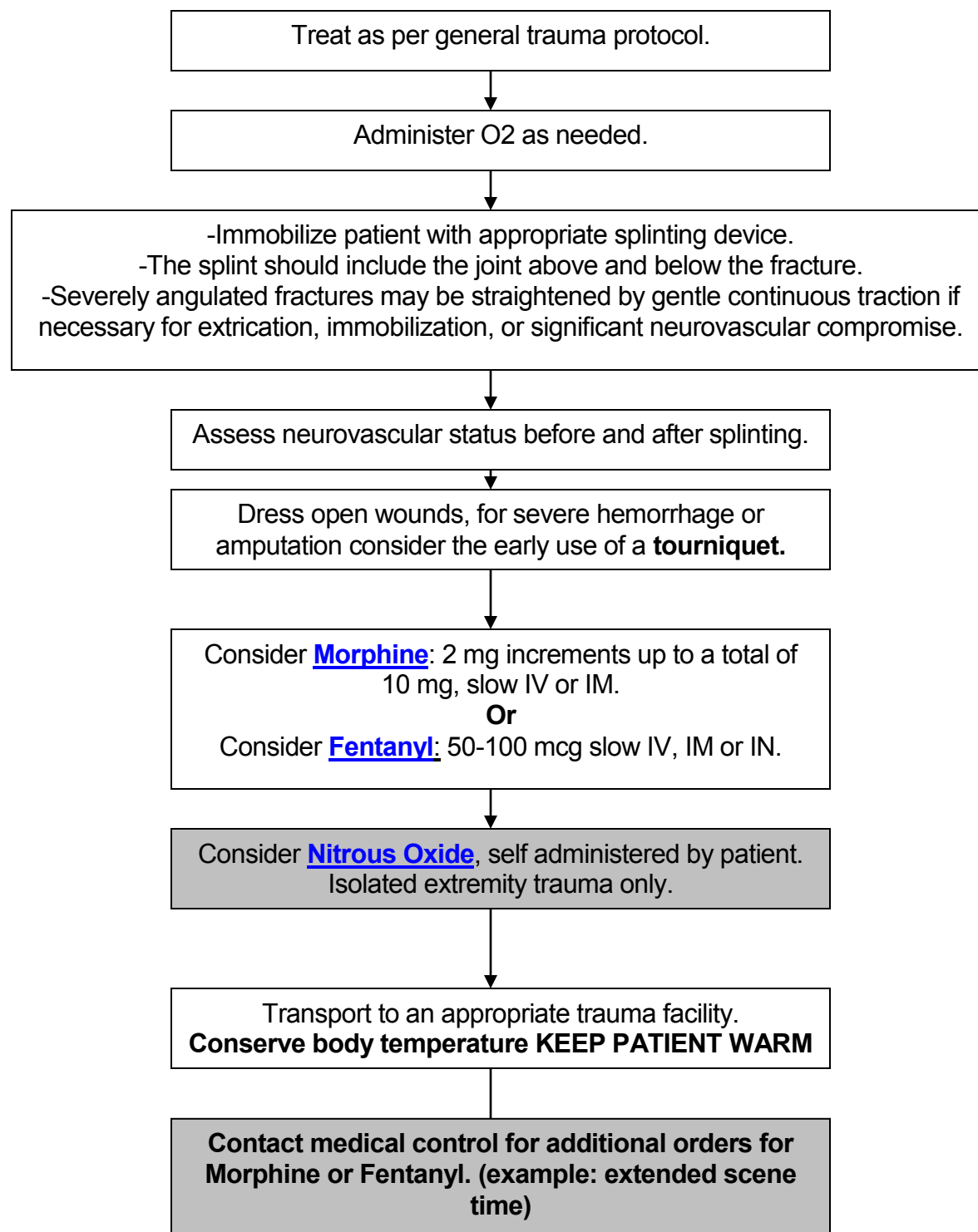
If vitals are stable, consider elevating head 30 degrees.

Transport to appropriate trauma facility.
Conserve body temperature KEEP PATIENT WARM

ORTHOPEDIC EXTREMITY TRAUMA

Assessment

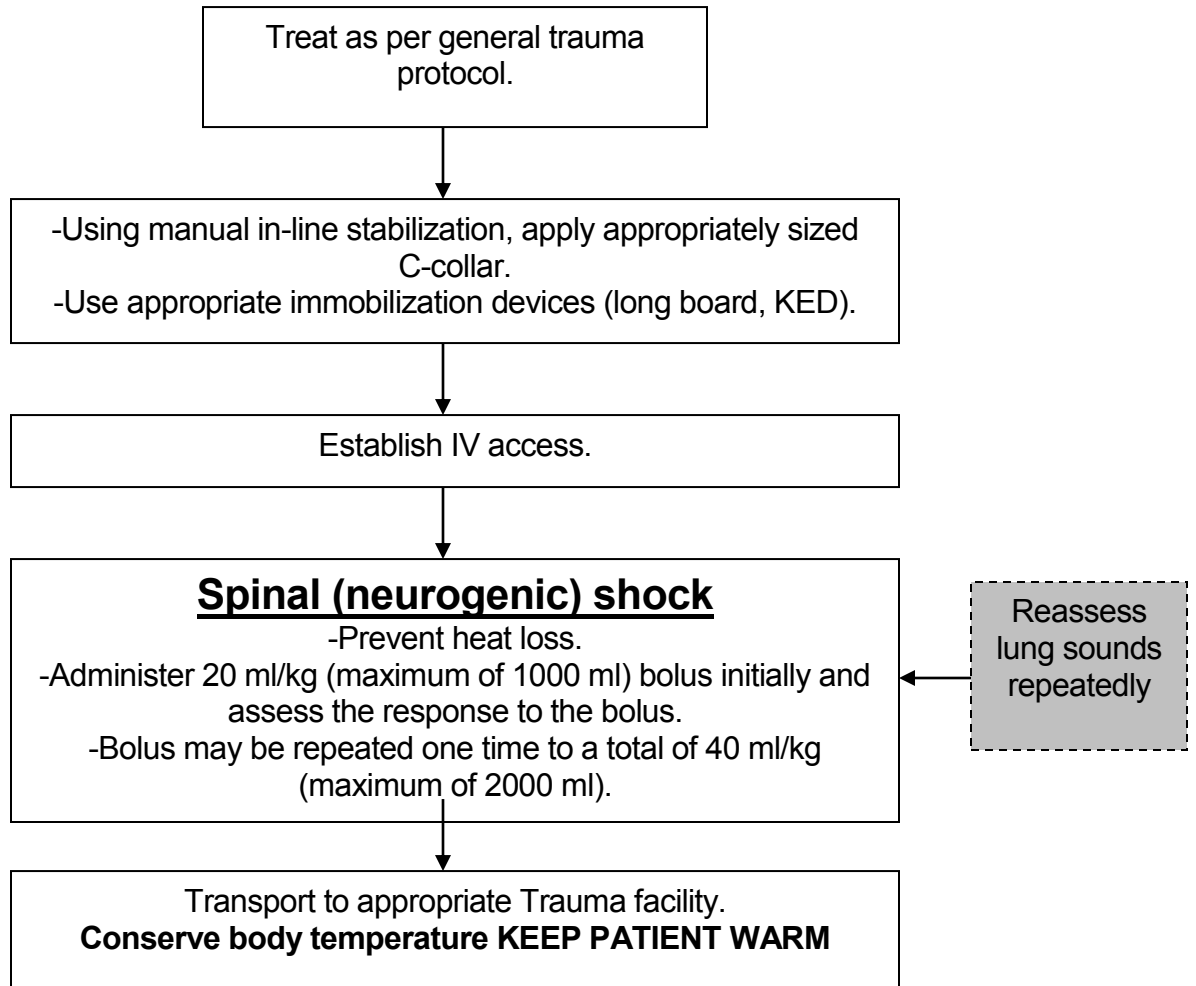
1. Assess per general trauma protocol.
2. Briefly assess the injured extremity noting neurovascular adequacy (color, temperature, deformity, open wounds, and distal sensation and movement).



SPINAL TRAUMA

Assessment

1. Assess per general trauma protocol.
2. Perform a brief neurological exam to include motor and sensory status.



ANALGESIC MEDICATION ADMINISTRATION

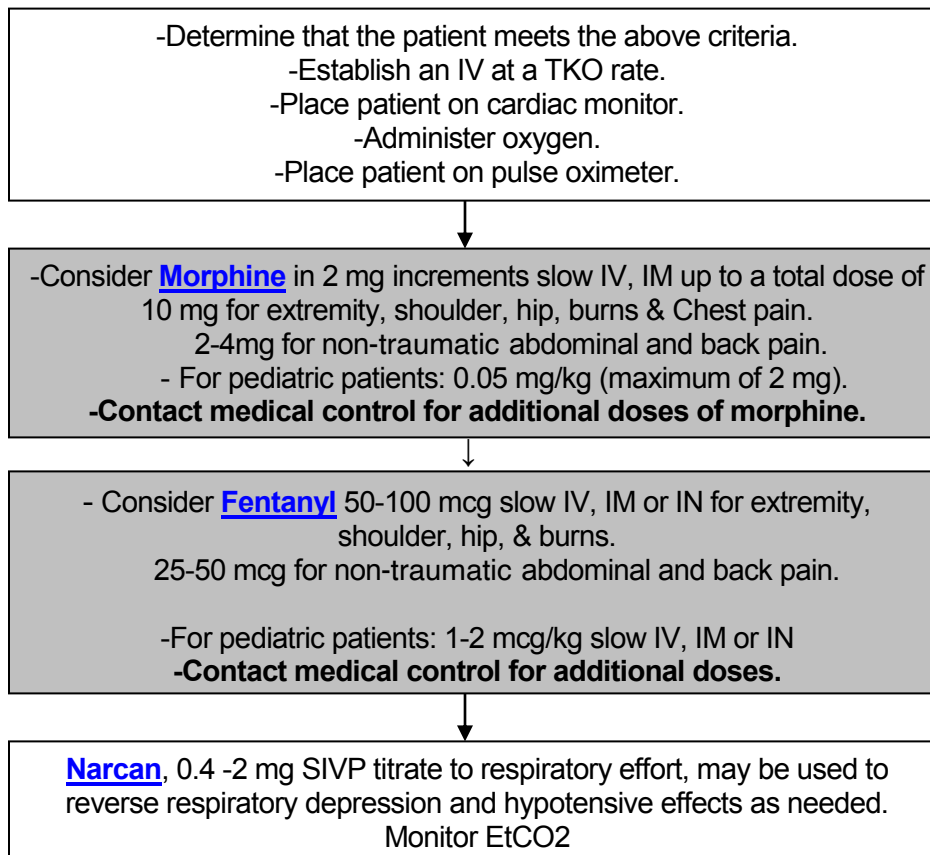
Indications

The relief of moderate to severe acute pain in patients with:

- A. An extremity injury
- B. A shoulder or hip injury
- C. Back and Abdominal pain (Non-Traumatic)
- D. Burns without airway or respiratory compromise
- E. Cardiac Chest Pain

Contraindications

- 1. Trauma
 - A. Head injury
 - B. Suspected spinal injury
 - C. Thoraco-abdominal trauma
- 2. Systems
 - A. Respiratory distress or compromise (asthma or COPD)
 - B. Cardiac dysrhythmia present
 - C. Altered mental status
 - D. Pregnancy
- 3. Drugs
 - A. Sensitivity or allergy to opiates
- 4. Hypotension and suspected shock
- 5. Age less than 12 months.



BIPAP

Indications

1. Short-term management of acute respiratory distress or failure in an awake cooperative patient.
2. Near drowning patients that are awake and cooperative.

Contraindications

1. Need for immediate intubation.
2. Unstable respiratory drive (inability to maintain their own airway.)
3. Ventilatory failure
4. Gastric distention
5. Claustrophobia

Precautions

1. Requires patient cooperation. (The major complication is the inability to tolerate the mask; in which case the mask should be removed and an alternate airway should be used.)
2. If patient complains of nausea, remove mask. The mask may be held in place manually. (Vomiting with the mask in place virtually assures aspiration.)
3. Adequate supply of oxygen is required.

Inform patient of the procedure.



Turn on the oxygen, hold the mask firmly against the patient and instruct the patient to take slow deep breaths, relaxing and allowing the machine to help.



-After the patient has tolerated the mask, the straps may be attached.
-Monitor the patient for comfort, anxiety, and nausea.



Settings for BiLevel (BL) should be
FIO2 set at 100%
IPAP 12 cm/Hg
EPAP 6cm/Hg
LC compensation on

The mask must be unstrapped if any nausea develops, but may be held in place manually by the patient with coaching.

CPAP

Indications

- 1 Short-term management of acute respiratory distress or failure in an awake cooperative patient.
- 2 Near drowning patients that are awake and cooperative.

Contraindications

- 1 Need for immediate intubation.
- 2 Unstable respiratory drive (inability to maintain their own airway.)
- 3 Ventilatory failure
- 4 Gastric distention
- 5 Claustrophobia

Precautions

1. Requires patient cooperation. (The major complication is the inability to tolerate the mask; in which case the mask should be removed and an alternate airway should be used.)
2. If patient complains of nausea, remove mask. The mask may be held in place manually. (Vomiting with the mask in place virtually assures aspiration.)
3. Adequate supply of oxygen is required.

Inform patient of the procedure.



Turn on the oxygen, hold the mask firmly against the patient and instruct the patient to take slow deep breaths, relaxing and allowing the machine to help.



-After the patient has tolerated the mask, the straps may be attached.
-Monitor the patient for comfort, anxiety, and nausea.

The mask must be unstrapped if any nausea develops, but may be held in place manually by the patient with coaching.

Double Sequential External Defibrillation

Indications

Refractory Ventricular Fibrillation or Pulseless Ventricular Tachycardia resistant to pre-defibrillation attempts and pharmacological efforts.

Contraindications

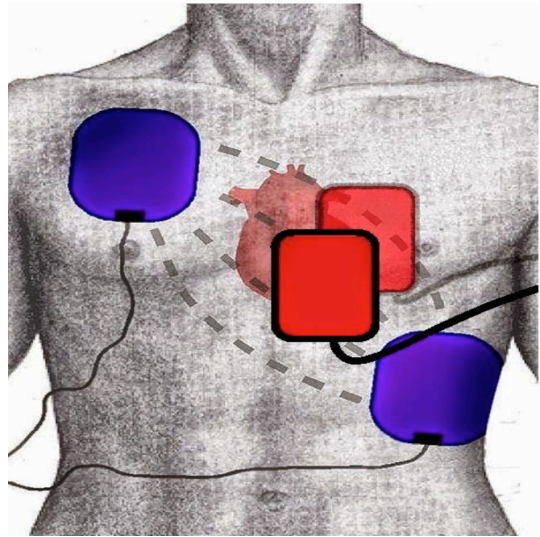
Pediatric cardiac arrest

After two attempts of defibrillation and the administration of an anti-arrhythmic, consider double sequential.

Place second set of pads on anterior and posterior regions from second monitor.

Simultaneously charge both monitors at 200 Joules.

Defibrillate the patient with both monitors simultaneously.



ROSC Achieved?

See Return of Spontaneous Circulation Protocol.

ROSC Not Achieved?

Return to needed cardiac arrest protocol.

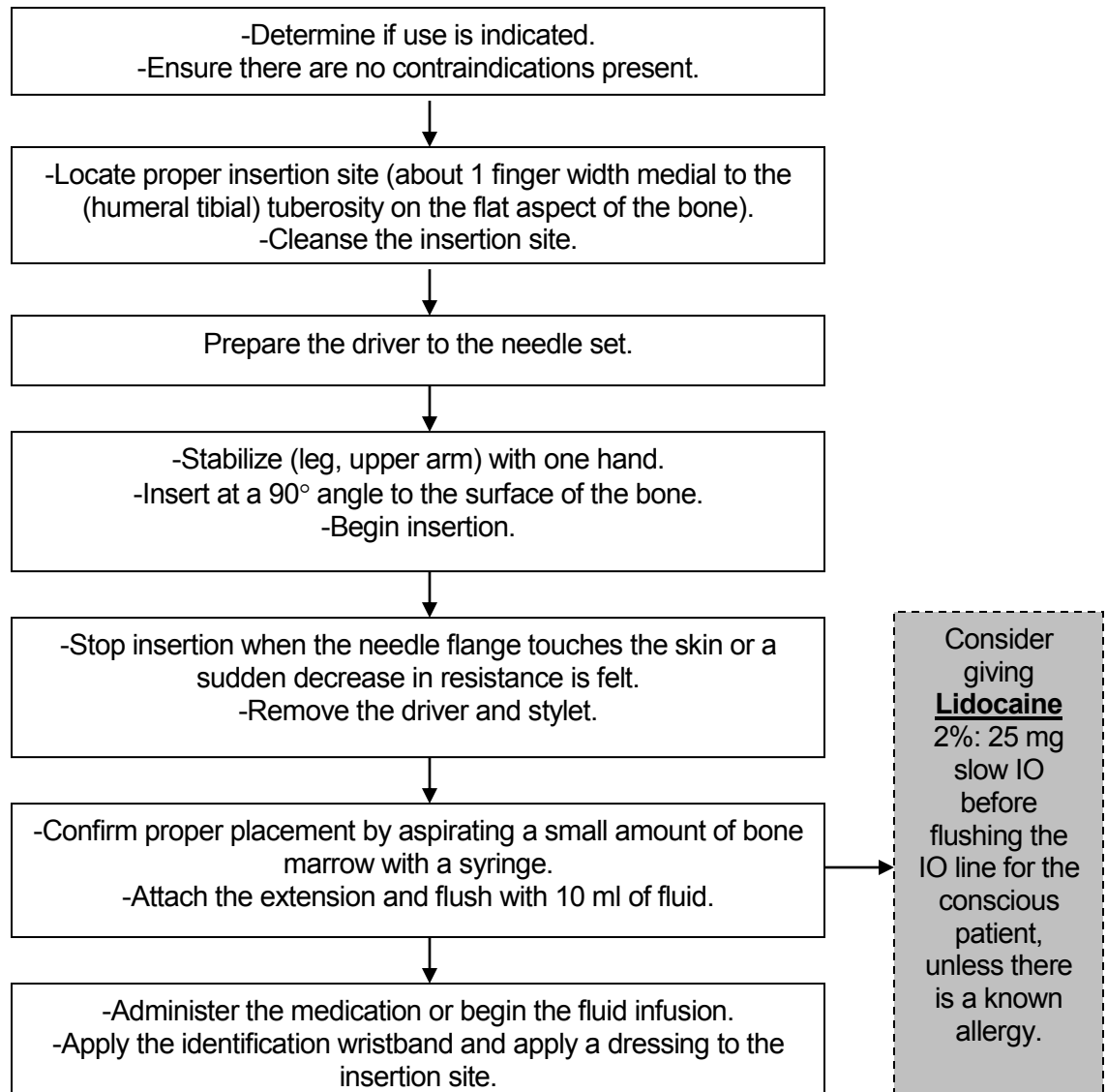
EZ-IO INSERTION PROCEDURE

Indications

1. For use whenever traditional IV access is not available, or after three unsuccessful IV attempts.
2. For use with patients between 3 –39 kg when peripheral IV access cannot be established in 2 attempts or 90 seconds.

Contraindications

1. Fracture of tibia, humerus or femur.
2. Previous orthopedic procedures (knee placements)
3. Any infection over the insertion site.
4. Inability to locate the three anatomical landmarks.
5. Excessive tissue over the insertion site.
6. Needles are **one time use only**.
7. **Never attempt a second insertion in the same leg.**



EZ-IO INSERTION CONTINUED

SHOULDER INSERTION

- In the event the lower extremities are unavailable for IO access, the head of the humerus may be used.
- Patients must be above the age of 8 and/or >40 kg.
- Patients should be in a supine position.



- Expose the shoulder and adduct the humerus.
- Place the patient's hand on the umbilicus and ensure the elbow remains in contact with the stretcher or ground.



- Palpate the humeral shaft with the thumb progressing superiorly toward the humeral head, palpating for the greater tubercle.
- As you near the shoulder you will feel a protrusion. This is the base of the greater tubercle, your insertion site.
- Confirm the insertion site by palpating the outer margins of the tubercle. This should be slightly anterior to the arms midline.
- Once the landmark is identified, follow the insertion procedure.

PEDIATRIC INSERTION

The tibial tuberosity may be difficult or impossible to palpate on very young patients. This insertion technique shall be used on patients 3-12 kg.

Insertion site is located 2 finger widths below the patella, then medial along the flat aspect of the tibia.



On patients 13-39 kg where the tibial tuberosity is palpable, the landmark is one finger width below the tibial tuberosity along the flat aspect of the tibia.



- Gently guide the needle in at a 90° angle.
- Do not push.**
- Allow the driver to do the work. Stop when you feel the pop.
- Excessive force can cause a widening of the needle entry point increasing the risk of extravasation.
- Flush the site with 5 ml of fluid.
- In patient's <40 kg, shoulder insertion is not allowed.**

IMPLANTABLE CARDIOVERTER - DEFIBRILLATOR (ICD)

DE-ACTIVATION

Indications

ICD malfunctioning causing **inappropriate shocks to rhythms other than VF/VT.**

Contraindications

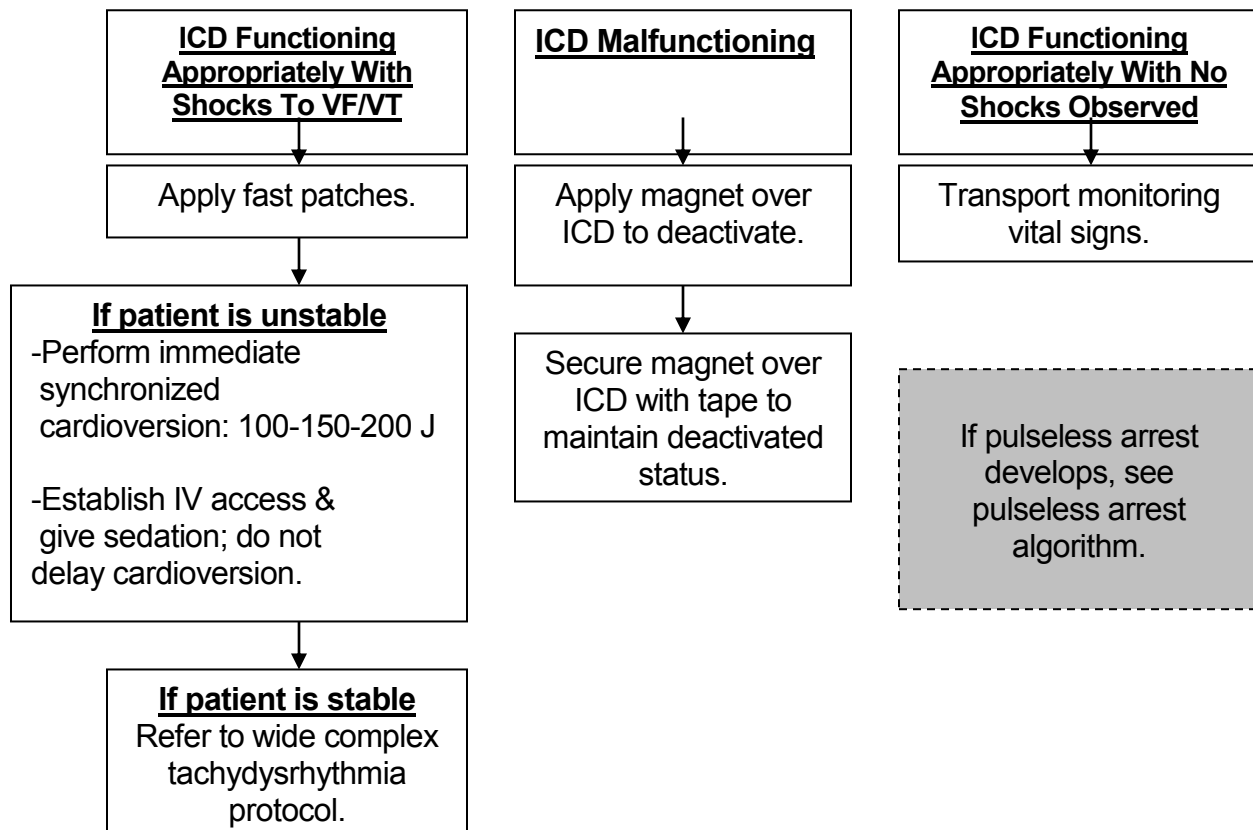
1. **ICD functioning properly and administering appropriate shocks to VF/VT.**
2. EMS not witnessing the shocks by the ICD.
3. Patient not on cardiac monitor.

Assessment

1. Assess and support ABC's.
2. Determine cardiac rhythm and monitor continuously.
3. Assess cardiac and respiratory systems and level of consciousness.

General Treatment

- Monitor blood pressure, oximetry, and administer oxygen as appropriate.
- Start an IV, TKO or as directed.



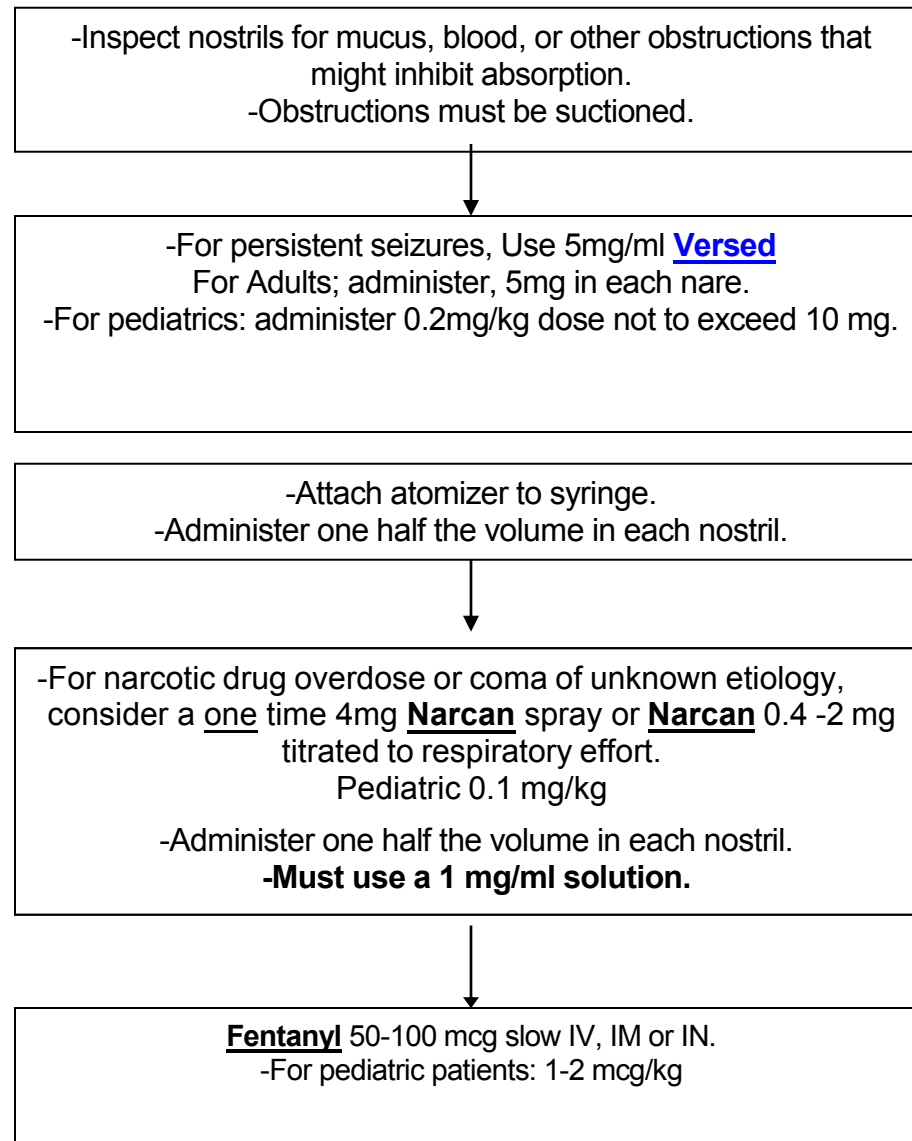
INTRANASAL ADMINISTRATION PROCEDURE

Indications

For the treatment of persistent seizure activity, narcotic drug overdose, Pain relief and sedation when IV access is unobtainable, or a life threatening condition exists.

Contraindications

1. When IV access is obtainable.
2. If unable to clear blood or mucus from nostrils.
3. Sensitivity to benzodiazepines.
4. 0.5 cc per nostril is considered well tolerated in adult and pediatric patients



NEEDLE / SURGICAL CRICOTHYROTOMY AIRWAY PROCEDURE

Indications

1. Upper airway obstruction that is not relieved by BLS measures or direct laryngoscopy with removal of object by magill forceps.
2. Severe facial trauma, which makes obtaining an airway by less invasive means impossible.

Contraindications

1. Neck swelling that obscures landmarks.
2. Not a prophylactic measure. Must actually need an airway and ventilation now.

Precautions

1. Contact medical control after procedure.

Technique

1. Locate the cricothyroid membrane between the thyroid and cricoid cartilage.
2. Cleanse the area with betadine or alcohol as time permits.
3. Using the scalpel provided in the kit, make a 2 cm vertical incision over the cricothyroid membrane.
4. Separate the incision and locate the cricothyroid membrane.
5. Using the scalpel, make an incision through the cricothyroid membrane.
6. Insert the dilator and tube into the trachea.
7. Remove the dilator and inflate the cuff of the tube.
8. Control any bleeding and apply sterile dressing around area of the incision.

Complications

1. Placement at the wrong level can cause laryngeal or tracheal damage.
2. Bleeding
3. Massive subcutaneous and/or mediastinal emphysema
4. Pneumothorax

PUSH DOSE EPINEPHRINE 1:100,000

Indications: Used to treat unremitting hypotension after appropriate fluid resuscitation in post-cardiac arrest, cardiogenic shock, or other medical etiology, in bradycardia if poor perfusion persists secondary to transcutaneous pacing and appropriate fluid resuscitation.

Contraindications: Cardiac Arrest, Trauma etiology, Pediatric patients or Known hypersensitivity

Precautions: Use caution in patients with suspected myocardial infarction or history of coronary artery disease. In patients taking digitalis, epinephrine may exacerbate ventricular ectopy.

Doses: 10-20mcg every 2-3 minutes if SBP < 90mmHg Route: IV/IO Protocol Pages: Bradycardia & Hypotension

Technique: Push Dose Epinephrine

1. Draw up 9mL of normal saline into an empty 10mL syringe
 - a. A 10mL flush may also be used after 1mL is wasted
2. Into the same syringe, draw up 1mL of epinephrine from the cardiac amp (1:10,000)
3. Syringe now contains 10mLs of Epinephrine at (1:100,000) or 10 mcg/mL
4. Dose is 1-2 mL every 2-3 minutes as needed
 - Onset- 1 minute
 - Duration- 5-10minutes
 - Dose-0.5ml-2ml every 2-5 minutes (5-20 mcg)

Mark or label syringe with the mix of 1:100,000 as soon as it's made up to prevent medication administration error.

PLEURAL DECOMPRESSION

PLACE OCCLUSIVE DRESSING

Indications

Tension pneumothorax should be considered when the following signs/symptoms are present:

1. Progressive severe respiratory distress.
2. Decreased or absent breath sounds on affected side.
3. Jugular vein distension.
4. Tracheal deviation away from the affected side.
5. Tympany to percussion on the affected side.
6. Cardiac compromise.

Contraindications

1. Not sure of diagnosis.
2. Clinical signs suggestive of diagnosis, but patient is not in shock and/or respiratory distress.

Precautions

1. Contact medical control after procedure.

Technique

1. Locate the second and third intercostal space in the mid-clavicular line on the affected side of the chest, or the fifth and sixth intercostal space in the **anterior**-axillary line.
2. Cleanse with betadine or alcohol preps as time allows.
3. Insert a 14-gauge catheter on the superior plane of the rib. Withdraw the needle and attach a flutter valve to the catheter. Secure catheter to the chest.
4. Re-evaluate respiratory status and vital signs.

Complications

1. Creation of a pneumothorax if one is not present.
2. Damage to lung or viscera.
3. Bleeding (intercostal vessels are below each rib).
4. Infection

CAPNOGRAPHY

Clinical Indications:

Capnography shall be used when available with the use of all invasive airway procedures including but not limited to endotracheal intubation, nasotracheal intubation, cricothyrotomy, or Blind Insertion Airway Devices (BIAD). Capnography shall also be used: In accordance with current ACLS recommendations for determining effectiveness of resuscitation efforts (10 mm/Hg or above), on all patients in respiratory distress, and all patients undergoing sedation.

Procedure:

1. Attach capnography sensors.
2. Note CO₂ level and waveform changes. These will be documented on each respiratory failure, cardiac arrest, or severe respiratory distress patient.
3. The capnometer shall remain in place with the airway and be monitored throughout the prehospital care and transport.
4. Any loss of CO₂ detection or waveform indicates an airway problem or device malfunction and should be documented.
5. The capnogram should be monitored as procedures are performed to verify or correct the airway problem.
6. Document the procedure and results on/with the Patient Care Report (PCR).

Sudden loss of waveform

- ET tube disconnected, dislodged, kinked or obstructed
- Loss of circulatory function



Bronchospasm ("Shark-fin" appearance)

- Asthma
- COPD

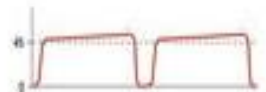


Decreasing EtCO₂

- ET tube cuff leak
- ET tube in hypopharynx
- Partial obstruction



Hypoventilation



Hyperventilation



CPR Assessment

- Attempt to maintain minimum of 10mmHg



Sudden increase in EtCO₂

- Return of spontaneous circulation (ROSC)



Decreased EtCO₂

- Apnea
- Sedation



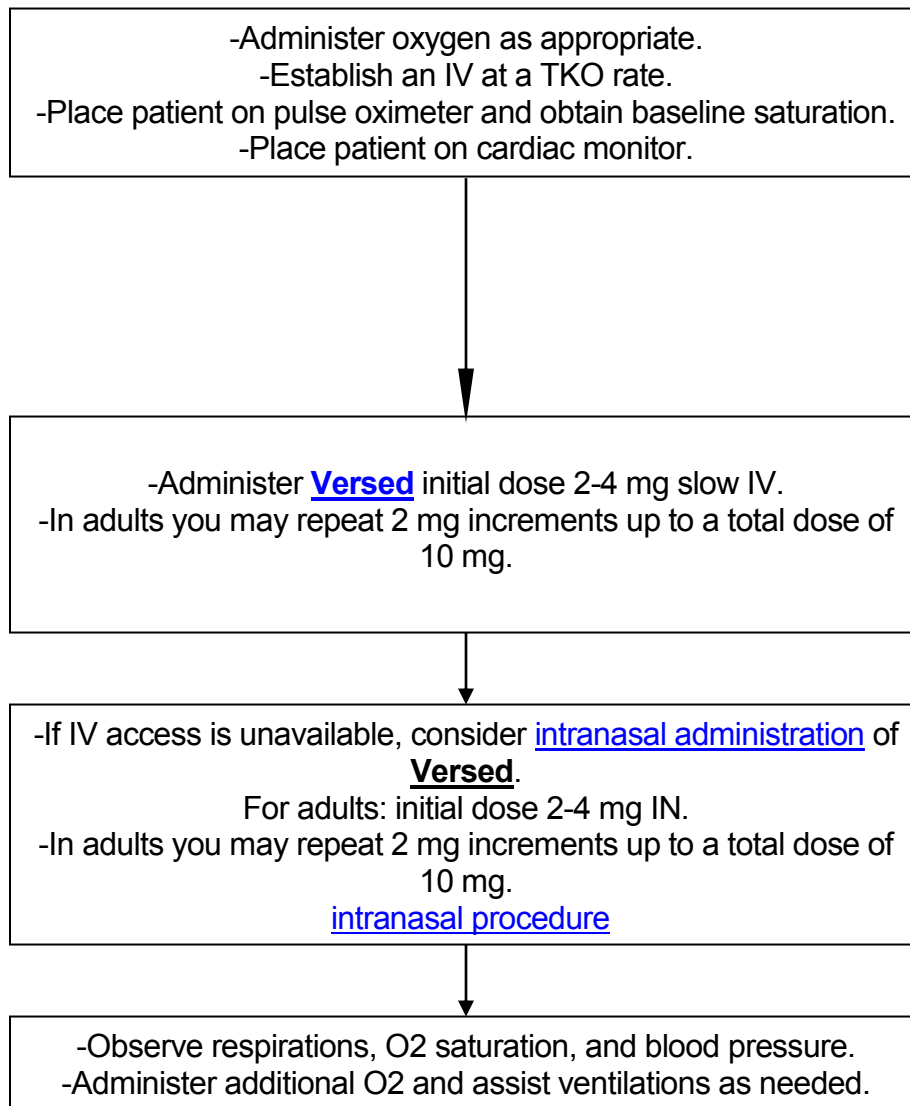
SEDATION

Indications

For patients about to undergo a painful procedure (cardioversion, conscious endotracheal intubation)

Contraindications

1. Known sensitivity or allergy to benzodiazepines
2. Shock



SPINAL CLEARANCE

Indications:

To allow Paramedics to perform field clearance from full spinal restrictions.

Contraindications

1. Any patient less than 12 years of age
2. Patients with chronic neck or back pain
3. Patients exhibiting signs of shock
4. Signs/symptoms of alcohol or drug intoxication
5. Altered Mental Status
6. Distracting Injuries

Assessment

All patients will be assessed by the following criteria. Only those of whom answer no to all requirements, may be cleared.

1. Does the patient have a GCS less than 15?
2. Does the patient complain of neck or back pain?
3. Is there tenderness, swelling, or deformity noted when the complete spine is palpated? You **must visualize and palpate the entire spine: neck to lower lumbar.**
4. Is there a distracting injury or distracting pain?
5. Are there signs/symptoms of alcohol or drug intoxication present?

-Full spinal precautions must take place if yes was answered to **any** of the above questions.

*Maintain a **high index of suspicion with older/elderly patients, as occult fractures are possible with benign mechanisms of injury or trauma.**

-If the patient meets the criteria to withhold spinal precautions, EMS providers may still elect to provide spinal precautions.

Examples of distracting injuries:

Extremity fractures, rib fractures, pelvic fractures, abdominal pain from trauma related injury, and other significant injuries.

Examples of signs/symptoms of intoxication:

GCS <15, slurred speech, dilated/constricted pupils, flushed skin, unsteady gait, irregular behavior, presence of paraphernalia or admitted use.

VITAL SIGNS

INDICATIONS:

1. A minimum of two sets of vitals will be obtained on every transported patient. This will include Pulse, Respirations, BP, Spo2 and alert and oriented status.
2. A minimum of one set of vitals will be taken on refusals.

Technique:

1. The first set of vitals a manual blood pressure and pulse will be taken on the following calls: chest pain, stroke, overdose, syncope, trauma, hypo/hypertension, anaphylaxis and any unconscious or critical patient.
2. A manual BP will be taken if the automated device is more than 30 points different than first set obtained (systolic or diastolic) in either direction.
3. A full set of vitals will be taken after each medication administration.
4. Capillary refill and/or an automated blood pressure device is expectable for pediatric patients.

ORTHOSTATIC VITAL SIGNS

Indications:

1. Syncope or Near Syncope (ie. dizzy, faint)

TECHNIQUE:

Patient in supine position for minimum of three minutes before taking lying blood pressure and pulse, then stand patient and immediately repeat blood pressure and pulse. If patient unable to stand, sit him upright with legs dangling over cot.

SPECIAL NOTES AND EVALUATION:

1. Orthostatic changes are not valid if:
 - a. Taken initially upright, then supine.
 - b. Taken upright with only head of cot elevated and legs not dangling.
 - c. If only pulse or blood pressure is checked by itself. Must have both blood pressure and pulses in both positions.
2. Significant orthostatic changes include:
 - a. Pulse increase greater than 20 beats per minute when moved upright.
 - b. Pressure (diastolic most significant) drop greater than 20 MM Hg when upright.
 - c. Pallor, diaphoresis, or faintness when upright.
3. Significant changes could indicate volume depletion (dehydration or blood loss), treat the symptoms, follow [DEHYDRATION](#) protocol if indicated.
4. Considerations:
 - a. Treat the symptoms.
 - b. Medications can often cause changes, usually indicated by pressure change but no pulse change.
 - c. May be due to slowing of compensatory mechanisms in the elderly.

KING AIRWAY LT-D

Indications

1. Patient who is apneic or unconscious and spontaneously breathing without an intact gag reflex, requiring airway management:
2. Two failed intubation attempts and one failed tube introducer attempt (see difficult airway algorithm).

Contraindications

1. Known esophageal disease (varicies, alcoholism, cirrhosis etc.).
2. Known ingestion of a caustic substance.
3. Patients who are conscious or who have an intact gag reflex.

If the patient regains consciousness or begins to fight the tube

1. You may remove the King Tube.
CAUTION: When removing the King Tube there is a strong possibility of emesis.
2. You may leave the King Tube in place and sedate the patient.

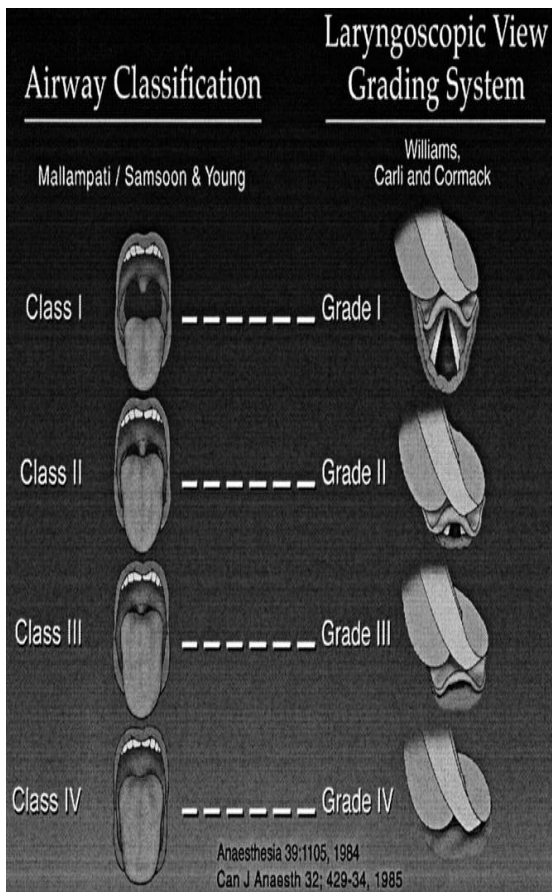
Endotracheal intubation

1. Endotracheal intubation provides a definitive airway. Every attempt following protocol should be made to secure the airway with an Endotracheal tube.
 - a. If the King Tube has been placed prior to your arrival and is functioning properly, the decision to leave it in place should be highly considered.
 - b. If deemed necessary by ALS personnel, an endotracheal tube may be inserted to take the place of the King Tube.

Additional notes

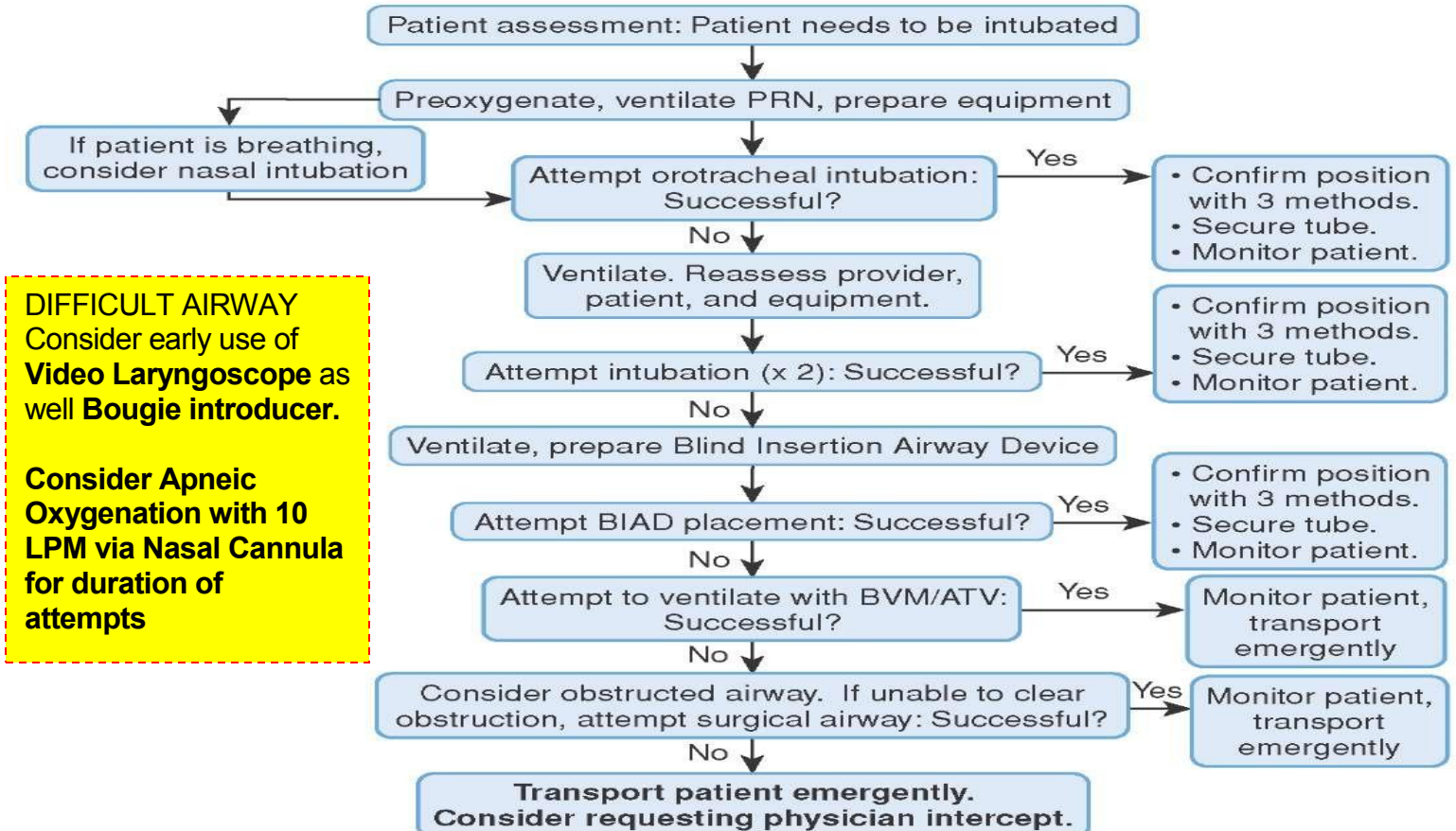
1. Placement of the King Tube should never delay:
 - a. CPR.
 - b. Basic airway management.
 - c. Use of the defibrillation unit.
 - d. Other necessary patient care.
2. Adequate ventilation should be assessed by the use of:
 - a. Standard checks for breath sounds and visual chest rise.
 - b. Capnography.
 - c. Pulse oximetry if available.

DIFFICULT AIRWAY ALGORITHM



- L=Look externally (facial trauma, large incisors, beard or moustache, and large tongue)
- E=Evaluate the 3-3-2 rule (incisor distance <3 fingerbreadths, hyoid/mental distance <3 fingerbreadths, thyroid-to-mouth distance <2 fingerbreadths)
- M=Mallampati (Mallampati score ≥ 3)
- O=Obstruction (presence of any condition that could cause an obstructed airway)
- N=Neck mobility (limited neck mobility).

For each patient, the airway assessment score was compared with the Cormack-Lehane laryngoscopic view seen during intubation (1=full view, 4=glottis not visualized). At intubation, 114 patients were classified as Cormack-Lehane grade 1 (defined by the authors as easy intubation), and 42 were classified as grade 2 or higher (defined by the authors as difficult intubation). Patients in the difficult-intubation group had significantly higher LEMON scores than did those in the easy-intubation group. Of the criteria used to calculate the score, only large incisors, inter-incisor distance <3 fingerbreadths, and thyroid-to-floor-of-mouth distance <2 fingerbreadths were associated significantly with difficult intubation.



MCI TRIAGE- START/JUMP START



START / Jump START Triage

START Triage is used to bring control to a seemingly overwhelming situation. Incidents which produce multiple casualties are rare but do occur and planning is paramount. A multiple casualty incident is defined as any incident where more casualties are present than initial response can reasonably handle. More response is needed for triage, treatment and transport than can arrive in a timely fashion.

Responders are also tasked with assuring / maintaining the scene safety as well as dealing with injury and illness. First arriving responders can become overwhelmed with patients presenting with a wide variety of injury and illness as well as those with no injury or illness.

JUMP START Triage is an objective pediatric patient multiple casualty triage tool that mirrors START in structure. It provides an objective structure to help assure responders triage children with their heads and not their hearts which can lead to over triage and diversion of precious resources from other patients who may need them more. Under triage is addressed as well by recognizing key differences between adult and pediatric physiology. *JUMP START should only be used with true multiple casualty incidents and disasters where resources for care are limited and should not be used for routine pre-hospital triage.*

Sorting / Triage:

Sort patients based on objective criteria in how they present. The severity of injury and therefore treatment / transport priority in START is color coded. Triage tags contain these colors so treatment and transport crews easily can see which patients have been triaged and to which level.

Color Codes:

GREEN: Minor injury / illness (walking wounded)
YELLOW: Delayed (can wait)
RED: Immediate
BLACK: Deceased

Scene Size Up:

1. Conduct a scene size up. Assure well being of responders. Determine or ensure scene safety before entering. If there are several patients with the same complaints consider HazMat, WMC or CO poisoning.
2. Take START system kit.
3. Determine number of patients. Communicate the number of patients and nature of the incident, establish command and establish a medical officer and triage officer if personnel available.

Triage:

Attempt to clear the walking wounded by announcing "if you are well enough to stand and walk from the scene do so now." Direct them to an area and consider a HazMat situation and follow decontamination procedure if indicated. If no evidence of contamination some of the walking wounded may be gathered to help you and contain the other GREEN TAG (walking wounded) in a separate area.

Start where you stand. Assess the first patient you encounter using the three objective criteria which can be remembered by RPM.

R: Respiratory
P: Perfusion
M: Mental Status

If your patient falls into the RED TAG category, stop, place RED TAG and move on to next patient. Attempt only to correct airway problems or treat uncontrolled bleeding before moving to next patient.

Treatment:

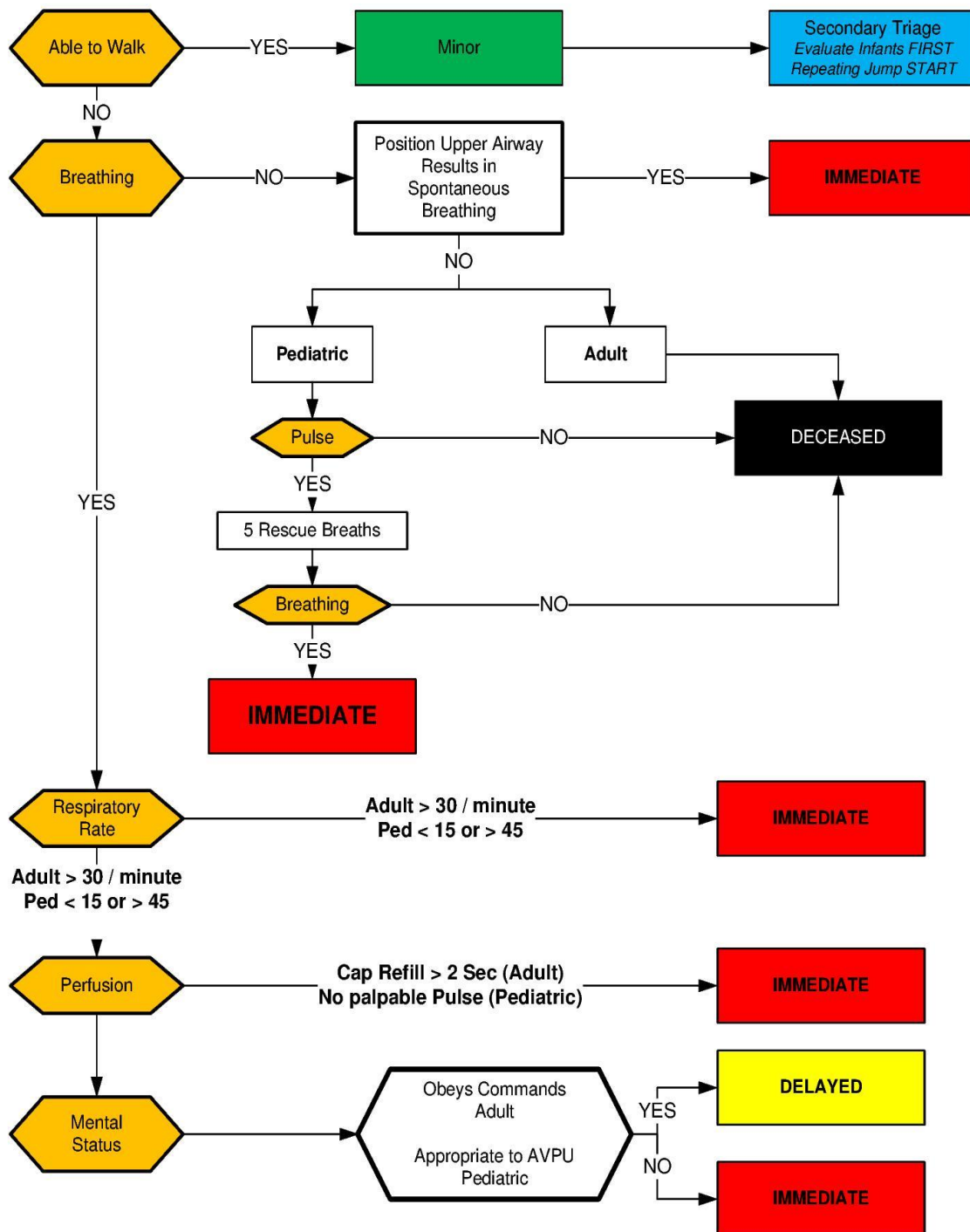
Once casualties are triaged focus on treatment can begin. You may need to move patients to treatment areas. RED TAGs are moved / treated first followed by YELLOW TAGs. BLACK TAGs should remain in place. As more help arrives then the triage / treatment process may proceed simultaneously.

Continue assessment of patients as they may be triaged up or down in severity as condition changes.

Pearls

- First evaluate all children who did not walk under their own power where possible and safety allows.
- All pre-hospital providers are encouraged to use the START algorithm any time there are more than 3 patients requiring evaluation, treatment or transport.

START / Jump START TRIAGE



Pearls

- First evaluate all children who did not walk under their own power.
- All prehospital providers are encouraged to use the START algorithm any time there are more than 3 patients requiring evaluation, treatment or transport.

Medical Emergency Triage Tag MT-137 Instructions



Know Simple Triage And Rapid Treatment (START)

NOTE: Diagonal tear-offs on upper corners may be used to indicate casualty positions at the MCI site, to attach to severed limbs, to place with personal effects, or for any additional use deemed necessary by local authorities.

FRONT

- A. Enter time and date of triage
- B. Enter name of patient (if conscious and coherent)
- C. Enter street address (if conscious and coherent)
- D. Enter city, state and zip (if conscious and coherent)
- E. Enter pertinent medical information and observations on blank lines
- F. Enter name of triage personnel on signature line (X)

BACK

- G. Indicate location of bodily injuries on front and back body diagrams
- H. Enter blood pressure, pulse, and respirations in vital signs chart with time taken. Indicate if pulse is full or weak, and if regular or irregular. Multiple vital signs may be taken.
- I. Enter intravenous (IV) and time
- J. Enter intramuscular (IM) and time

TRANSPORT PATIENT BY PRIORITY LEVELS

0 BLACK (TERMINAL - Dead or Expected)

I RED (CRITICAL - Primary Care)

II YELLOW (SERIOUS - Secondary Care)

III GREEN (NON-SERIOUS - Minor Care)

ALBUTEROL (PROVENTIL®, VENTOLIN®)

CLASS OF DRUG

Sympathomimetic, Beta 2 selective adrenergic bronchodilator

INDICATIONS

1. Albuterol is used to treat reversible airway obstruction caused by:
 - a. Wheezing associated with [asthma](#)
 - b. COPD

CONTRAINDICATIONS

1. Hypersensitivity

DRUG INTERACTION

1. Beta adrenergic agents - potentiates the effects
2. MAO inhibitors - may lead to hypertensive crisis
3. Beta adrenergic blockers - decreases the effectiveness

ADMINISTRATION

Nebulizer

Adult: [2.5mg] in 3 ml of sterile NS given as inhalation therapy over 5-15 minutes, may be repeated x 2.

Pediatric: [1.25-2.5 mg] in 3 ml of sterile NS given as inhalation therapy over 5-15 minutes, may be repeated x 2.

SPECIAL NOTES

1. Most side effects are dosage related.

ALBUTEROL & ATROVENT (Duoneb®)

PHARMACOLOGY & ACTIONS:

- A potent relatively selective beta 2 - adrenergic bronchodilator.
- Has occasional beta 1 overlap with clinically significant cardiac effects.

INDICATIONS:

- Bronchial [asthma](#)
- Reversible bronchial spasm that occurs with chronic pulmonary disease.

CONTRAINDICATIONS:

- Stop treatment if pulse increases by 20 bpm, frequent PVC's develop, any tachyarrhythmias other than sinus tachycardia appear, chest pain, apnea, nausea or vomiting, or increased shortness of breath occur.
- **Patients with allergies to soy or peanuts should not be administered Atrovent**

ADMINISTRATION:

- Nebulizer dosage for adults & children over 1 year - 2.5 mg mixed in 3 ml of saline for a concentration of .83 mg/ml.
- Infants under 1 year: Nebulizer dosage 0.03 ml/kg max dose 1ml.
- Nebulization:
 - Oxygen flow should be set at 6 Lpm.
 - Patients should inhale slowly & exhale passively through nose.

SIDE EFFECTS & SPECIAL NOTES:

- Patients with COPD should be monitored carefully for CO2 retention & decreased levels of consciousness.
- Paradoxical bronchospasm may occur with excessive administration.
- Skeletal muscle tremors
- Albuterol should be used with caution in pregnancy.
- Continually assess patient's respiratory rate, effort, and lung sounds.

ADENOCARD (ADENOSINE)

Indications: [PSVT](#)

Contraindications: Second or third degree heart block
Known hypersensitivity to the drug
Sick-sinus syndrome or symptomatic bradycardia
Atrial fibrillation or flutter

Side Effects: Dyspnea,
Chest pressure/pain
Dizziness
Facial flushing
Arrhythmias including blocks

Precautions: Patients with asthma
Children

Dose: Initial Dose: 6 mg given as a rapid I.V. push
Second: 12mg given as a rapid I.V. push
May repeat 12 mg dose, once.

Route: IV should be established in the AC, Adenocard should be administered into the medication port closest to the heart and followed by flushing the line.

AMIODARONE

Indications:	Ventricular Fibrillation Pulseless Ventricular Tachycardia Wide complex Tachydysrhythmias Premature ventricular contractions
Contraindications:	Known hypersensitivity to the drug or any of its components including iodine Sinus bradycardia Second or third degree heart block
Precautions:	Exacerbation of presenting arrhythmia can occur with concurrent use of other antiarrhythmic medications (whatever antiarrhythmic you start with, stay with) CHF Children
Side Effects:	Prolonged QT interval Hypotension Bradycardia
Doses:	<u>VF/Pulseless VT</u> : initial dose is 300 mg IV. Repeat doses at 150 mg every 3-5 min up to 2.2 g <u>Wide complex tach</u> : 150 mg mixed in a 100 ml bag of NS, given over 10 min (15 mg/min). May repeat at 150 mg every 10 minutes up to 2.2g. <u>Premature ventricular contractions (unstable)</u> : 150 mg mixed in a 100 ml bag of NS, given over 10 minutes (15 mg/min). May repeat same dose every 10 min up to a 2.2 g total.
Route:	IV, IO

ASPIRIN

Indications: [Chest pain](#) suggestive of AMI

Contraindications: Known hypersensitivity to the drug
Patients with bleeding disorders
Children <12 with flu-like symptoms
Peptic ulcer

Precautions: GI bleeding or upset stomach
Patients with asthma

Side Effects: Heartburn
Nausea
Wheezing
GI bleeding
Reye's syndrome in children

Dose: 324 mg

Route: Oral/must be chewed

ATROPINE

Indications: [Bradycardia](#)
Organophosphate/Carbonates poisoning

Contraindications: Tachycardia/Tachydysrhythmias

Side Effects: Palpitations
Tachycardia
PVCs
HTN
Pupil dilation
Nausea/vomiting

Doses: Unstable bradycardia: 0.5 mg initially, then 1 mg IV every 3-5 minutes to a total dose of 3mg.

Type II Second Degree AV block and Third Degree AV block:
0.5 mg IV while awaiting pacer. May repeat to a total dose of 3 mg. If ineffective begin pacing.

Organophosphate/Carbonate poisoning: administer [Atropine](#) 2 mg SIVP. If no tachycardia or pupil dilation, give second dose of 2mg SIVP.

Pediatric Dose: Refer to Broselow Tape

Route: IV, IO, ET tube, IM (only for organophosphate poisoning)

BENADRYL (DIPHENHYDRAMINE)

Indications: [Allergic Reactions](#)
Anaphylaxis with respiratory distress
Anaphylactic shock
Extrapyramidal Symptoms (EPS)
Dystonic Reactions

Precautions: Patient with asthma
Pregnancy
Lactating females

Side Effects: Drowsiness
Headache
Palpitations
Dizziness
Dilated pupils
Thickening of bronchial secretions

Dose: 25-50 mg

Pediatric Dose: 1 to 2 mg/kg

Routes: IV, IO or IM

CALCIUM CHLORIDE 10%

Indications: [Calcium channel blocker](#) overdoses with hypotension. Examples of calcium channel blockers are Amlodipine (Norvasc), and most drugs ending in “ipine,” that are used to reduce vascular resistance and arterial pressure

Contraindications: [Hypercalcemia](#)
Ventricular Fibrillation
Digitalis toxicity

Precautions: Patients on digitalis
Children
Renal disease and respiratory disease
Make sure IV is completely patent because it will cause severe tissue necrosis.
Will precipitate with NaHCO₃

Side Effects: Pain or burning at IV site
Necrosis of tissue
Shortened QT
Heart block
Arrhythmias

Dose: 500 to 1000 mg (5-10 ML) over 5-10 minutes

Route: IV, IO

DILTIAZEM HCL (CARDIZEM®)

CLASS OF DRUG

Calcium Channel Blocker; Coronary Vasodilator, Antidysrhythmic

INDICATIONS

1. Symptomatic [Atrial Fibrillation](#) or Atrial Flutter with rapid ventricular response
2. Symptomatic Paroxysmal Supraventricular Tachycardia

CONTRAINDICATIONS

1. Sick sinus syndrome except in the presence of a functioning ventricular pacemaker.
2. Patients with second- or third degree AV block except in the presence of a functioning ventricular pacemaker.
3. Patients with severe hypotension, cardiogenic shock or severe CHF
4. Patients who have demonstrated hypersensitivity to the drug
5. Patients with atrial fibrillation or atrial flutter associated with an accessory bypass tract such as in WPW syndrome or short PR syndrome
6. Patients with ventricular tachycardia
7. Pregnancy
8. Children under the age of 15

WARNING- DILTIAZEM use in severe [congestive heart failure](#); patients with reduced ventricular function may not be able to counteract the inotropic and chronotropic effects of diltiazem, the result being an even higher compromise of function.

DRUG INTERACTION

Additive effects in prolonging AV conduction when using beta-blockers or digitalis concomitantly with Diltiazem HCl.

ADMINISTRATION (For patients over age 70 decrease dose by 5mg)

0.25 mg/kg max 20mg SIVP over 2 minutes, continue to monitor the PT, if hypotension develops, DC medication and consider a fluid bolus. Repeat 12 lead ECG if rhythm converts. If no response after 10 minutes, consider a **second dose of 0.35mg/kg max 20mg SIVP** over 2 minutes. If hypotension develops DC medication and consider a fluid bolus. Repeat 12 lead ECG if the rhythm converts.

SPECIAL NOTES

**Pediatric: Not usually used.*

**DO NOT TREAT BRADYCARDIA WITH ATROPINE, DILTIAZEM WILL WEAR OFF.*

**For patients over age 70 decrease dose by 5mg.*

Dextrose: D10 & Oral Glucose

Indications: Hypoglycemia (< 60 mg/dL) and Altered Mental Status (GCS < 14)

Contraindications: Hyperglycemia

Delirium tremens
Cranial/spinal hemorrhage
CHF

Precautions: May cause necrosis if infiltrated into tissue

Side Effects: Local venous irritation
Pulmonary edema
Hyperglycemia

Dose: For **adults**, give **D10 (250mL) 25g** or equivalent IV.

Pediatric Dose: Children **greater** than 8 years of age should receive **2 mL/kg of D10 IV**
Children **less** than 8 years of age should receive **1 mL/kg of D10 IV**

****Refer to the Broselow tape for weight estimation as needed.****

Oral Dose: 15g oral glucose may be considered for awake, cooperative patients with a patent airway and the ability to swallow

DOPAMINE

Indications:	Cardiogenic shock (hypotension) Type II second degree AV block Third degree AV block
Contraindications:	Hypersensitivity VF Tachydysrhythmias
Precautions:	Peripheral vascular disease Will cause tissue necrosis with extravasation of IV
Side Effects:	Headache Palpitations Tachycardia HTN Ectopic beats Wide QRS complex Nausea/vomiting Necrosis of tissue
Doses:	<u>Type II second degree & Third degree AV blocks:</u> 2-10 mcg/kg/min <u>Hypotension of cardiac origin:</u> 2-10 mcg/kg/min. Titrate to a blood pressure of 100.
Route:	IV drip only

EPINEPHRINE 1:1,000

Indications: Asthma
Mild anaphylaxis
Anaphylaxis with respiratory distress
Anaphylactic shock
Children with unstable bradydysrhythmias

Contraindications: Over 40 years old **AND**
History of CAD
Heart rate over 120

Must contact Medical Control prior to administering to above patients

Precautions: Pregnancy
HTN

Side Effects: Tremors
Anxiety
Palpitations
Tachycardia
HTN
Dysrhythmia
Dyspnea

Doses: Severe Asthma: 0.3 mg IM

Mild anaphylaxis/Anaphylaxis with respiratory distress: 0.3 – 0.5 mg IM

Anaphylactic shock: 0.3 – 0.5 mg max given via injection in the venous plexus at the base of the tongue.

Pediatric Dose: Refer to Broselow Tape

Route: IV, IO, IM, IN, ET

EPINEPHRINE 1:10,000

Indications: Asystole/PEA
Persistent or recurrent VF/VT
Unstable brady dysrhythmias in children
Pediatric Anaphylaxis

Contraindications: None in pulseless arrest situations

Precautions: None in pulseless arrest situations

Doses: Pulseless Arrest: 1 mg IV/IO every 3-5 minutes or 2 mg ET

Pediatric/Neonatal Dose:

Anaphylaxis 0.01 mg/kg

Route: IM (Refer to Broselow Tape)

Cardiac 0.01 mg/kg

Route: IV, IO, and ET (Refer to Broselow Tape)

Route: IM, IV, IO, ET x 2

FENTANYL

Indications:	Pain from the following: An extremity injury A shoulder or hip injury Back pain and Abdominal pain (non-traumatic) Burns without airway or respiratory compromise - (Contact Med Control)
Contraindications:	Known hypersensitivity
Precautions:	Respiratory depression Must be given slow or it can cause muscle rigidity (stone chest or rigid chest syndrome)
Side Effects:	Muscle rigidity Bradycardia Respiratory depression Laryngospasm
Dose:	50-100 mcg slow IV for extremity pain 25-50 mcg slow IV for non-traumatic back or abdominal pain Must contact medical control for use with burns.
Pediatric	1 to 2 mcg/kg IV, IN
Dose: Route:	IV, IO, IN

GLUCAGON

Must transport patient after administration.

Indications: Hypoglycemia (< 60 mg/dL) and Altered Mental Status (GCS <14) **when unable to obtain IV access**

Contraindications: **IV/IO access**
Known sensitivity

Precautions: Only effective in patients with sufficient amounts of glucose stored in the liver (glycogen). So use with caution in patients with prolonged fasting, starvation, or chronic hypoglycemia.

Secondary hypoglycemia can occur, and is counteracted by sufficient carbohydrate intake.

Pregnancy

Side Effects: Nausea
Vomiting
Hypotension
Hyperglycemia

Adult Dose: 1 mg

Pediatric Dose: Pediatrics **greater** than **25kg** give **1 mg**
Pediatrics **less** than **25kg** give **0.5 mg**

Route: IM

Ketamine

Indications: Extreme agitation/Hyper-agitated delirium syndrome

Contraindications: Known sensitivity, hypoxia, head or airway trauma

Precautions: Use with caution in intoxicated patients or patients who have already received sedating medications.

Time to onset: IM dosing 3-4 minutes

Side effects: 1. Blood pressure and pulse are frequently elevated.

2. Can result in patient dizziness and nausea/vomiting.

3. Laryngospasm and hypersalivation.

4. Emergence Phenomenon in 10-20% of adults.

5. Possible intercerebral and interocular pressure increase.

Route: IM only

Pediatric use not approved at the time.

Dose: 1. <70 kg (<154lb): 200mg IM

2. 70kg to 100kg (154lb to 220lb); 300mg IM

3. 100kg to 130kg (220lb to 286lb); 400mg IM

4. >130kg (>286lb); 500mg IM

5. If the medication is given as an adjunct (i.e. the patient already has been given sedating medication or is intoxicated with sedating substances such as alcohol or benzodiazepines), a lower dose will be needed. 2 mg/kg is recommended as the dose in this situation.

Dissociation could occur at 3-5 mg/kg IM. May require repeat dosing based on frequent assessment. Emergence reactions are typically self limited, and can be treated with benzodiazepines. Pretreatment not proven to be effective in preventing this. Patients require constant monitoring.

Medical control should be contacted for additional doses or subsequent treatment with any further potentially sedating medications. This protocol is for a one time dose of Ketamine only.

LIDOCAINE

Indications: VENTRICULAR FIBRILLATION or PULSELESS VENTRICULAR TACHYCARDIA

Frequent PVC's of 6 or more per minute

For anesthesia prior to fluid and drug administration through IO on a conscious patient.

Contraindications: High-degree heart blocks (2nd degree type II, 3rd degree, Bi-fascicular blocks)
Supraventricular dysrhythmias
Adams-Stokes syndrome
Wolff-Parkinson-White syndrome

Side Effects: Hypotension
Heart block
Dizziness
Convulsions

Dose: VF or Pulseless VT: 1 to 1.5 mg/kg initial dose. Repeat at 0.5 to 0.75 mg/kg, max 3 doses or 3 mg/kg. If down the ET tube, double the dose.

IO insertion: 25 mg slow IO before flushing the IO line

Pediatric Dose: Refer to Broselow Tape

Routes: IV, ET, IO

MAGNESIUM SULFATE

Indications: Torsades de pointes
Eclampsia
Asthma
COPD

Contraindications: Renal disease
Acute surgical abdomen
Nausea/Vomiting

Side Effects: Sharply lowered blood pressure
Respiratory depression

Precautions: Pregnancy

Doses: Torsades: 1 to 2 g SIV/IO

Eclampsia: 2 to 4 g SIVP.

Severe Asthma: Consider 1 gm drip over 10 minutes in 100 mL bag. Recheck vital signs, and then give the Second 1g dose over 10 minutes.

Severe COPD: Consider 1gm drip over 10 min in 100 ML bag. Re-evaluate recheck vitals and then give the second 1g dose over 10 minutes.

Routes: IV, IO

METHYLPREDNISOLONE (MEDROL®, SOLU-MEDROL®)

CLASS OF DRUG

Glucocorticoid; anti-inflammatory; immunosuppressant

INDICATIONS

1. Asthma/COPD Exacerbation
2. Anaphylaxis

CONTRAINDICATIONS

1. Hypersensitivity

DRUG INTERACTION

1. Potential hypokalemia may increase risk of digitalis toxicity
2. **May increase insulin requirement**
3. Additive hypokalemia with diuretic

ADMINISTRATION

2 mg/kg IV up to 62.5 mg

SPECIAL NOTES

Use on patients with a history of diabetes.

MORPHINE SULFATE

Indications:	Chest pain unrelieved by NTG Pain from orthopedic trauma or burns Back pain & Abdominal pain (Non-Traumatic) CHF
Contraindications:	Known sensitivity Systolic BP < 100
Precautions:	Pregnancy Respiratory depression
Side Effects:	Nausea/vomiting Respiratory depression Lowered BP
Dose:	2 mg increments SIVP until the pain is resolved or a total dose of 10 mg is given. 2-4 mg for non-traumatic abdominal or back pain For chest pain: 4 mg can be given initially, followed by 2 mg increments <u>Must contact medical control for use with burns per Med Director.</u>
Pediatric Dose:	0.05 mg/kg slow IV/IN to a maximum dose of 2 mg.
Route:	IV, IM

NARCAN (NALOXONE)

Indications: Narcotic overdoses
Coma or confusion of unknown origin
Neonatal resuscitation

Contraindications: Known sensitivity

Precautions: Pregnancy
Rapid administration or high dose may cause Severe withdrawal effects in patients dependent on narcotics.

Side Effects: Abrupt administration may cause nausea, vomiting, sweating, tachycardia, increased blood pressure, seizures

Dose:

Adult: 0.4 mg increments to a max of 2 mg SIVP, **titrated** to achieve satisfactory respiratory effort. Monitor **EtCO₂**.

If unable to obtain IV access consider IN administration using a one time **4 mg Narcan** nasal spray or **Narcan 0.4 -2 mg** titrated to respiratory effort.

Pediatric: Refer to Broselow Tape

Neonatal resuscitation: 0.1 mg/kg

Routes: IV, IO, ET, IN

NITROGLYCERIN

Indications: Chest pain
CHF

Contraindications: Systolic BP <100
Pulse rate < 50 and >140
Known sensitivity
Cerebral hemorrhage

Precautions: Pregnancy

Side Effects: Headache
Postural hypotension
Flushing
Dizziness

Dose: Chest Pain: 0.4 mg every five minutes until chest pain is relieved or systolic blood pressure becomes <100 or 3 doses are given.

****Patients taking NTG on a regular basis, may be given NTG prior to obtaining IV access.***

CHF: 0.4 mg. May be repeated every five minutes until distress is relieved or systolic blood pressure <100 or 3 doses.

Route: Sublingual

NITROUS OXIDE

Indications: Pain associated with isolated orthopedic extremity trauma

Contraindications: Multi-systems trauma
Chest trauma
Head trauma
COPD

Dose: Self-administered

Route: Inhalation

SODIUM BICARBONATE

Indications:	Near drowning with prolonged submersion Tricyclic antidepressant overdose with wide complex tachycardia VF/Pulseless VT
Contraindications:	Respiratory Alkalosis Metabolic Alkalosis
Precautions:	May cause alkalosis or tetany
Dose:	1 meq/kg Must contact medical control prior to administering for tricyclic antidepressant overdoses that is not in cardiac arrest.
Route:	IV, IO

TERBUTALINE SULFATE

INDICATIONS:

- ❑ Moderate to severe bronchial asthma
- ❑ Reversible bronchospasm in COPD

ADMINISTRATION:

- ❑ **SQ only**

DOSAGE:

ADULT:

0.25 mg (0.25 ml)

PEDIATRIC:

0.01 mg/kg **Maximum Dose = 0.25 mg**

THERAPEUTIC EFFECTS:

- Bronchodilator through selective beta 2 stimulation
- Relatively little effect on heart rate
- Relaxes bronchial smooth muscle

RELATIVE CONTRAINDICATIONS:

- Angina
- Dysrhythmias
- Hypertension
- Hypersensitivity to the drug

SIDE EFFECTS:

- Palpitations
- Tremor
- Drowsiness
- Headache
- Sweating
- Nausea
- Muscle cramping

SPECIAL NOTES / RESTRICTIONS:

- May repeat dosage in 15-30 minutes
- Do **not** give with other sympathomimetic drugs

TRANEXAMIC ACID (TXA)

Indications:	Adult patients within 3 hours of traumatic injury with suspected hemorrhage. Signs/Symptoms of shock (hypotension, tachycardia, pallor, etc.)
Contraindications:	Pediatric patients Isolated traumatic brain injury Known thrombotic disease (DVT, PE) Known allergy
Side Effects:	Dizziness Nausea Hypotension (If administered too fast) Visual disturbance (blurred vision, color distortion)
Dose:	1 gram in 100mL of NS over 10 minutes
Route:	IV drip
Classification:	Anti-Fibrinolytic

VERSED (MIDAZOLAM)

Indications: Seizures
Sedation
Excited Delirium

Contraindications: Known sensitivity to the drug or benzodiazepines
Shock
Pregnancy

Precautions: COPD
CHF

Side Effects: Respiratory depression
Apnea
Hypotension
Nausea/vomiting

Dose: Seizures: seizure activity is present, 5-10 mg SIVP, may repeat in 10-15 minutes. You may also consider IN administration adult 5 mg in each nare.

Sedation: 2-4 mg may repeat 2 mg increments up to a total dose of 10 mg. If IV access is unavailable consider IN administration at a dose of 2-4mg.

Excited Delirium: 5 mg IM may repeat up to a total dose of 10 mg. If IV access is unavailable consider IN administration. Contact Medical control immediately for higher dosing. Be prepared to aggressively maintain airway.

Pediatric Dose: Seizures: 0.2 mg/kg IN, not to exceed 10 mg.

Sedation: 0.05 mg/kg slow IV (maximum initial dose 2-4 mg).
0.2 mg/kg IN, not to exceed 10 mg.

Route: IV, IO, IM, IN

XOPENEX

Indications: Asthma/COPD

Contraindications: Known sensitivity
Heart rate greater than 150

Side Effects: Increased heart rate
Increased blood pressure

Dose: 1.25 mg nebulized. May repeat to a total of 3 treatments or heart rate exceeds 150 bpm.

Pediatric Dose: For children 1.25 mg mixed in 3 ml of NS nebulized.

Route: Nebulized

ZOFRAN (ONDANSETRON)

Classification:

Anitemetic

Actions/Pharmacology:

Blocks the effects of serotonin at 5-HT₃ receptor sites (selective antagonist) located in vagal nerve terminals and the chemoreceptor trigger zone in the CNS.

Indications:

Nausea and vomiting, regardless of mechanism.

Contraindications:

Age < 4 years

Parkinson's patients receiving Apomorphine

Side and Adverse effects:

Headache

Sedation

Constipation

Diarrhea

Transient blindness

May cause transient dizziness after IV administration

May cause prolonged QT interval and/or Torsade de Pointes

May cause transient oculogyric crisis and dystonic reactions

Adult Administration:

4mg IM or orally dissolving tablet (ODT)

4mg slow IVP over 2 to 5 minutes

Must contact medical control for additional doses.

Pediatric Administration:

Must be 4 years of age and older

2mg IM or ODT

2mg slow IVP over 2 to 5 minutes

Must contact medical control for additional doses.

Interactions:

When taken with apomorphine, may cause hypotension and ALOC

Precautions:

Tablet should not be handled with moist or wet hands

May cause prolonged QT interval and/or Torsade de Pointes

Patients receiving IV administration should be placed on cardiac monitor

Special Considerations:

Tablet should be placed onto tongue and allowed to dissolve.

Reference Pages

Tricyclic Antidepressant List, Common

The TCAs include the following agents which are predominantly [serotonin](#) and/or [norepinephrine reuptake inhibitors](#):

- [Amitriptyline](#) (Tryptomer, Elavil)
- [Amitriptylinoxide](#) (Amioxid, Ambivalon, Equilibrin)
- [Butriptyline](#) (Evadyne)
- [Clomipramine](#) (Anafranil)
- [Demexiptiline](#) (Deparon, Tinoran)
- [Desipramine](#) (Norpramin, Pertofrane)
- [Dibenzepin](#) (Noveril, Victoril)
- [Dimetacrine](#) (Istonil, Istonyl, Miroistonil)
- [Dosulepin/Dothiepin](#) (Prothiaden)
- [Doxepin](#) (Adapin, Sinequan)
- [Imipramine](#) (Tofranil, Janimine, Praminiil)
- [Imipraminoxide](#) (Imiprex, Elepsin)
- [Lofepramine](#) (Lomont, Gamanil)
- [Melitracen](#) (Deanxit, Dixeran, Melixeran, Trausabun)
- [Metapramine](#) (Timaxel)
- [Nitroxazepine](#) (Sintamil)
- [Nortriptyline](#) (Pamelor, Aventyl, Norpress)
- [Noxiptiline](#) (Agedal, Elronon, Nogedal)
- [Pipofezine](#) (Azafen/Azaphen)
- [Propizepine](#) (Depressin, Vagran)
- [Protriptyline](#) (Vivactil)
- [Quinupramine](#) (Kevopril, Kinupril, Adeprim, Quinuprine)

As well as the following atypical compounds:

- [Amineptine](#) (Survector, Maneon, Directim) - [Norepinephrine-dopamine reuptake inhibitor](#)
- [Iprindole](#) (Prondol, Galatur, Tetran) - [5-HT₂ receptor antagonist](#)
- [Opipramol](#) (Insidon, Pramolan, Ensidon, Oprimol) - [σ receptor agonist](#)
- [Tianeptine](#) (Stablon, Coaxil, Tatinol) - [Selective serotonin reuptake enhancer](#)
- [Trimipramine](#) (Surmontil) - 5-HT₂ receptor antagonist

Calcium Channel Blockers

Calcium channel blockers are called CCBs for short. They are also called calcium channel antagonists. CCBs are the most widely prescribed medicines in the United States today. They can be used alone or in a combination pill with an **ACE inhibitor**

When are calcium channel blockers prescribed?

In addition to treating those with high blood pressure, calcium channel blockers are also used for people who have heart disease. Because they allow the blood vessels that supply the heart to open wider, more oxygen-rich blood can get to the heart. They are also used to treat **arrhythmia**.

Blockers

The following table lists some common brand and generic names for calcium channel blockers.

Brand	Generic
Adalat CC	Nifedipine
Calan SR	verapamil HCl
Cardene	nicardipine HCl
Cardene SR	nicardipine HCl
Cardizem CD	diltiazem HCl
Cardizem SR	diltiazem HCl
Covera-HS	verapamil HCl
Dilacor XR	diltiazem HCl
DynaCirc	Isradipine
DynaCirc CR	Isradipine
Isoptin SR	verapamil HCl
Nimotop	Nimodipine
Norvasc	amlodipine besylate
Plendil	Felodipine
Procardia XL	Nifedipine
Sular	Nisoldipine
Tiamate	diltiazem HCl
Tiazac	diltiazem HCl
Vascor	bepridil HCl
Verelan	verapamil HCl
Verelan PM	

OUTSIDE THE HOSPITAL DO-NOT-RESUSCITATE (OHDNR) ORDER

Out of Hospital DNR FAQ

OUT OF HOSPITAL DNR (OHDNR) FREQUENTLY ASKED QUESTIONS

Is my existing DNR form now illegal?

No, any existing DNR form is still valid. It does not carry liability protection provided for the form developed in the statute or regulation

Can the next of kin act as a surrogate and sign the new purple OHDNR form?

No, only the patient or "patient representative" may sign the form. There is no language allowing surrogate in the statute.

What is a patient representative?

As defined, a patient representative is, (1) An attorney in fact designated in a durable power of attorney for health care for a patient determined to be incapacitated under sections 404.800 to 404.872, RSMo; or (2) A guardian or limited guardian appointed under Chapter 475, RSMo, to have responsibility for an incapacitated patient."

Are there instruction that will tell me how to complete the form?

Yes, the form's reverse side has instruction on how to complete the form. The form can be obtained after August 30, 2009 from www.dhss.mo.gov/ems/forms.html or from the Bureau of EMS by calling (573) 751-6356.

The form on the website is purple, but I don't have a color printer. What should I do?

Complete the form on your computer and print it on purple card stock.

Is there a specific color of purple designated?

No, any purple paper is accepted

Can my OHDNR form be used while I am in the hospital?

The form is developed only to be used outside the hospital. That decision would need to be made by the hospital

What is a patient?

"Patient", a person eighteen years of age or older who is not incapacitated, as defined in section 475.010, RSMo, and who is otherwise competent to give informed consent to an outside the hospital do-not-resuscitate order at the time such order is issued, and who, with his or her attending physician, has executed an outside the hospital do-not-resuscitate order under sections 190.600 to 190.621. A person who has a patient's representative shall also be a patient for the purposes of sections 190.600 to 190.621, if the person or the person's patient's representative has executed an outside the hospital do-not-resuscitate order under sections 190.600 to 190.621"

What is a “health care facility”?

"Health care facility", any institution, building, or agency or portion thereof, private or public, excluding federal facilities and hospitals, whether organized for profit or not, used, operated, or designed to provide health services, medical treatment, or nursing, rehabilitative, or preventive care to any person or persons. Health care facility includes but is not limited to ambulatory surgical facilities, health maintenance organizations, home health agencies, hospices, infirmaries, renal dialysis centers, long-term care facilities licensed under sections 198.003 to 198.186, RSMo, medical assistance facilities, mental health centers, outpatient facilities, public health centers, rehabilitation facilities, and residential treatment facilities;

Can a photocopy or fax be used instead of the original?

Yes, maybe photocopied or faxed, and this photocopy or other complete facsimile of the original OHDNR order may be used for any purpose for which the original OHDNR order may be used.

If you have questions in your local area, please contact your local ambulance service provider. For General questions you may contact the Bureau of EMS at (573)751-6356

PEDIATRIC EMERGENCY FORM



Children's Mercy Hospitals and Clinics
2401 Gillham Road
Kansas City, MO 64108

Patient Name: Zzpmtest, Cools
Location: Hematology Oncology Clinic
Attending Provider: August, MD, Keith J
DOB/Age/Sex: 01/24/2002 11 years Female

Admit Date:
Disch Date:
MRN: 01436200
Acct: 341998912

Pediatric Emergency Form

Document Name: Pediatric Emergency Form
Event Date: 01/20/2014 14:17 CST
Result Status: In Progress
Reviewed By:
Signed By:
Signed Date/Time:

PEDIATRIC EMERGENCY FORM---EMS PLEASE READ

POTENTIAL LIFE THREATENING EVENTS	MANAGEMENT
* * * * * *	* * * * * *

CODE STATUS- CONFIRM CODE STATUS WITH FAMILY

ADVERSE REACTIONS/ALLERGIES

Adverse Reaction/Allergy: amoxicillin Type: Allergy/Hypersensitivity Severity: Stop Substance:
Moderate Reaction:
Adverse Reaction/Allergy: morphine Type: Allergy/Hypersensitivity Severity: Unknown Reaction:

Baseline Vital Signs

T: _ C HR: _ RR: _ O2 SAT: _

BP-Systolic range: _ BP-Diastolic Range: _

Baseline Neurologic Status: _

Baseline Physical Findings: _

Baseline Ancillary Findings:(lab/xray/ECG): _

Prosthesis/Appliances/Advanced Technology: _

MEDICATIONS

Current medications as of 01/20/2014 14:17

Vyvanse 40 mg oral capsule 40 mg (1 capsule) Take with food. by mouth once a day (in the morning)

Chart Request ID: 19447730

Patient Name: Zzpmtest, Cools

Print Date/Time: 1/20/2014 14:19

MRN: 01436200

Page 1 of 2
COPY

Pediatric Emergency Form

Document Name: Pediatric Emergency Form Signed By:
Event Date: 01/20/2014 14:17 CST Signed Date/Time:
Result Status: In Progress
Reviewed By:

Prevacid 30 mg oral delayed release capsule 30 mg (1 capsule) by mouth every day
Nebulizer Machine Use as directed. Indication: Asthma/RAD w/o status (493.90)
Prevacid 30 mg oral delayed release capsule 30 mg (1 capsule) by mouth every day (**Prescription Called-in to Pharmacy**)
clonidine 0.3 mg oral tablet 0.3 mg (1 tablet) by mouth 4 times a day 30 day(s) (**Prescription Called-in to Pharmacy**)
clonidine 0.3 mg oral tablet 0.3 mg (1 tablet) by mouth 4 times a day 30 day(s) (**Prescription Called-in to Pharmacy**)
ibuprofen 100 mg oral tablet 150 mg (1.5 tablet) by mouth every 6 hours as needed for Fever or Mild Pain (**Printed Prescription Provided**)

Problem List:

1. _
2. _
3. _
4. _
5. _

Synopsis of Problems: _

Comments on child, family or other specific medical issues: _

Preferred Hospital: *Children's Mercy Hospital and Clinics*

Physician List		
PCP:	Phone:	Fax:
Physician:	Specialty:	Phone:
Physician:	Specialty:	Phone:
Physician:	Specialty:	Phone:
Physician:	Specialty:	Phone:
Physician:	Specialty:	Phone:

AFTER HOUR CONTACT NUMBER FOR CHILDREN'S MERCY HOSPITAL
LOCAL CALL 816-234-3000
TOLL FREE CALL 1-866-512-2168

Copyright, 2013. The Children's Mercy Hospital. All rights reserved

Performed by: Trisha L Williams RN

Chart Request ID: 19447730
Patient Name: Zzpmtest, Cools

Print Date/Time: 1/20/2014 14:19
MRN: 01436200

Page 2 of 2
COPY

Children's Mercy Hospital & Clinics | Kansas City

TPOPP for EMS

What is TPOPP?

Transportable Physician Orders for Patient Preferences (TPOPP) is a program designed to improve the quality of care people receive at the end of life by translating patient/resident goals and preferences into medical orders for treatment.

What is the TPOPP form?

The TPOPP form is a bright pink piece of paper that can communicate patient preferences regarding life-sustaining medical treatment. When signed by a patient (or a recognized decision maker) and a physician licensed in Kansas or Missouri, it becomes a medical order. Providers at all sites of care should follow the medical orders.

What types of medical orders are on the form?

The form includes medical orders and patient preferences regarding:

- ☐ CPR
- ☐ Implementing comfort care
- ☐ Defibrillation and cardioversion
- ☐ Intubation and mechanical ventilation
- ☐ Transport decisions

Who might have a TPOPP form?

TPOPP is designed for persons who have advanced progressive chronic illness or are terminally ill. EMS may see TPOPP forms for persons who:

- ☐ Live independently;
- ☐ Reside in a long-term care or assisted living facility;
- ☐ Receive hospice care.

Does every person with advanced chronic or terminal illness have a form?

No. TPOPP is voluntary and based on the belief that an individual has the right to make his or her own health care decisions. TPOPP forms can be used by persons of any age with an advanced progressive illness or who is terminally ill.

How is a TPOPP form completed?

The TPOPP form is completed after having a conversation with the patient (or recognized decision maker) and a member of the healthcare team such as a physician, nurse, social worker or chaplain. When the patient (or recognized decision maker) and a physician sign the form it is complete. Both signatures are necessary. The signatures verify that the preferences on the form are consistent with the patient's values and treatment goals.

What do you do with a completed form?

The TPOPP form travels with the patient between care settings. The original form should accompany the patient and be presented to each provider (first responder, hospital or other care setting).

Where is the form kept?

EMS personnel should look for or ask for the form. It should be conspicuously located:

- ☐ In the front of the patient's medical chart at care facilities;
- ☐ On the refrigerator, by a phone, or bedside when the patient resides at home;

The TPOPP form should go with the patient when the person is transferred from the hospital or health care setting.

How is the form used by EMS?

EMS should administer the level of treatment ordered in Section A and in Section B when they arrive on scene. EMS will transfer the individual to the emergency room if ordered or contact medical control for further instructions.

What sections apply to EMS providers?

Section A orders apply only when the person is in cardiopulmonary arrest. If the "Attempt Resuscitation/CPR" box is checked then CPR should be performed. If the "Do Not Attempt Resuscitation (DNAR/no CPR/Allow Natural Death)" box is checked, CPR should not be performed. Comfort measures are provided and CPR is not attempted. Section B orders apply to emergency medical circumstances for a person who has a pulse and/or is breathing. These range from full treatment to comfort measures only. If all life-sustaining treatments are desired the "Full Treatment" box is checked. However, if the person desires some limitation, then either the "Comfort Measures Only" box or the "Limited Additional Interventions" box is checked. EMS first will administer the level of emergency medical services ordered and then contact medical control as needed. Comfort care is always provided regardless of indicated level of EMS treatment. If Attempt Resuscitation in Section A is checked, then Full Treatment in Section B is checked. Full resuscitation implies full treatment. Section D validates the order with the dated signatures of the physician and patient (or recognized decision maker).

When can EMS honor the TPOPP form?

EMS personnel can honor the TPOPP form in any situation in which a valid TPOPP form is presented for the patient requiring emergency medical care.

Will the person still have or need a signed Outside the Hospital Do Not resuscitate (DNR) form?

Your organization/agency will determine the specific policies associated with TPOPP. Remember, that if DNAR is selected in Section A and the form is valid, you have a physician's order not to resuscitate.

Is a copy or faxed copy of the form acceptable?

Yes – A copy, faxed copy, or electronic representation of the TPOPP form is acceptable provided it has the person's name and signatures of the physician and patient (or recognized decision maker).

Where can I get more information?

Visit the website of the Center for Practical Bioethics at www.practicalbioethics.org, send an e-mail to TPOPP@practicalbioethics.org or call the TPOPP Managing Director at 816-979-1366.

TPOPP FORM

- For Educational Purposes Only -

FORM SHALL ACCOMPANY PERSON WHEN TRANSFERRED OR DISCHARGED												
Kansas – Missouri Transportable Physician Orders for Patient Preferences (TPOPP) This Physician Order set is based on the patient's current medical condition and preferences. Any section not completed indicates full treatment for that section. Photocopy or fax copy of this form is valid.												
Last Name:	First Name:	Middle Initial:										
Date of Birth:	Last 4 SSN:	Gender: M F										
A. CHECK ONE	CARDIOPULMONARY RESUSCITATION (CPR): Person has no pulse and is not breathing. If patient is not in cardiopulmonary arrest, follow orders in B and C. ☐ Attempt Resuscitation/CPR (Selecting CPR in Section A requires selecting Full Treatment in Section B) ☐ Do Not Attempt Resuscitation (DNAR/no CPR/Allow Natural Death)											
B. CHECK ONE	MEDICAL INTERVENTIONS: Person has pulse and/or is breathing. ☐ Comfort Measures Only. Treat with dignity and respect. Keep clean, warm, and dry. Use medication by any route, positioning, wound care and other measures to relieve pain and suffering. Use oxygen, suction and manual treatment of airway obstruction as needed for comfort. Transfer to hospital only if comfort needs cannot be met in current location. TREATMENT GOAL: ATTEMPT TO MAXIMIZE COMFORT THROUGH SYMPTOM MANAGEMENT ONLY. ☐ Limited Additional Interventions. In addition to care described in Comfort Measures Only, use medical treatment, antibiotics, and IV fluids as indicated. Do not intubate. May use non-invasive positive airway pressure. Generally avoid intensive care. Transfer to hospital only if treatment needs cannot be met in current location. TREATMENT GOAL: ATTEMPT TO RESTORE FUNCTION WITH TREATMENTS FOR REVERSIBLE CONDITIONS. ☐ Full Treatment. In addition to care described in Comfort Measures Only and Limited Additional Interventions, use intubation, advanced airway interventions, mechanical ventilation, and defibrillation/cardioversion as indicated. Transfer to hospital if indicated. Includes intensive care. TREATMENT GOAL: ATTEMPT TO PROLONG LIFE BY ALL MEDICALLY EFFECTIVE MEANS. Additional Orders: _____											
C. CHECK ONE	MEDICALLY ADMINISTERED NUTRITION: Offer food by mouth if feasible and desired. ☐ No medically administered nutrition, including feeding tubes. ☐ Medically administered nutrition, including feeding tubes, for trial period: _____ ☐ Long term medically administered nutrition, including feeding tubes Additional Orders: _____											
D. CHECK ALL THAT APPLY	INFORMATION AND SIGNATURES Discussed with: ☐ Patient/Resident ☐ Agent/DPOA healthcare ☐ Parent of minor ☐ Legal guardian ☐ Health care surrogate ☐ Other <i>(specify)</i> : _____ Signature of patient or recognized decision maker By signing this form, the recognized decision maker acknowledges that this request regarding above treatment measures is consistent with the known desires, and with the best interest, of the individual who is the subject of the form. <table style="width: 100%; border: none;"> <tr> <td style="width: 33%; border: none;">Print name:</td> <td style="width: 33%; border: none;">Signature <i>(required)</i>:</td> <td style="width: 33%; border: none;">Relationship <i>(write "self" if patient)</i>:</td> </tr> <tr> <td style="border: none;">Address:</td> <td colspan="2" style="border: none;">Phone:</td> </tr> </table> Signature of physician My signature below indicates to the best of my knowledge that these orders are consistent with the patient's medical condition and preferences. <table style="width: 100%; border: none;"> <tr> <td style="width: 60%; border: none;">Print physician Practitioners: Go to www.practicalbioethics.org for TPOPP resources</td> <td style="width: 40%; border: none;">Physician phone:</td> </tr> <tr> <td style="border: none;">Physician signature <i>(required)</i>:</td> <td style="border: none;">Date:</td> </tr> </table>		Print name:	Signature <i>(required)</i> :	Relationship <i>(write "self" if patient)</i> :	Address:	Phone:		Print physician Practitioners: Go to www.practicalbioethics.org for TPOPP resources	Physician phone:	Physician signature <i>(required)</i> :	Date:
Print name:	Signature <i>(required)</i> :	Relationship <i>(write "self" if patient)</i> :										
Address:	Phone:											
Print physician Practitioners: Go to www.practicalbioethics.org for TPOPP resources	Physician phone:											
Physician signature <i>(required)</i> :	Date:											

HIPAA PERMITS DISCLOSURE TO HEALTH CARE PROFESSIONALS AND PROXY DECISION MAKERS AS NECESSARY FOR TREATMENT

Apneic Oxygenation

Prior to intubation apply a nasal cannula and flow at 8-10 LPM

In an experiment by Frumin et al., patients were pre-oxygenated, intubated, paralyzed, and placed on an anesthesia machine that provided 1.0 FiO₂ and no ventilations. These patients were maintained in this apneic state for between 18 and 55 min. none of these patients desaturated below 98%, despite being paralyzed and receiving no breaths. Although their CO₂ levels rose, their oxygenation was maintained due to apneic oxygenation. Oxygen was absorbed from the patients' alveoli by pulmonary blood flow; this established a gradient for the continued pull of oxygen from the endotracheal tube and anesthesia circuit. In another study, Teller et al. showed that pharyngeal insufflation with oxygen significantly extended the time to desaturation during apnea.

Numerous studies on apneic oxygenation during brain death testing confirm that even without any respiratory effort, oxygen saturation can be maintained. If a continuous path of oxygen is maintained from the pharynx to the glottis during the apneic period of RSI, the patient will continue to oxygenate. This has led us to perform a jaw thrust in all high-risk patients during their apneic period. In some cases, we also place nasopharyngeal airways to augment the passage of oxygen. These techniques, combined with high-flow O₂ from a High flow cannula, NRB mask, NIV mask, or the facemask of a BVM, will allow continued apneic oxygenation.

NIH Stroke Scale

NIH STROKE SCALE

Patient Identification: _____

Pt. Date of Birth: ____/____/____

Hospital: _____ (____-____)

Date of Exam: ____/____/____

Interval: ☐ Baseline ☐ 2 hours post treatment ☐ 24 hours post onset of symptoms ± 20 minutes ☐ 7-10 days
☐ 3 months ☐ Other: _____ (____)

Time: ____:____ ☐ am ☐ pm

Person Administering Scale: _____

Administer stroke scale items in the order listed. Record performance in each category after each subscale exam. Do not go back and change scores. Follow directions provided for each exam technique. Scores should reflect what the patient does, not what the clinician thinks the patient can do. The clinician should record answers while administering the exam and work quickly. Except where indicated, the patient should not be coached (i.e., repeated requests to patient to make a special effort).

Instructions	Scale Definition	Score
1a. Level of Consciousness: The investigator must choose a response if a full evaluation is prevented by such obstacles as an endotracheal tube, language barrier, orotracheal trauma/bandages. A 3 is scored only if the patient makes no movement (other than reflexive posturing) in response to noxious stimulation.	0 = Alert; keenly responsive. 1 = Not alert; but arousable by minor stimulation to obey, answer, or respond. 2 = Not alert; requires repeated stimulation to attend, or is obtunded and requires strong or painful stimulation to make movements (not stereotyped). 3 = Responds only with reflex motor or autonomic effects or totally unresponsive, flaccid, and areflexic.	_____
1b. LOC Questions: The patient is asked the month and his/her age. The answer must be correct - there is no partial credit for being close. Aphasic and stuporous patients who do not comprehend the questions will score 2. Patients unable to speak because of endotracheal intubation, orotracheal trauma, severe dysarthria from any cause, language barrier, or any other problem not secondary to aphasia are given a 1. It is important that only the initial answer be graded and that the examiner not "help" the patient with verbal or non-verbal cues.	0 = Answers both questions correctly. 1 = Answers one question correctly. 2 = Answers neither question correctly.	_____
1c. LOC Commands: The patient is asked to open and close the eyes and then to grip and release the non-paretic hand. Substitute another one step command if the hands cannot be used. Credit is given if an unequivocal attempt is made but not completed due to weakness. If the patient does not respond to command, the task should be demonstrated to him or her (pantomime), and the result scored (i.e., follows none, one or two commands). Patients with trauma, amputation, or other physical impediments should be given suitable one-step commands. Only the first attempt is scored.	0 = Performs both tasks correctly. 1 = Performs one task correctly. 2 = Performs neither task correctly.	_____
2. Best Gaze: Only horizontal eye movements will be tested. Voluntary or reflexive (oculocephalic) eye movements will be scored, but caloric testing is not done. If the patient has a conjugate deviation of the eyes that can be overcome by voluntary or reflexive activity, the score will be 1. If a patient has an isolated peripheral nerve palsy (CN III, IV or VI), score a 1. Gaze is testable in all aphasic patients. Patients with ocular trauma, bandages, pre-existing blindness, or other disorder of visual acuity or fields should be tested with reflexive movements, and a choice made by the investigator. Establishing eye contact and then moving about the patient from side to side will occasionally clarify the presence of a partial gaze palsy.	0 = Normal. 1 = Partial gaze palsy; gaze is abnormal in one or both eyes, but forced deviation or total gaze paresis is not present. 2 = Forced deviation, or total gaze paresis not overcome by the oculocephalic maneuver.	_____

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05/23/2025

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N I H STROKE SCALE

Patient Identification: _____

Pt. Date of Birth: ____/____/____

Hospital: _____

Date of Exam: ____/____/____

Interval: ☐ Baseline ☐ 2 hours post treatment ☐ 24 hours post onset of symptoms ± 20 minutes ☐ 7-10 days
☐ 3 months ☐ Other: _____

3. Visual: Visual fields (upper and lower quadrants) are tested by confrontation, using finger counting or visual threat, as appropriate. Patients may be encouraged, but if they look at the side of the moving fingers appropriately, this can be scored as normal. If there is unilateral blindness or enucleation, visual fields in the remaining eye are scored. Score 1 only if a clear-cut asymmetry, including quadrantanopia, is found. If patient is blind from any cause, score 3. Double simultaneous stimulation is performed at this point. If there is extinction, patient receives a 1, and the results are used to respond to item 11.	0 = No visual loss. 1 = Partial hemianopia. 2 = Complete hemianopia. 3 = Bilateral hemianopia (blind including cortical blindness).	_____
4. Facial Palsy: Ask – or use pantomime to encourage – the patient to show teeth or raise eyebrows and close eyes. Score symmetry of grimace in response to noxious stimuli in the poorly responsive or non-comprehending patient. If facial trauma/bandages, orotracheal tube, tape or other physical barriers obscure the face, these should be removed to the extent possible.	0 = Normal symmetrical movements. 1 = Minor paralysis (flattened nasolabial fold, asymmetry on smiling). 2 = Partial paralysis (total or near-total paralysis of lower face). 3 = Complete paralysis of one or both sides (absence of facial movement in the upper and lower face).	_____
5. Motor Arm: The limb is placed in the appropriate position: extend the arms (palms down) 90 degrees (if sitting) or 45 degrees (if supine). Drift is scored if the arm falls before 10 seconds. The aphasic patient is encouraged using urgency in the voice and pantomime, but not noxious stimulation. Each limb is tested in turn, beginning with the non-paretic arm. Only in the case of amputation or joint fusion at the shoulder, the examiner should record the score as unstable (UN), and clearly write the explanation for this choice.	0 = No drift; limb holds 90 (or 45) degrees for full 10 seconds. 1 = Drift; limb holds 90 (or 45) degrees, but drifts down before full 10 seconds; does not hit bed or other support. 2 = Some effort against gravity; limb cannot get to or maintain (if cued) 90 (or 45) degrees, drifts down to bed, but has some effort against gravity. 3 = No effort against gravity; limb falls. 4 = No movement. UN = Amputation or joint fusion, explain: _____ 5a. Left Arm _____ 5b. Right Arm _____	_____
6. Motor Leg: The limb is placed in the appropriate position: hold the leg at 30 degrees (always tested supine). Drift is scored if the leg falls before 5 seconds. The aphasic patient is encouraged using urgency in the voice and pantomime, but not noxious stimulation. Each limb is tested in turn, beginning with the non-paretic leg. Only in the case of amputation or joint fusion at the hip, the examiner should record the score as unstable (UN), and clearly write the explanation for this choice.	0 = No drift; leg holds 30-degree position for full 5 seconds. 1 = Drift; leg falls by the end of the 5-second period but does not hit bed. 2 = Some effort against gravity; leg falls to bed by 5 seconds, but has some effort against gravity. 3 = No effort against gravity; leg falls to bed immediately. 4 = No movement. UN = Amputation or joint fusion, explain: _____ 6a. Left Leg _____ 6b. Right Leg _____	_____

Rev 10/1/2003

N I H STROKE SCALE

Patient Identification. _____

Pt. Date of Birth _____

Hospital _____

Date of Exam _____

Interval: ☐ Baseline ☐ 2 hours post treatment ☐ 24 hours post onset of symptoms ± 20 minutes ☐ 7-10 days
☐ 3 months ☐ Other _____

7. Limb Ataxia: This item is aimed at finding evidence of a unilateral cerebellar lesion. Test with eyes open. In case of visual defect, ensure testing is done in intact visual field. The finger-nose-finger and heel-shin tests are performed on both sides, and ataxia is scored only if present out of proportion to weakness. Ataxia is absent in the patient who cannot understand or is paralyzed. Only in the case of amputation or joint fusion, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice. In case of blindness, test by having the patient touch nose from extended arm position.	0 = Absent. 1 = Present in one limb. 2 = Present in two limbs. UN = Amputation or joint fusion, explain: _____	_____
8. Sensory: Sensation or grimace to pinprick when tested, or withdrawal from noxious stimulus in the obtunded or aphasic patient. Only sensory loss attributed to stroke is scored as abnormal and the examiner should test as many body areas (arms [not hands], legs, trunk, face) as needed to accurately check for hemisensory loss. A score of 2, "severe or total sensory loss," should only be given when a severe or total loss of sensation can be clearly demonstrated. Stuporous and aphasic patients will, therefore, probably score 1 or 0. The patient with brainstem stroke who has bilateral loss of sensation is scored 2. If the patient does not respond and is quadriplegic, score 2. Patients in a coma (item 1a=3) are automatically given a 2 on this item.	0 = Normal; no sensory loss. 1 = Mild-to-moderate sensory loss; patient feels pinprick is less sharp or is dull on the affected side; or there is a loss of superficial pain with pinprick, but patient is aware of being touched. 2 = Severe to total sensory loss; patient is not aware of being touched in the face, arm, and leg.	_____
9. Best Language: A great deal of information about comprehension will be obtained during the preceding sections of the examination. For this scale item, the patient is asked to describe what is happening in the attached picture, to name the items on the attached naming sheet and to read from the attached list of sentences. Comprehension is judged from responses here, as well as to all of the commands in the preceding general neurological exam. If visual loss interferes with the tests, ask the patient to identify objects placed in the hand, repeat, and produce speech. The intubated patient should be asked to write. The patient in a coma (item 1a=3) will automatically score 3 on this item. The examiner must choose a score for the patient with stupor or limited cooperation, but a score of 3 should be used only if the patient is mute and follows no one-step commands.	0 = No aphasia; normal. 1 = Mild-to-moderate aphasia; some obvious loss of fluency or facility of comprehension, without significant limitation on ideas expressed or form of expression. Reduction of speech and/or comprehension, however, makes conversation about provided materials difficult or impossible. For example, in conversation about provided materials, examiner can identify picture or naming card content from patient's response. 2 = Severe aphasia; all communication is through fragmentary expression; great need for inference, questioning, and guessing by the listener. Range of information that can be exchanged is limited; listener carries burden of communication. Examiner cannot identify materials provided from patient response. 3 = Mute, global aphasia; no usable speech or auditory comprehension.	_____
10. Dysarthria: If patient is thought to be normal, an adequate sample of speech must be obtained by asking patient to read or repeat words from the attached list. If the patient has severe aphasia, the clarity of articulation of spontaneous speech can be rated. Only if the patient is intubated or has other physical barriers to producing speech, the examiner should record the score as untestable (UN), and clearly write an explanation for this choice. Do not tell the patient why he or she is being tested.	0 = Normal. 1 = Mild-to-moderate dysarthria; patient slurs at least some words and, at worst, can be understood with some difficulty. 2 = Severe dysarthria; patient's speech is so slurred as to be unintelligible in the absence of or out of proportion to any dysphasia, or is mute/anarthric. UN = Intubated or other physical barrier, explain: _____	_____

Rev 10/1/2003

N I H STROKE SCALE

Patient Identification. _____

Pt. Date of Birth ____/____/____

Hospital _____ (____-____)

Date of Exam ____/____/____

Interval: ☐ Baseline ☐ 2 hours post treatment ☐ 24 hours post onset of symptoms ± 20 minutes ☐ 7-10 days
☐ 3 months ☐ Other _____ (____)

11. Extinction and Inattention (formerly Neglect): Sufficient information to identify neglect may be obtained during the prior testing. If the patient has a severe visual loss preventing visual double simultaneous stimulation, and the cutaneous stimuli are normal, the score is normal. If the patient has aphasia but does appear to attend to both sides, the score is normal. The presence of visual spatial neglect or anosagnosia may also be taken as evidence of abnormality. Since the abnormality is scored only if present, the item is never untestable.	0 = No abnormality. 1 = Visual, tactile, auditory, spatial, or personal inattention or extinction to bilateral simultaneous stimulation in one of the sensory modalities. 2 = Profound hemi-inattention or extinction to more than one modality; does not recognize own hand or orients to only one side of space.	_____
---	--	--------------

Rev 10/1/2003

You know how.

Down to earth.

I got home from work.

**Near the table in the dining
room.**

**They heard him speak on the
radio last night.**

MAMA

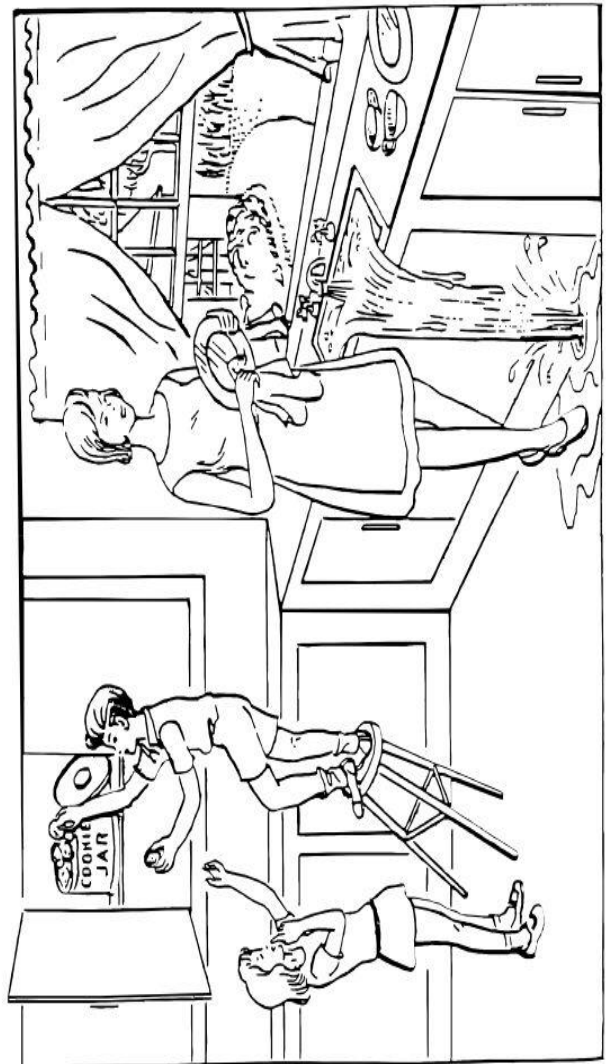
TIP – TOP

FIFTY – FIFTY

THANKS

HUCKLEBERRY

BASEBALL PLAYER



Restraint Checklist

Patient's Name: _____

Report Number: _____

1. Reason for restraint (check all that apply):

- ☐ Patient attempting to hurt self
- ☐ Patient attempting to hurt others
- ☐ Patient attempting to remove medical necessary devices
- ☐ Patient under arrest by Law enforcement (If yes then Law Enforcement Officer MUST accompany the patient and attending paramedic in the back of the ambulance.)

2. Attempted verbal assurance/redirection?

- ☐ Yes
- ☐ No

3. Attempted environmental modification?(i.e. remove patient from stressful environment)

- ☐ Yes
- ☐ No

4. Received medical control order for restraints?

- ☐ Yes
- ☐ No

5. Time and Type of restraint applied (circle all that apply):

Date: ____/____/____ Time: _____

Limb restraints: RUE, LUE, LLE, RLE

Chemical restraint: Yes No If Yes: Drug Used &

Dose: _____

6. Vital signs and extremity neurovascular exam should be taken every 10 min.

7. Transport Position (Patient should NOT be in prone position)

- ☐ Supine position for transport
- ☐ Lateral recumbent position for transport